Can Religion Bridge Ethnic Divides?

Pastors, Chiefs, and a Field Experiment on Collective Action

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Ethnic Fragmentation Weakens Public Goods Provision

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$\Rightarrow$ Who, then, can bridge ethnic divides?

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- Yet evidence on whether religious leaders can sustain cooperation in ethnically divided societies remains scarce.

Research Questions

1. Can religious leaders promote cooperation in ethnically divided societies?

2. If so, does this ability come from their different *scope of authority*, or their different *source of enforcement*?

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Sample: 1,040 residents across 336 work groups.

Outcomes: Daily work records, transcripts of meetings, and survey outcomes.

Context Experiment

Contribution

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1. **Ethnic fragmentation and collective action**
   - Fragmentation is associated with lower growth, civic participation, and public-goods. (Easterly and Levine, 1997, Alesina, Baqir, and Easterly, 1999, Alesina and La Ferrara (2000, 2005). Miguel and Gugerty, 2005, Habyarimana et al., 2007, Baldwin and Huber, 2010, Glennerster et al., 2013)
   - Cooperation outside formal enforcement depends on shared rules, local monitoring, and social sanctions. (Ostrom (1990, 2000, 2010). Baland and Platteau, 1996, Dietz, Ostrom, and Stern, 2003, Tang, 1992, Platteau and Seki, 2007)
   - Cross-cutting identities can change the boundaries around which cooperation is organized. (Miguel, 2004, Posner, 2004, Clingingsmith et al., 2009, Dunning and Harrison, 2010, Sambanis and Shayo, 2013, Blouin and Mukand, 2019, Bazzi et al., 2019, Depetris-Chauvin et al., 2020)

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This paper: Experimentally compare religious and ethnic leaders across homogeneous and fragmented groups within the same public good activity.

$\Rightarrow$ Ethnic fragmentation changes which leaders can promote collective action.

Contribution

2. **External and internal sources of enforcement**
   - Social image makes behavior responsive to others' judgment. (Bénabou and Tirole, 2006, Andreoni and Bernheim, 2009, DellaVigna et al., 2012, Schram and Charness, 2015, Bursztyn and Jensen, 2017, Breza et al., 2026)
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$\Rightarrow$ Chiefs activate ethnic peer monitoring, while pastors invoke divine monitoring.

Contribution

3. **Leadership and governance**
   - Leaders coordinate, communicate, set examples, sanction, and shape expectations. (Bianco and Bates, 1990, Hermalin (1998, 2012). Caillaud and Tirole, 2007, Dewan and Myatt, 2008, Bolton et al., 2013, Jack and Recalde, 2015, Kosfeld and Rustagi, 2015, Antonakis et al., 2022, Bhalotra et al., 2023, Boudreau et al., 2025)
   - Who leads matters. Traditional authorities can supply public goods and state capacity. (Besley, 2005, Caselli and Morelli, 2004, Jones and Olken, 2005, Acemoglu et al., 2014, Baldwin, 2016, Dal Bó et al., 2017, Balan et al., 2022, Henn and Tanutama, 2023, Baldwin et al., 2025)
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- Focus on two widespread leader types in developing countries.
- Vary group composition and contribution visibility around the same leaders.

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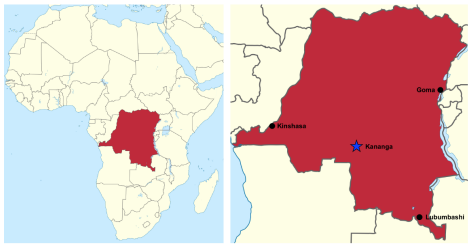
- Focus on two widespread leader types in developing countries.
- Vary group composition and contribution visibility around the same leaders.

$\Rightarrow$ Effective leadership depends on audiences and disclosure regimes.

Outline

1. **Setting**
2. Experiment
3. Results
4. Conclusion

Kananga, D.R. Congo

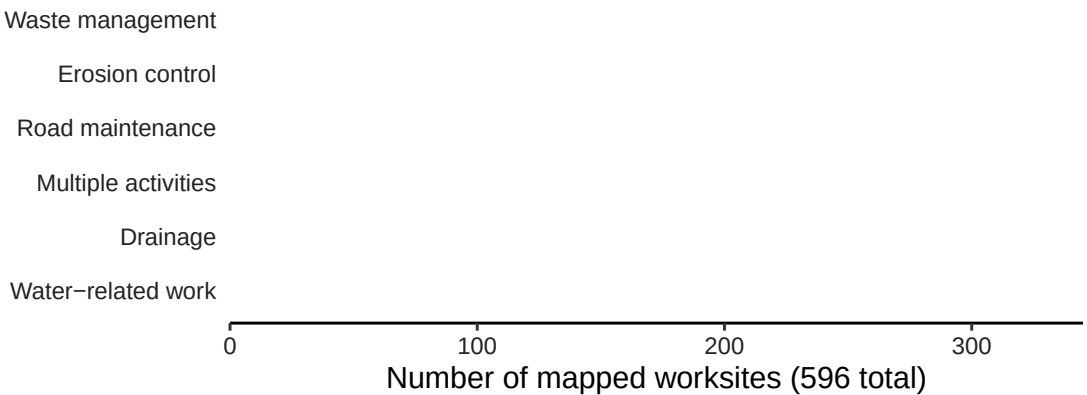


- A major inland city of roughly **two million inhabitants**.
- Capital of **Kasaï-Central Province**.
- Degraded roads, severe erosion, and limited water, electricity, and sanitation.
- Local tax revenue is less than one dollar per person. *Salongo* instead mobilizes residents' **labor and materials** for roads, drainage, and cleaning. (Balan et al., 2022)

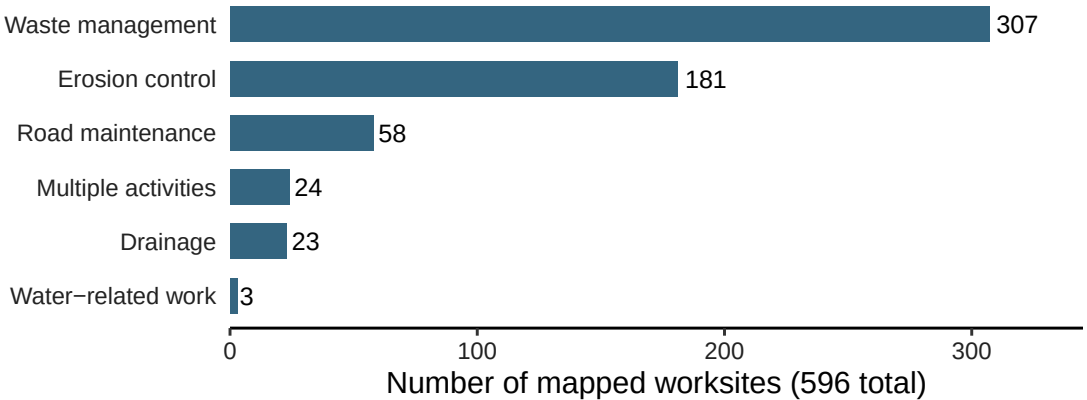
A Very Weak State Relies on Informal Taxation



What projects did Salongo workers implement?



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A Very Weak State Relies on Informal Taxation



Prêcher par l'exemple

Lors du lancement de salongo, il a indiqué la voie à tous ses administrés. Grand sabre à la main, vêtu de jeans et de baskets, il a sarclé une entendue d'herbes au gouvernorat. Comme pour dire, la charité bien ordonnée commence par soi-même.



Le maire de Kananga, les bourgmestres, la police, les taxi-motards, les médias l'Office des Voiries et Drainage et l'Office des Routes ont été tous instruits pour la réussite de l'opération. *« Si nous ne rendons pas notre milieu propre, personne ne le fera à notre place »*, dit le gouverneur.

Avenue chiefs link local communities to city government



1. **Urban local notables, embedded in city administration.**
   - **Principal responsibilities:** organize salongo, mediate disputes, and communicate grievances to government.
   - **48 pp more likely to know the quartier chief** than ordinary residents.

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Neighborhood composition Recruitment profile Question details

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3. **Stronger standing among coethnics.**
   - Rates of trust, moral standing, and motivational appeal are higher among majority-origin residents (the chief's coethnics). Full profile

Neighborhood composition Recruitment profile Question details

Pastors' standing extends beyond ethnic origins

1. **Christian churches are important local institutions.**
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2. **Networks organized around multi-ethnic church communities.**
   - Pastors have a **22 pp higher share of friends from their own church** than chiefs. Churches bring together residents from different ethnic groups.



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1. **Christian churches are important local institutions.**
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2. **Networks organized around multi-ethnic church communities.**
   - Pastors have a **22 pp higher share of friends from their own church** than chiefs. Churches bring together residents from different ethnic groups.
3. **Favorable standing across ethnic origins.**
   - Rates of trust and favorable assessments of moral standing and motivational appeal extend across ethnic origins.



Full pastor profile

Church composition | Recruitment profile | Question details

Outline

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Salongo: community work and civic-work contests



1. **Salongo is community work for local public goods.**

Salongo: community work and civic-work contests



1. **Salongo is community work for local public goods.** We study a five-week city civic-work contest with daily contribution tracking.
   - Related programs operate in Kinshasa, Togo, Rwanda, and Cameroon. Examples

Salongo: community work and civic-work contests

Daily work



2. **Two activities run in parallel.**

Salongo: community work and civic-work contests

Daily work



Work teams



2. **Two activities run in parallel.** Four assigned members meet in debriefing meetings to discuss past and future work.

Salongo: community work and civic-work contests



3. **The city recognizes highly engaged teams.**

Salongo: community work and civic-work contests

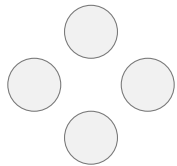




3. **The city recognizes highly engaged teams.** Three teams receive public recognition at a neighborhood ceremony.

The experiment changes the fourth member of each team

Within each neighborhood



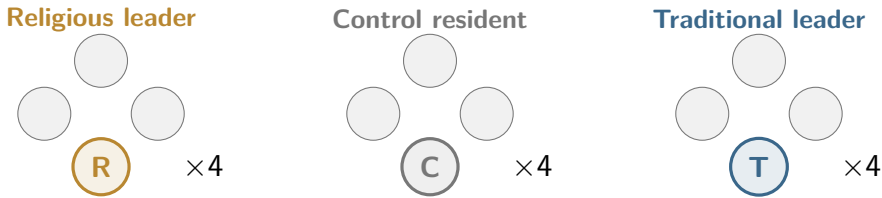
$\times 12$

- We form teams of four members in each neighborhood.

Recruitment | Assignment balance | Meeting delivery

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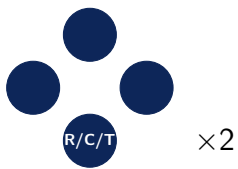
- Randomly assign the fourth seat to **R, C, or T**: Leaders join as members, with no script or special organizer role.

Recruitment | Assignment balance | Meeting delivery

Composition varies independently of the fourth member

Within each R / C / T condition, in each neighborhood

Homogeneous



Colors denote origins. **blue is the majority**.

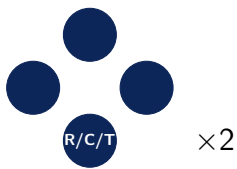
- **Homogeneous:** the three ordinary residents come from the **ethnic majority**. The chief and fixed control participant also belong to that **majority**.

Group formation

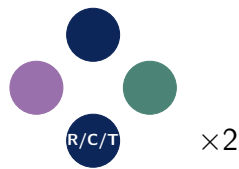
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Within each R / C / T condition, in each neighborhood

Homogeneous



Mixed



Colors denote origins. **blue is the majority**.

- **Mixed:** the three residents come from different origins.

Outcome Data

- **Main outcome: Salongo contribution.** Daily contributions through labor, cash, or in-kind support, averaged over six days after the first meeting.
- **Meetings:** recorded debriefing transcripts.
- **Surveys:** baseline, before each meeting's discussion, and endline. Measure identities, beliefs, social ties and views of leaders.

Hours and effort | Measurement timeline | Outcome windows

▸ Timeline | ▸ Work quantity

Compare leader assignments within the same stratum

$$Y_i = \lambda_{s(i)} + \rho Y_{i0} + \sum_{c \in \{H,M\}} (\tau_c^T T_{g(i)} + \tau_c^R R_{g(i)}) \mathbf{1}\{c(i) = c\} + X_i'\beta + \varepsilon_i.$$

- Y_i: share of days contributing in the six days after the first meeting.
- Y_{i0}: the same measure in the six days before any meeting.
- τ_c^T, τ_c^R: chief (T) and pastor (R) effects vs. control, in homogeneous (H) or mixed (M) groups.
- $\lambda_{s(i)}$: neighborhood $\times$ composition $\times$ disclosure fixed effects.
- X_i: 11 baseline controls and spillover exposure. LASSO

Outline

1. Setting
2. Experiment
3. **Results**
   - **Main results**
   - Mechanisms
   - Alternative explanations
4. Conclusion

Hypotheses

Chiefs: authority bounded by ethnicity

	Mixed	Homogeneous
Chiefs		

Hypotheses

Chiefs: authority bounded by ethnicity

	Mixed	Homogeneous
Chiefs		
		$\tau_H^T > 0$

- **Leadership:** where their authority is recognized, leaders can encourage cooperation by monitoring contributions and rewarding or sanctioning them. (Bianco and Bates, 1990)

Hypotheses

Chiefs: authority bounded by ethnicity

Mixed | **Homogeneous**

Chiefs



$\tau_H^T > \tau_M^T$

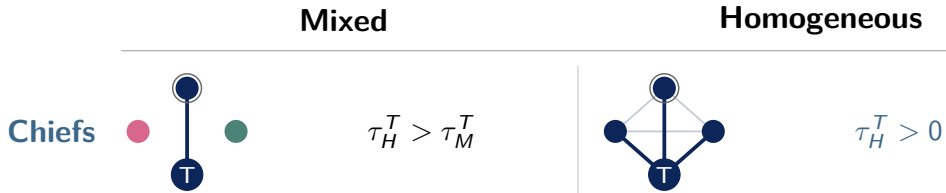


$\tau_H^T > 0$

- **Leadership:** where their authority is recognized, leaders can encourage cooperation by monitoring contributions and rewarding or sanctioning them. (Bianco and Bates, 1990)
- **Ethnic scope:** stronger within-group social sanctions (Miguel and Gugerty, 2005) suggest greater chief influence among coethnics.

Hypotheses

Chiefs: authority bounded by ethnicity



- **Leadership:** where their authority is recognized, leaders can encourage cooperation by monitoring contributions and rewarding or sanctioning them. (Bianco and Bates, 1990)
- **Ethnic scope:** stronger within-group social sanctions (Miguel and Gugerty, 2005) suggest greater chief influence among coethnics.

Hypotheses

Pastors: activates cross-ethnic religious identity

	Mixed	Homogeneous
Pastors		

- **Cross-ethnic scope:** religion transcends ethnic boundaries. Pastors should promote work in both homogeneous and mixed groups. (McCauley, 2014, Henrich et al., 2010)

Hypotheses

Pastors: activates cross-ethnic religious identity

	Mixed		Homogeneous	
Pastors		$\tau_M^R > 0$		$\tau_H^R > 0$

- **Cross-ethnic scope:** religion transcends ethnic boundaries. Pastors should promote work in both homogeneous and mixed groups. (McCauley, 2014, Henrich et al., 2010)

Hypotheses

Pastors: activates cross-ethnic religious identity

	Mixed	Homogeneous
Pastors	 $\tau_M^R > 0$	 $\tau_M^R > \tau_H^R$?

- **Cross-ethnic scope:** religion transcends ethnic boundaries. Pastors should promote work in both homogeneous and mixed groups. (McCauley, 2014, Henrich et al., 2010)
- **Group identity:** group composition may shift attention between religious and ethnic identity, making the homogeneous-group prediction ambiguous. (Akerlof and Kranton, 2000, Benjamin, Choi, and Fisher, 2016)

Hypotheses

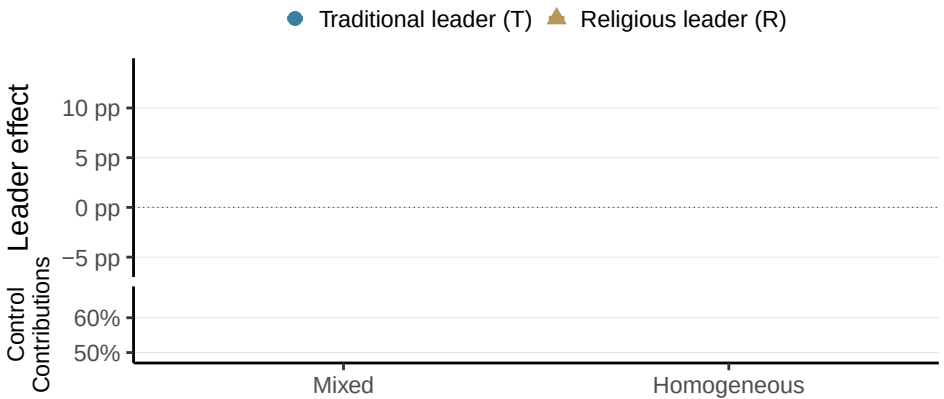
Pastors: activates cross-ethnic religious identity

	Mixed	Homogeneous
Pastors	 $\tau_M^R > 0$	 $\tau_M^R > \tau_H^R$?

Are pastors and ethnic cohesion **substitutes**?

- **Cross-ethnic scope:** religion transcends ethnic boundaries. Pastors should promote work in both homogeneous and mixed groups. (McCauley, 2014, Henrich et al., 2010)
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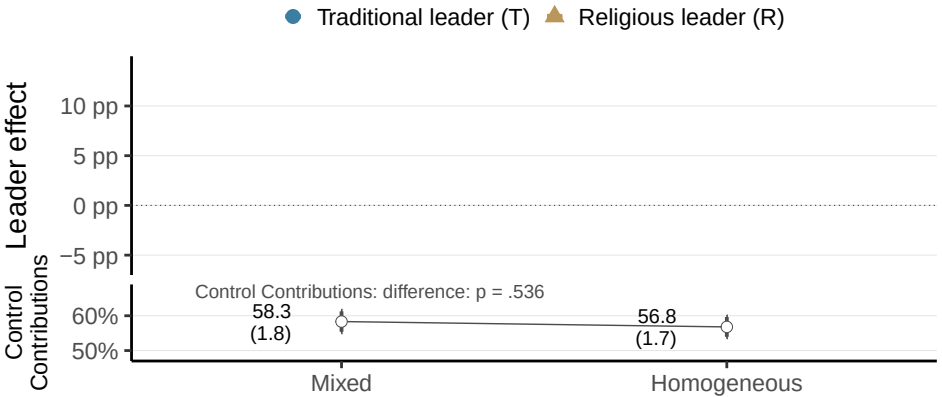
Chiefs mobilize homogeneous groups, pastors mixed groups



Post−LASSO OLS. N = 1,040 residents, 336 groups. Labels: effect (SE). * p<.10, ** p<.05, *** p<.01.

▸ Alternatives ▸ All contrasts

Chiefs mobilize homogeneous groups, pastors mixed groups

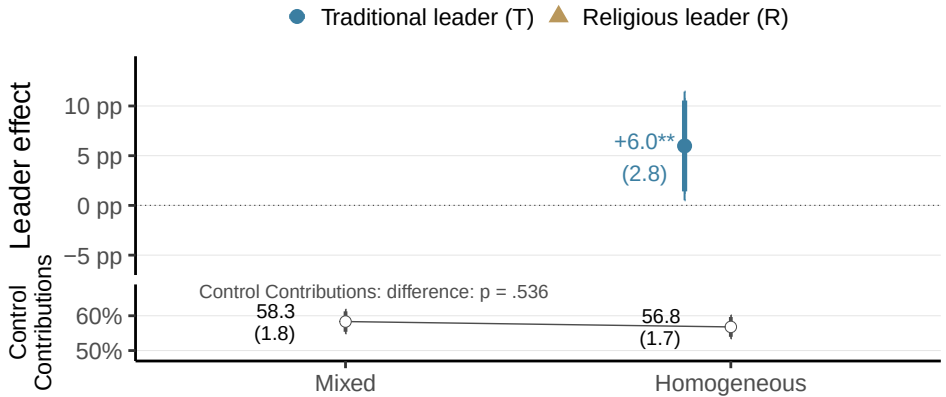


Post−LASSO OLS. N = 1,040 residents, 336 groups. Labels: effect (SE). * p<.10, ** p<.05, *** p<.01.

- **Control Contributions:** 58.3% in mixed vs. 56.8% in homogeneous groups: 3.50 vs. 3.41 contribution-days out of six ($p = .536$).

▸ Alternatives ▸ All contrasts

Chiefs mobilize homogeneous groups, pastors mixed groups

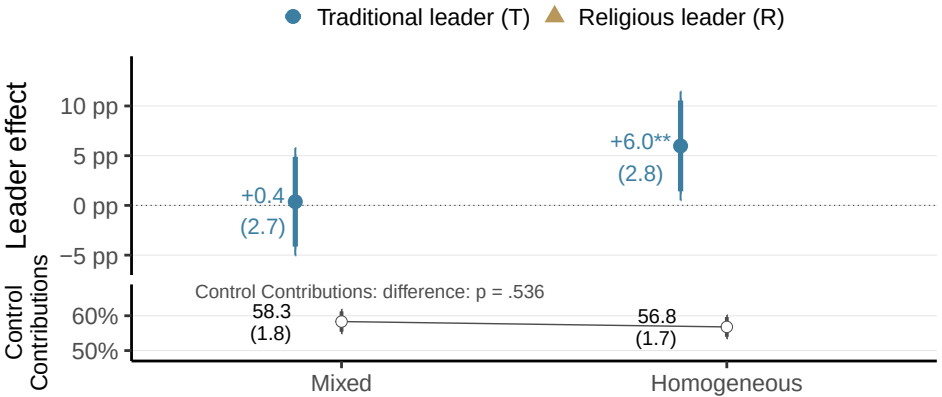


Post−LASSO OLS. N = 1,040 residents, 336 groups. Labels: effect (SE). * p<.10, ** p<.05, *** p<.01.

- **Chiefs, homogeneous groups:** +6.0 pp ($p = .032$), or +0.36 contribution-days per person: roughly one extra day across three residents.

▸ Alternatives ▸ All contrasts

Chiefs mobilize homogeneous groups, pastors mixed groups

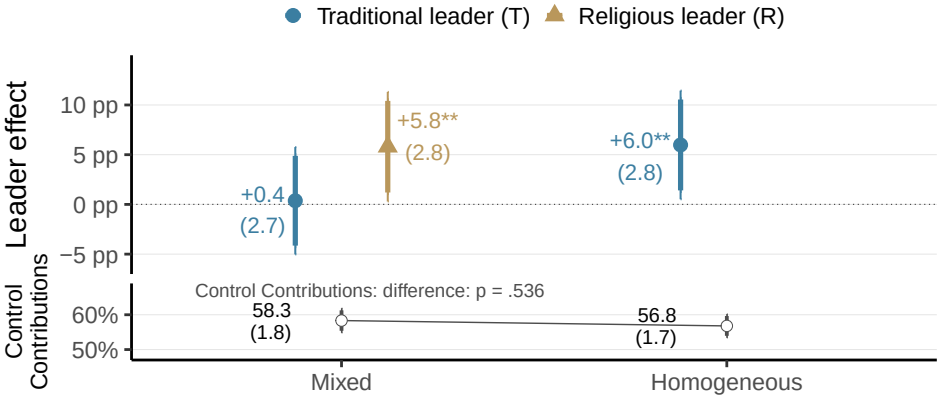


Post−LASSO OLS. N = 1,040 residents, 336 groups. Labels: effect (SE). * p<.10, ** p<.05, *** p<.01.

- **Chiefs, mixed groups:** no detectable effect (0.4 pp, $p = .893$).

▸ Alternatives ▸ All contrasts

Chiefs mobilize homogeneous groups, pastors mixed groups

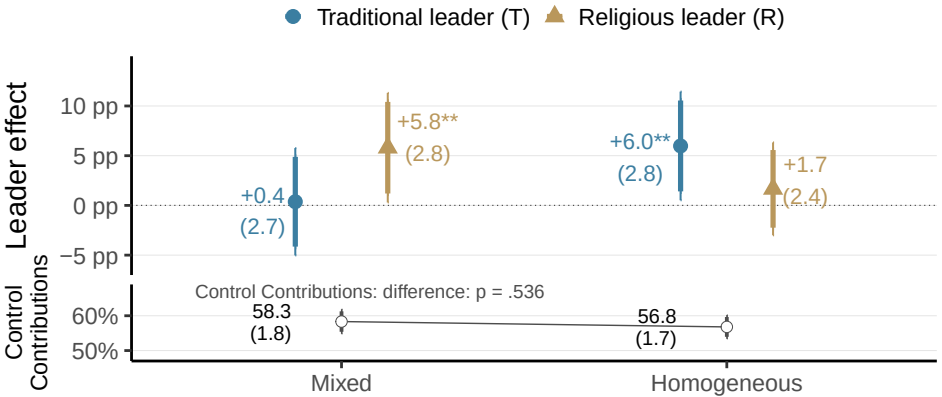


Post−LASSO OLS. N = 1,040 residents, 336 groups. Labels: effect (SE). * p<.10, ** p<.05, *** p<.01.

- **Pastors, mixed groups:** +5.8 pp ($p = .039$): +0.35 day per person, or about one extra contribution-day across three residents.

▸ Alternatives ▸ All contrasts

Chiefs mobilize homogeneous groups, pastors mixed groups



Post−LASSO OLS. N = 1,040 residents, 336 groups. Labels: effect (SE). * p<.10, ** p<.05, *** p<.01.

- **Pastors, homogeneous groups:** no detectable effect (1.7 pp, $p = .484$).

▸ Alternatives ▸ All contrasts

How does ethnic cohesion shape leaders' effects?

	Mixed	Homogeneous
Chiefs	 $\tau_H^T > \tau_M^T$	 $\tau_H^T > 0$
	Chiefs and ethnic cohesion are **complements**. ✓	
Pastors	 $\tau_M^R > 0$	 $\tau_M^R > \tau_H^R$
	Pastors and ethnic cohesion are **substitutes**. ✓	

▸ Alternative explanations

▸ 15 min: Pastor ethnic identity

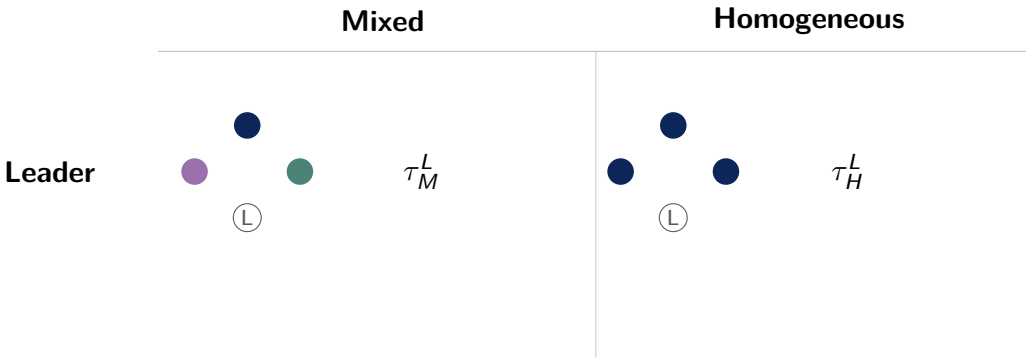
Outline

1. Setting
2. Experiment
3. **Results**
   - Main results
   - **Mechanisms**
   - Alternative explanations
4. Conclusion

Mechanisms

1. **Scope of authority: who can mobilize whom?**
   - **Chiefs: ethnic legitimacy.** Can mobilize via ethnic networks.
   - **Pastors: religious appeals.** Can mobilize across ethnic lines (conditional on the strength of ethnic identity).
2. **Source of enforcement: how? External or internal monitoring?**
   - **Chiefs: external monitoring.** Activating ethnic peer enforcement.
   - **Pastors: internal monitoring.** Activating divine judgment.

How does ethnic cohesion shape leaders' effects?



- **Main experimental contrast:** how does the leader effect change between mixed and homogeneous groups?

Ethnic attachment: the intensive margin

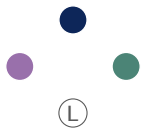
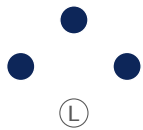
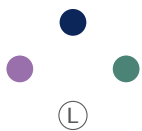
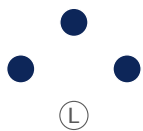
Baseline question: rank six identities by importance, including ethnicity.



Question details

Ethnic attachment: the intensive margin

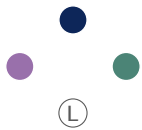
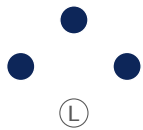
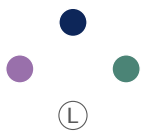
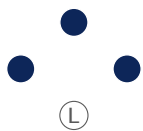
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	Mixed	**Homogeneous**
Strong ethnic attachment	 $\tau^L_{M,\text{strong}}$	 $\tau^L_{H,\text{strong}}$
Weak ethnic attachment	 $\tau^L_{M,\text{weak}}$	 $\tau^L_{H,\text{weak}}$

Question details

Ethnic attachment: the intensive margin

Baseline question: rank six identities by importance, including ethnicity.

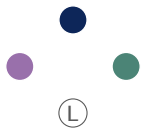
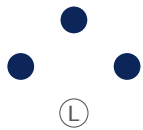
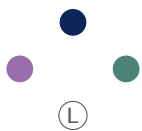
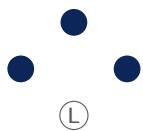
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- **Heterogeneous effects:** do leader effects vary with ethnic attachment?

Question details

Ethnic attachment: the intensive margin

Baseline question: rank six identities by importance, including ethnicity.

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Weak ethnic attachment	 $\tau^L_{M,\text{weak}}$	 $\tau^L_{H,\text{weak}}$

- **Heterogeneous effects:** do leader effects vary with ethnic attachment?
  - ↳ Group identity: mean available score among ordinary members; split at its median into weak and strong attachment.

Question details

Mechanisms

1. **Scope of authority: who can mobilize whom?**
   - **Chiefs: ethnic legitimacy.** Can mobilize via ethnic networks.
   - **Pastors: religious appeals.** Can mobilize across ethnic lines (conditional on the strength of ethnic identity).
2. **Source of enforcement: how? External or internal monitoring?**
   - **Chiefs: external monitoring.** Activating ethnic peer enforcement.
   - **Pastors: internal monitoring.** Activating divine judgment.

Chiefs and strong ethnic attachment are complements

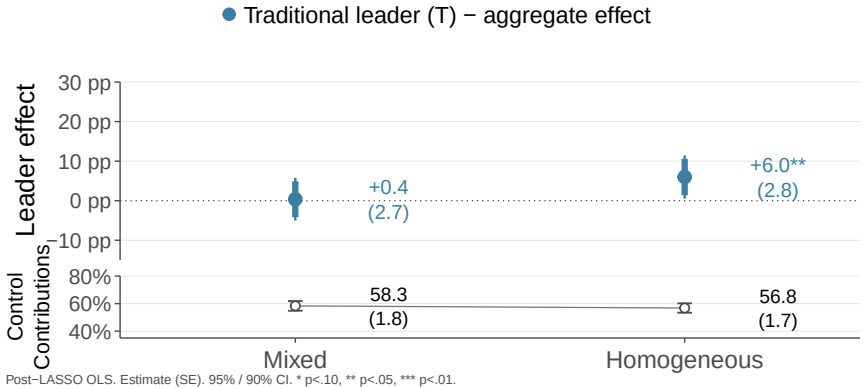
	Mixed	Homogeneous
Strong ethnic attachment		 Stronger effect
Weak ethnic attachment		 Weaker effect

- **Complements in homogeneous groups:** stronger ethnic attachment should reinforce the chief's authority through shared origins.

Mechanisms

1. **Scope of authority: who can mobilize whom?**
   Test 1: Heterogeneous effects by group ethnic identity
   - **Chiefs: ethnic legitimacy.** Can mobilize via ethnic networks.
     ↳ Chiefs should have larger effects in homogeneous groups with strong ethnic attachment.
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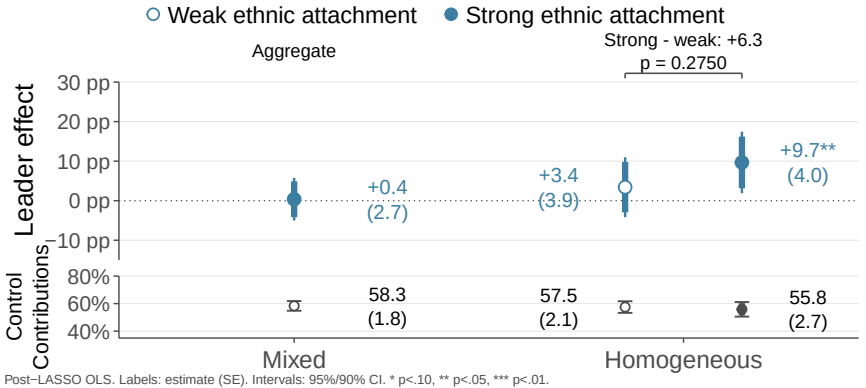
Post−LASSO OLS. Estimate (SE). 95% / 90% CI. * p<.10, ** p<.05, *** p<.01.

- **Aggregate chief effects:** 0.4 pp in mixed and 6.0 pp in homogeneous groups. Does strong ethnic attachment predict greater influence?

Question details | Individual identity | Full group results

HTE specification

Chiefs and strong ethnic attachment are complements



Post−LASSO OLS. Labels: estimate (SE). Intervals: 95%/90% CI. * p<.10, ** p<.05, *** p<.01.

- **Homogeneous groups:** effects rise from 3.4 to 9.7 pp with attachment, consistent with complementarity; the difference is imprecise ($p = .275$).

Question details | Individual identity | Full group results

HTE specification

Chiefs and strong ethnic attachment are complements

	Mixed	Homogeneous
Strong ethnic attachment		 Stronger effect
Weak ethnic attachment		 Weaker effect

✓ **Complements in homogeneous groups:** stronger ethnic attachment reinforces the chief's authority through shared origins.

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Chiefs and strong ethnic attachment are complements

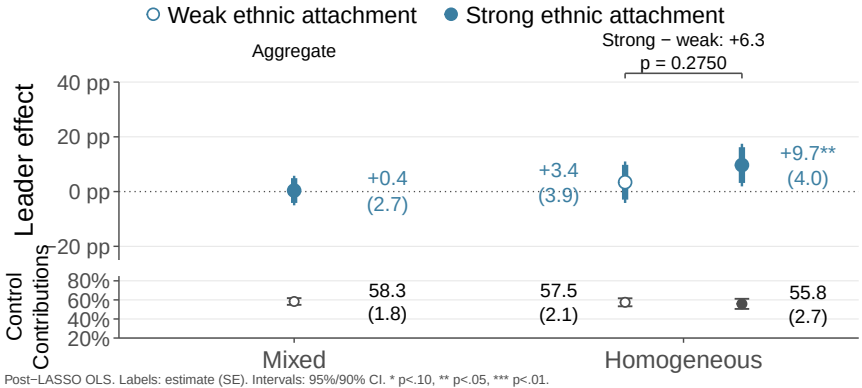
	Mixed	Homogeneous
Strong ethnic attachment	 Strong coethnic effect	Stronger effect
Weak ethnic attachment	 Weak coethnic effect	Weaker effect

- **Complements for coethnics:** stronger ethnic attachment should reinforce chiefs' influence among residents who share their origin.

Mechanisms

1. **Scope of authority: who can mobilize whom?**
   Test 2: Heterogeneous effects by coethnicity
   - **Chiefs: ethnic legitimacy.** Can mobilize via ethnic networks.
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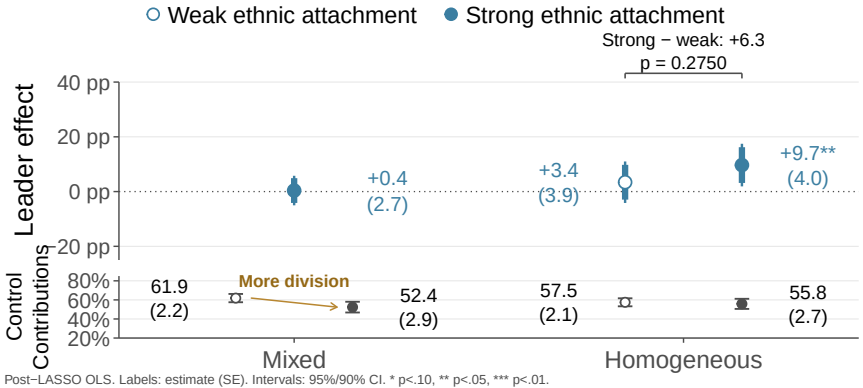
Chiefs and strong ethnic attachment are complements



Post−LASSO OLS. Labels: estimate (SE). Intervals: 95%/90% CI. * p<.10, ** p<.05, *** p<.01.

- **Mixed groups:** does stronger ethnic attachment predict a stronger chief effect?

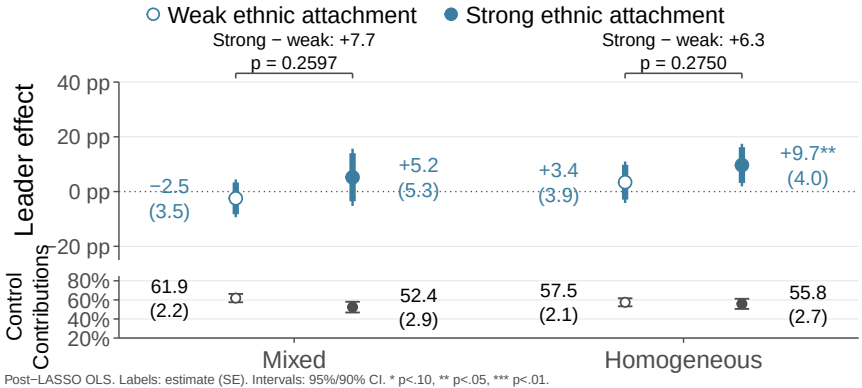
Chiefs and strong ethnic attachment are complements



Post−LASSO OLS. Labels: estimate (SE). Intervals: 95%/90% CI. * p<.10, ** p<.05, *** p<.01.

- **More division:** control contributions fall from 61.9% to 52.4% with stronger attachment to competing ethnic identities ($p = 0.0093$).

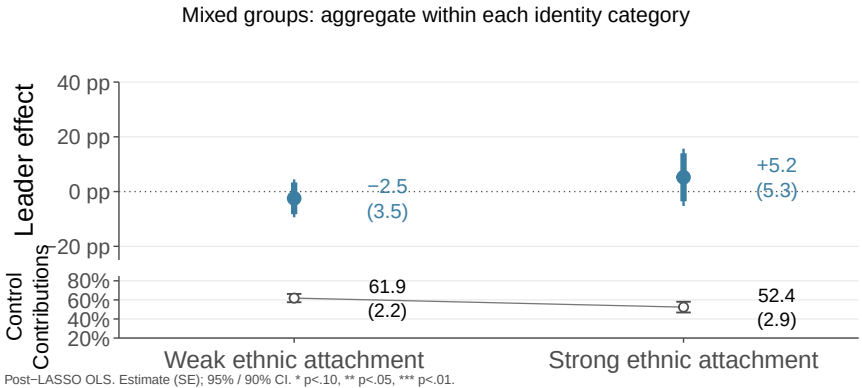
Chiefs and strong ethnic attachment are complements



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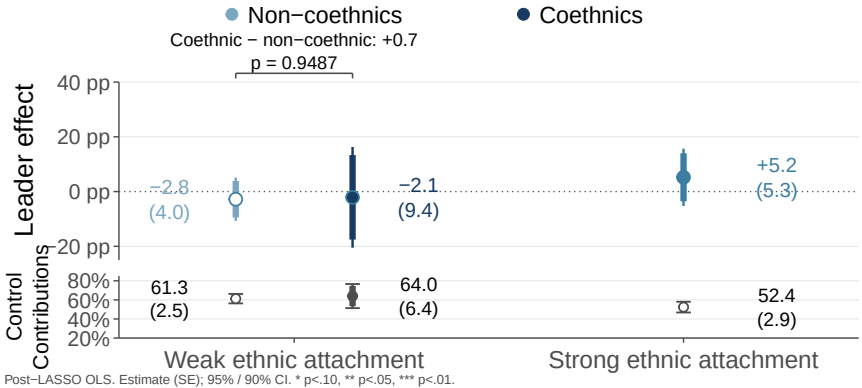
- **A weak positive effect:** +5.2 pp under strong attachment, but imprecise. Is it driven by the chief's coethnics?

Chiefs and strong ethnic attachment are complements



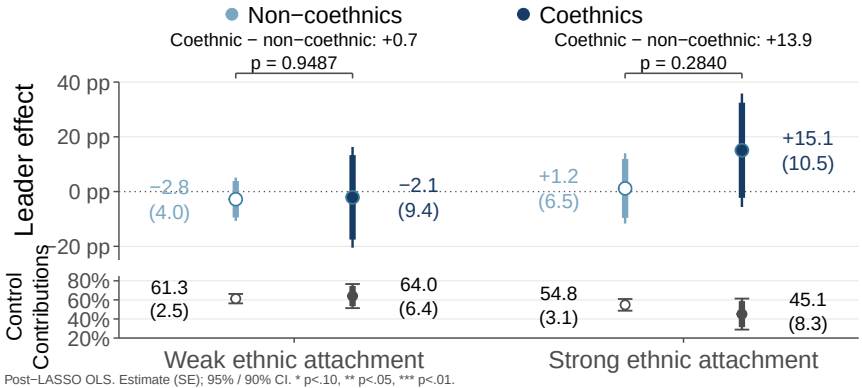
- **Zoom in on mixed groups:** next, separate coethnics from non-coethnics within each attachment category.

Chiefs and strong ethnic attachment are complements



- **Weak attachment:** neither coethnics nor non-coethnics respond detectably; their effects are similar ($p = .949$).

Chiefs and strong ethnic attachment are complements



Post−LASSO OLS. Estimate (SE); 95% / 90% CI. * p<.10, ** p<.05, *** p<.01.

- **Strong attachment:** the positive estimate is concentrated among coethnics (15.1 pp), but the coethnic difference is imprecise ($p = .284$).

Chiefs and strong ethnic attachment are complements

	Mixed	Homogeneous
Strong ethnic attachment	 Strong coethnic effect	Stronger effect
Weak ethnic attachment	 Weak coethnic effect	Weaker effect

✓ **Complements for coethnics:** stronger ethnic attachment reinforces chiefs' influence among residents who share their origin.

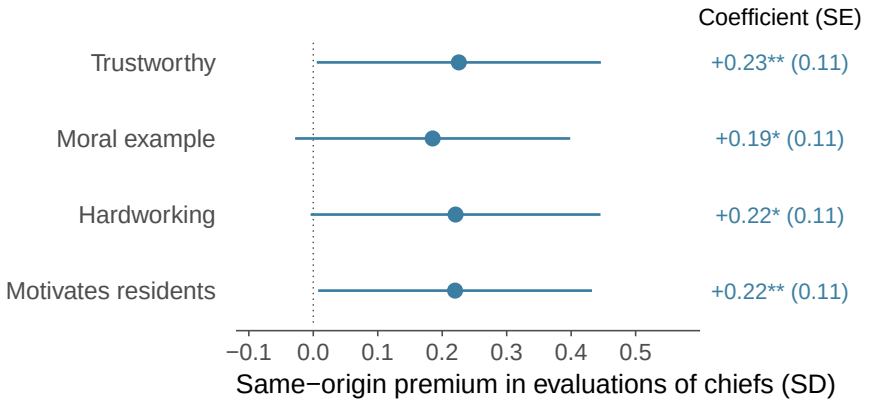
Mechanisms

1. **Scope of authority: who can mobilize whom?**
   Test 2: Heterogeneous effects by coethnicity
   - **Chiefs: ethnic legitimacy.** Can mobilize via ethnic networks.
     ✓ Chiefs influence coethnics even in mixed groups.
   - **Pastors: religious appeals.** Can mobilize across ethnic lines (conditional on the strength of ethnic identity).
2. **Source of enforcement: how? External or internal monitoring?**
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Mechanisms

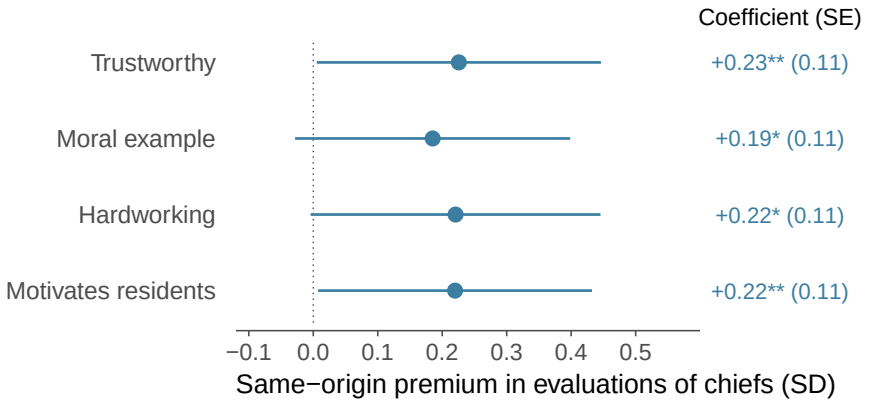
1. **Scope of authority: who can mobilize whom?**
   Test 3: Coethnicity, evaluations, and chief influence
   - **Chiefs: ethnic legitimacy.** Can mobilize via ethnic networks.
     ↳ Coethnics should rate chiefs higher; better ratings should predict greater influence.
   - **Pastors: religious appeals.** Can mobilize across ethnic lines (conditional on the strength of ethnic identity).
2. **Source of enforcement: how? External or internal monitoring?**
   - **Chiefs: external monitoring.** Activating ethnic peer enforcement.
   - **Pastors: internal monitoring.** Activating divine judgment.

Shared ethnicity, strong evaluations, and chief effectiveness



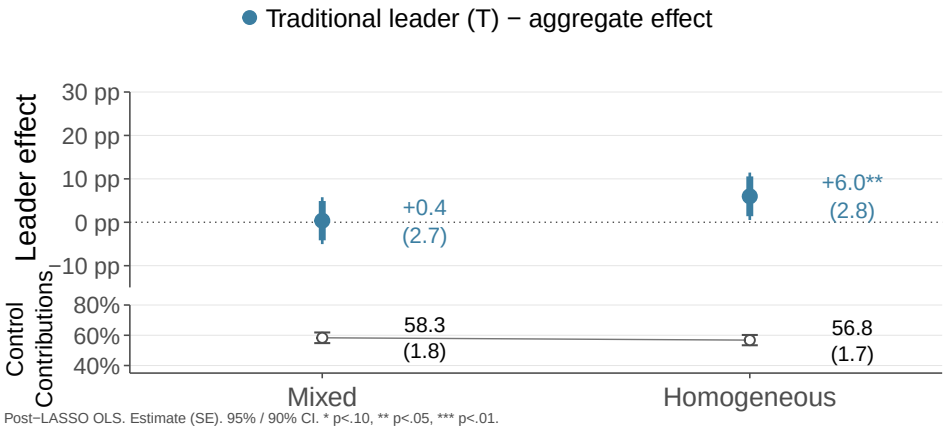
- **Coethnics give chiefs stronger evaluations across all four dimensions:** adjusted differences of 0.19–0.23 SD.

Shared ethnicity, strong evaluations, and chief effectiveness



- **Evaluation index:** combine the four ratings into one score; split at the median into weak and strong evaluations.

Shared ethnicity, strong evaluations, and chief effectiveness

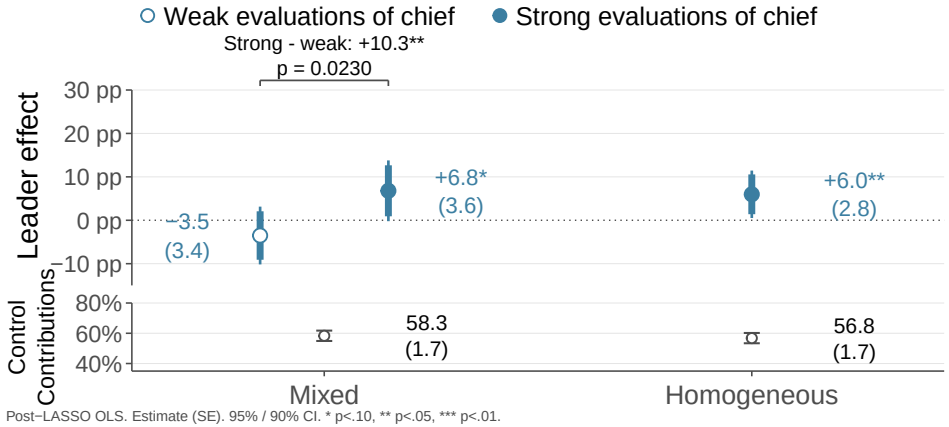


Post−LASSO OLS. Estimate (SE). 95% / 90% CI. * p<.10, ** p<.05, *** p<.01.

- **Aggregate chief effects:** 0.4 pp in mixed versus 6.0 pp in homogeneous groups. Do strong evaluations predict greater chief influence than weak evaluations?

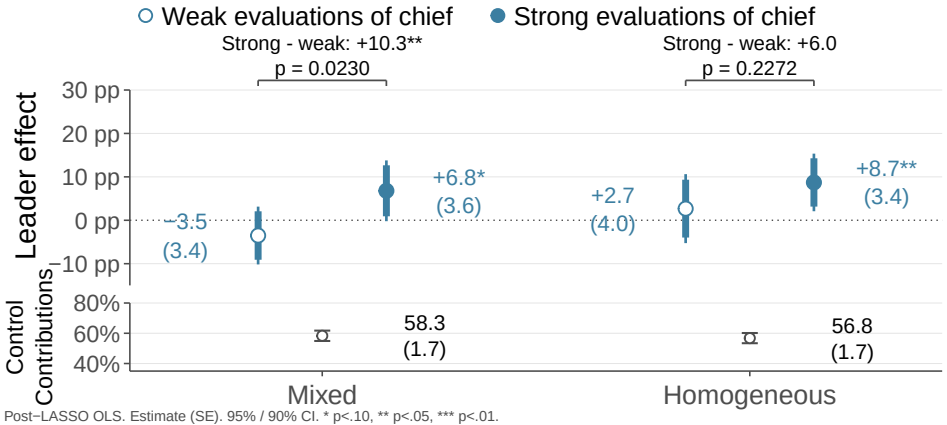
HTE specification ▸ Other ratings Question details

Shared ethnicity, strong evaluations, and chief effectiveness



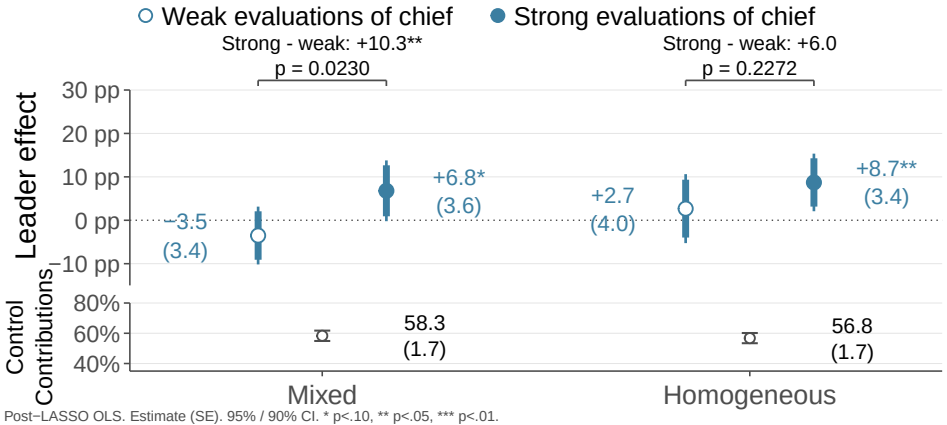
- **Mixed groups:** strong versus weak evaluations predict a 10.3 pp increase in chief effects ($p = .023$).

Shared ethnicity, strong evaluations, and chief effectiveness



- **Homogeneous groups:** strong versus weak evaluations predict a positive but imprecise 6.0 pp increase in chief effects ($p = .227$).

Shared ethnicity, strong evaluations, and chief effectiveness



Post−LASSO OLS. Estimate (SE). 95% / 90% CI. * p<.10, ** p<.05, *** p<.01.

- **Pastors do not follow the same pattern:** strong evaluations do not imply stronger pastor effects.

Mechanisms

1. **Scope of authority: who can mobilize whom?**
   Test 3: Coethnicity, evaluations, and chief influence
   - **Chiefs: ethnic legitimacy.** Can mobilize via ethnic networks.
     ✓ Coethnics rate chiefs higher; strong evaluations predict stronger chief effects.
   - **Pastors: religious appeals.** Can mobilize across ethnic lines (conditional on the strength of ethnic identity).
2. **Source of enforcement: how? External or internal monitoring?**
   - **Chiefs: external monitoring.** Activating ethnic peer enforcement.
   - **Pastors: internal monitoring.** Activating divine judgment.

Mechanisms

1. **Scope of authority: who can mobilize whom?**
   - **Chiefs: ethnic legitimacy.** Can mobilize via ethnic networks.
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2. **Source of enforcement: how? External or internal monitoring?**
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Pastors and strong ethnic attachment are substitutes

	Mixed	Homogeneous
Strong ethnic attachment		 Weaker effect
Weak ethnic attachment		 Stronger effect

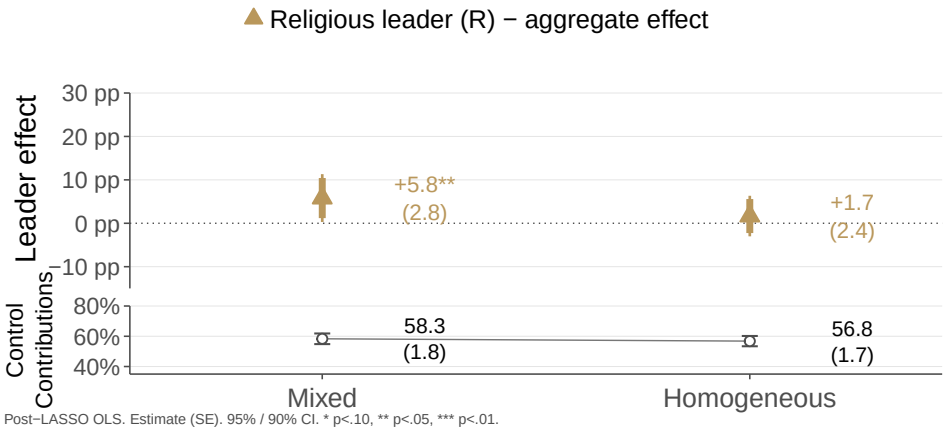
- **Substitutes in homogeneous groups:** stronger shared ethnic attachment gives ethnic obligations precedence, leaving less scope for religious appeals.

Mechanisms

1. **Scope of authority: who can mobilize whom?**
   Test 1: Heterogeneous effects by ethnic identity
   - **Chiefs: ethnic legitimacy.** Can mobilize via ethnic networks.
   - **Pastors: religious appeals.** Can mobilize across ethnic lines (conditional on the strength of ethnic identity).
     ↳ Pastors should have an effect in homogeneous groups with weak ethnic attachment.

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   - **Chiefs: external monitoring.** Activating ethnic peer enforcement.
   - **Pastors: internal monitoring.** Activating divine judgment.

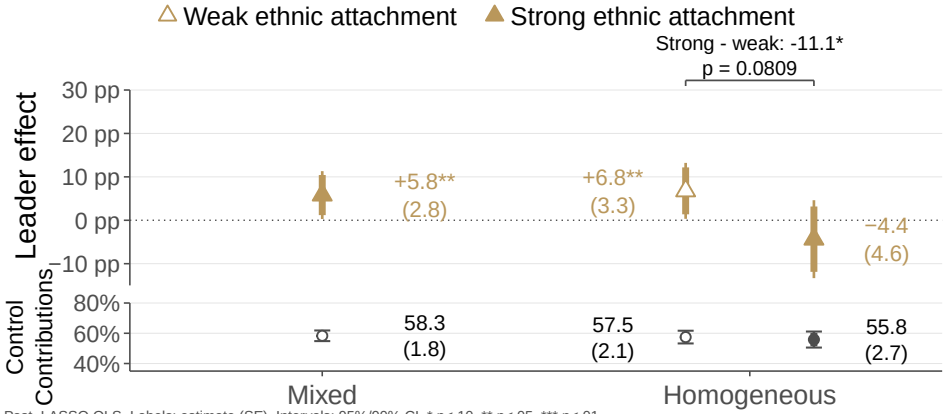
Pastors and strong ethnic attachment are substitutes



Post−LASSO OLS. Estimate (SE). 95% / 90% CI. * p<.10, ** p<.05, *** p<.01.

- **Aggregate pastor effects:** 5.8 pp in mixed versus 1.7 pp in homogeneous groups. Does strong shared ethnic attachment limit religious appeals?

Pastors and strong ethnic attachment are substitutes



Post−LASSO OLS. Labels: estimate (SE). Intervals: 95%/90% CI. * p<.10, ** p<.05, *** p<.01.

- **Homogeneous groups:** effects fall from 6.8 to −4.4 pp ($p = .089$), consistent with shared ethnic obligations displacing religious appeals.

Pastors and strong ethnic attachment are substitutes

	Mixed	Homogeneous
Strong ethnic attachment		 Weaker effect
Weak ethnic attachment		 Stronger effect

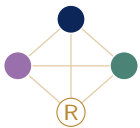
✓ **Substitutes in homogeneous groups:** stronger shared ethnic attachment gives ethnic obligations precedence, leaving less scope for religious appeals.

Mechanisms

1. **Scope of authority: who can mobilize whom?**
   Test 1: Heterogeneous effects by ethnic identity
   - **Chiefs: ethnic legitimacy.** Can mobilize via ethnic networks.
   - **Pastors: religious appeals.** Can mobilize across ethnic lines (conditional on the strength of ethnic identity).
     ✓ Pastors have an effect in homogeneous groups with weak ethnic attachment.

2. **Source of enforcement: how? External or internal monitoring?**
   - **Chiefs: external monitoring.** Activating ethnic peer enforcement.
   - **Pastors: internal monitoring.** Activating divine judgment.

Pastors and ethnic attachment can also be complements

	Mixed	Homogeneous
Strong ethnic attachment	 Stronger effect	Weaker effect
Weak ethnic attachment	 Weaker effect	Stronger effect

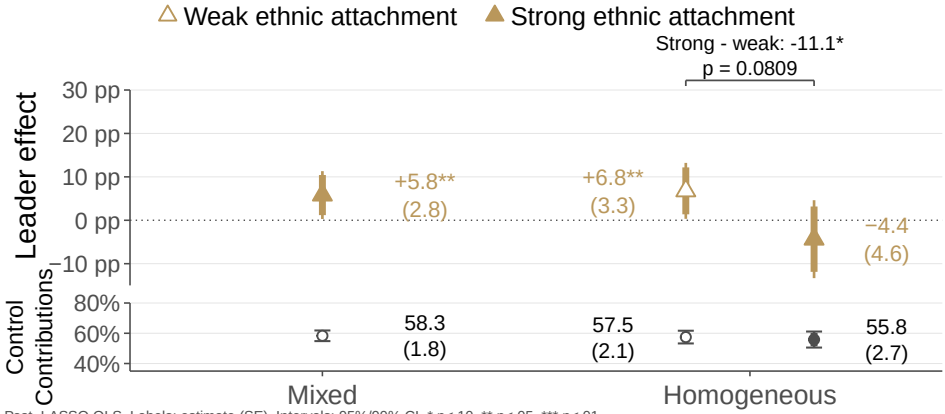
- **Complements in mixed groups:** stronger ethnic attachment creates more division, giving pastors more scope to bridge it through shared religion.

Mechanisms

1. **Scope of authority: who can mobilize whom?**
   Test 2: Heterogeneous effects by ethnic identity
   - **Chiefs: ethnic legitimacy.** Can mobilize via ethnic networks.
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     ↳ Pastors should have stronger effects when ethnic identity is the most divisive.

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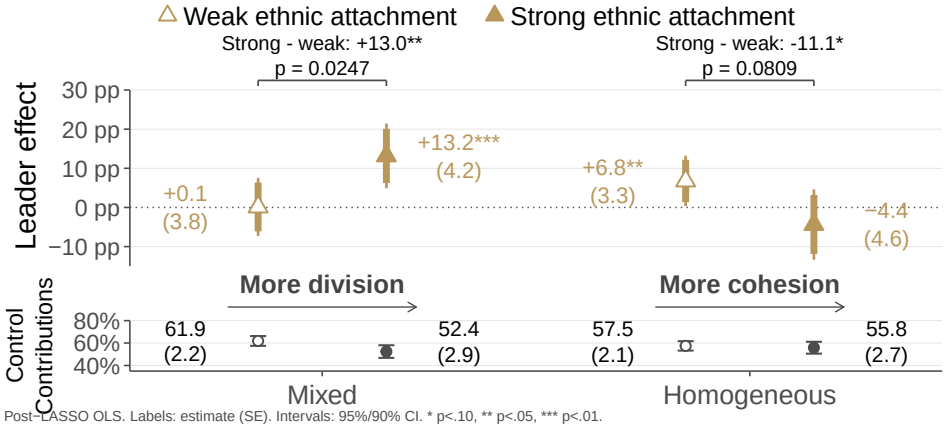
Pastors and ethnic attachment can also be complements



Post−LASSO OLS. Labels: estimate (SE). Intervals: 95%/90% CI. * p<.10, ** p<.05, *** p<.01.

- **Mixed groups:** stronger ethnic identification may deepen divisions. Does it strengthen pastors' influence?

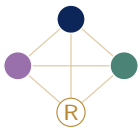
Pastors and ethnic attachment can also be complements



Post-LASSO OLS. Labels: estimate (SE). Intervals: 95%/90% CI. * p<.10, ** p<.05, *** p<.01.

- **Mixed groups:** effects rise from 0.1 to 13.2 pp ($p = .025$): pastors mobilize most where strongly held ethnic identities divide the group.

Pastors and ethnic attachment can also be complements

	Mixed	Homogeneous
Strong ethnic attachment	 Stronger effect	Weaker effect
Weak ethnic attachment	 Weaker effect	Stronger effect

✓ **Complements in mixed groups:** stronger ethnic attachment creates more division, giving pastors more scope to bridge it through shared religion.

Mechanisms

1. **Scope of authority: who can mobilize whom?**
   Test 2: Heterogeneous effects by ethnic identity
   - **Chiefs: ethnic legitimacy.** Can mobilize via ethnic networks.
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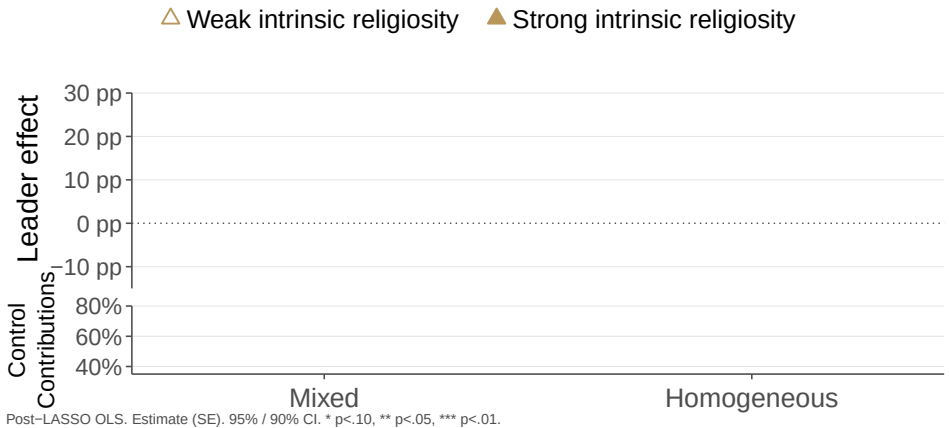
2. **Source of enforcement: how? External or internal monitoring?**
   - **Chiefs: external monitoring.** Activating ethnic peer enforcement.
   - **Pastors: internal monitoring.** Activating divine judgment.

Mechanisms

1. **Scope of authority: who can mobilize whom?**
   Test 3: Heterogeneous effects by intrinsic religiosity
   - **Chiefs: ethnic legitimacy.** Can mobilize via ethnic networks.
   - **Pastors: religious appeals.** Can mobilize across ethnic lines (conditional on the strength of ethnic identity).
     ↳ Stronger religiosity should increase responsiveness to pastors.

2. **Source of enforcement: how? External or internal monitoring?**
   - **Chiefs: external monitoring.** Activating ethnic peer enforcement.
   - **Pastors: internal monitoring.** Activating divine judgment.

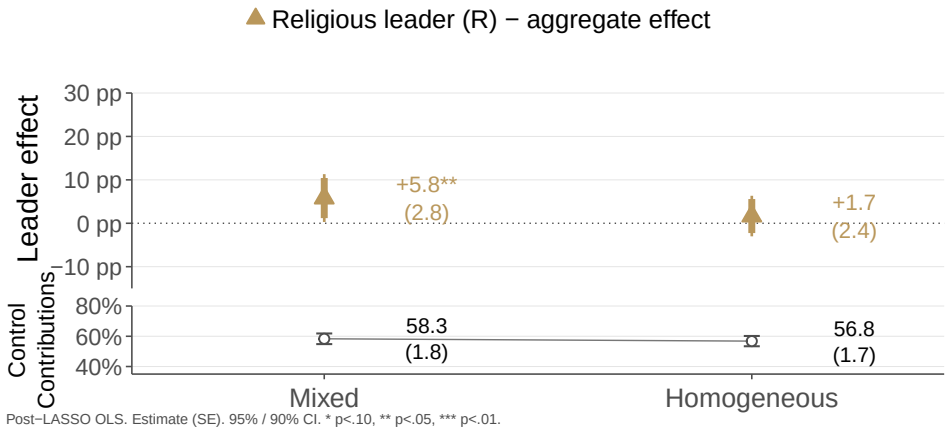
Pastors and religiosity are complements



Post−LASSO OLS. Estimate (SE). 95% / 90% CI. * p<.10, ** p<.05, *** p<.01.

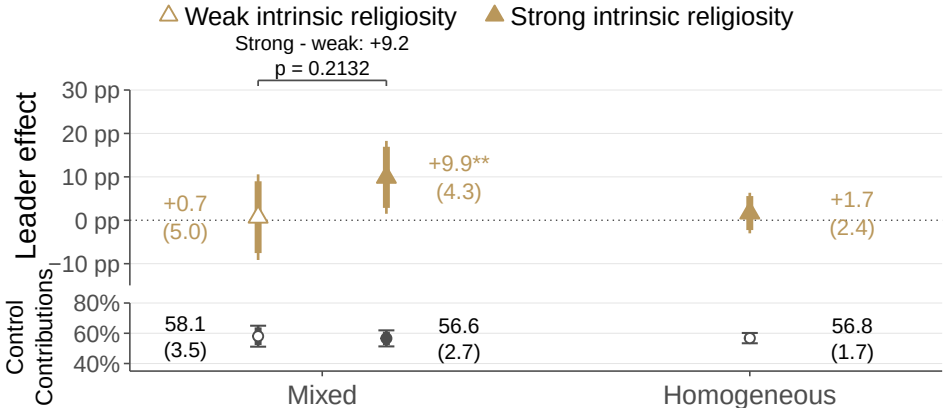
- **Religiosity:** religion may be a key part of a person's identity, strengthening responsiveness to pastors.

Pastors and religiosity are complements



- **Religiosity (Allport and Ross, 1967):** internal religious commitment versus external motives for social church-going.

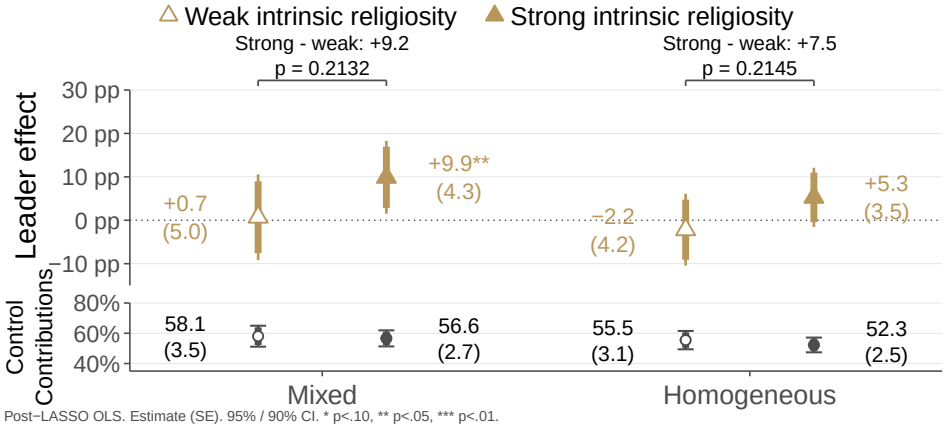
Pastors and religiosity are complements



Post−LASSO OLS. Estimate (SE). 95% / 90% CI. * p<.10, ** p<.05, *** p<.01.

- **Intrinsic religiosity:** mixed-group effects rise from 0.7 to 9.9 pp, consistent with religious motivation; the difference is imprecise ($p = .213$).

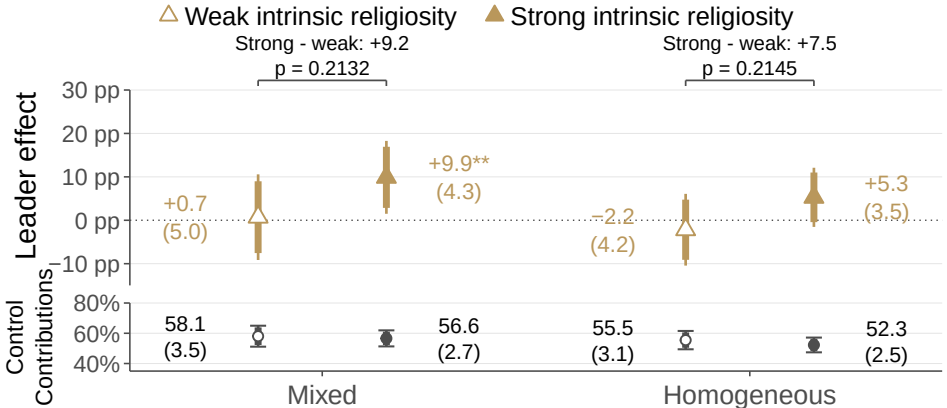
Pastors and religiosity are complements



- **Homogeneous groups:** effects rise from -2.2 to $+5.3$ pp with intrinsic religiosity; the difference is imprecise ($p = .214$).

HTE specification | Intrinsic religiosity | Extrinsic religiosity | Question details

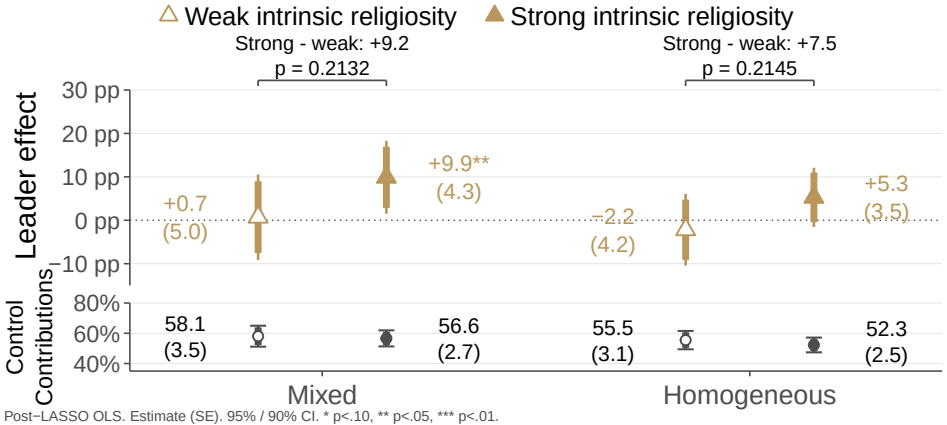
Pastors and religiosity are complements



Post−LASSO OLS. Estimate (SE). 95% / 90% CI. * p<.10, ** p<.05, *** p<.01.

- **Extrinsic religiosity:** strong extrinsic religiosity does not imply stronger pastor effects.

Pastors and religiosity are complements



Post−LASSO OLS. Estimate (SE). 95% / 90% CI. * p<.10, ** p<.05, *** p<.01.

- **Chiefs:** strong intrinsic religiosity does not imply stronger chief effects.

Mechanisms

1. **Scope of authority: who can mobilize whom?**
   Test 3: Heterogeneous effects by intrinsic religiosity
   - **Chiefs: ethnic legitimacy.** Can mobilize via ethnic networks.
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     ✓ Stronger religiosity increases responsiveness to pastors.

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   - **Chiefs: external monitoring.** Activating ethnic peer enforcement.
   - **Pastors: internal monitoring.** Activating divine judgment.

Mechanisms

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2. **Source of enforcement: how? External or internal monitoring?**
   Test 1: Analysis of Meeting Transcripts
   - **Chiefs: external monitoring.** Activating ethnic peer enforcement.
     ↳ Chiefs should invoke social accountability.
   - **Pastors: internal monitoring.** Activating divine judgment.
     ↳ Pastors should invoke divine rewards.

Who talks about what, and when?

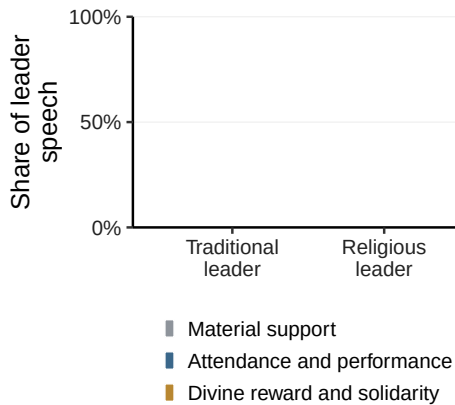
Treatment-blind text analysis

- **Topic model** of meeting transcripts.
  (Blei, Ng, and Jordan, 2003, Quinn et al., 2010)

Who talks about what, and when?

Treatment-blind text analysis

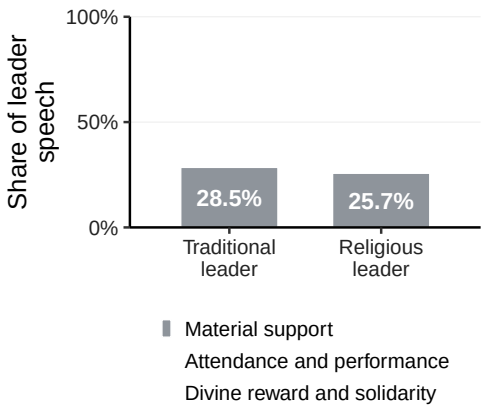
- **Topic model** of meeting transcripts. (Blei, Ng, and Jordan, 2003, Quinn et al., 2010)
- **Blind to leader roles and treatment assignments.**



Who talks about what, and when?

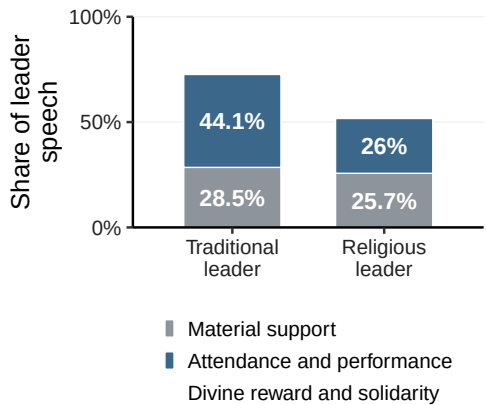
First, the practical needs of the work

Resident: "We ask for help so that we can maintain our streets: we need tools and empty sacks."



Who talks about what, and when?

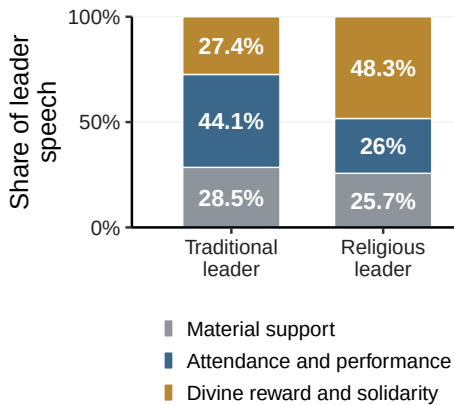
Chief: "[. . .] We must improve our performance to win the contest. [. . .] We must encourage those who did not come and go and see them to find out why they were absent. We must help one another so that our group does better."



Who talks about what, and when?

Pastor: "Do not think that the work we do is for a man: we do it for God. [...] In James it is written: 'What good is it for a person to say they have faith but have no works? Can such faith save a person?' [...] Please devote yourselves to this work so that our country may improve. God in heaven will be pleased with us."

James 2:14



Speech evidence

Who talks about what, and when?

Differences between leaders

Attendance/performance:
Chiefs +18.1 pp; $q = .009$.

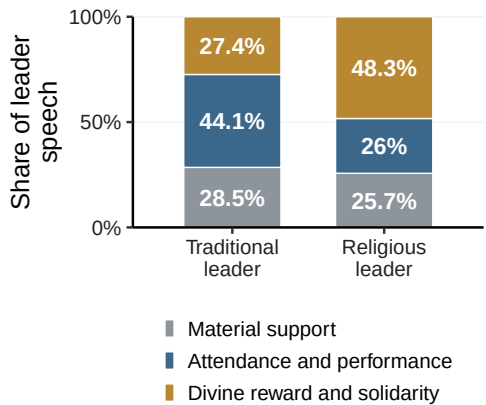
Divine reward/solidarity:
Pastors +20.9 pp; $q = .009$.

Neither chiefs nor pastors detectably change their appeals across environments.

Chiefs' attendance appeals: $p = .631$.
Pastors' divine appeals: $p = .990$.

Across environments

Speech evidence



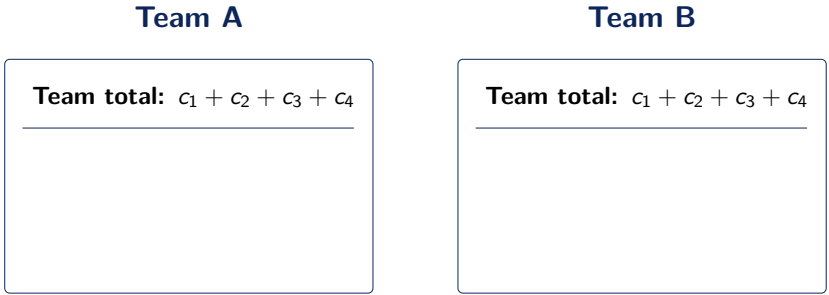
Mechanisms

1. **Scope of authority: who can mobilize whom?**
   - **Chiefs: ethnic legitimacy.** Can mobilize via ethnic networks.
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   - **Chiefs: external monitoring.** Activating ethnic peer enforcement.
     ✓ Chiefs do invoke social accountability.
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Mechanisms

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   - **Chiefs: ethnic legitimacy.** Can mobilize via ethnic networks.
   - **Pastors: religious appeals.** Can mobilize across ethnic lines (conditional on the strength of ethnic identity).
2. **Source of enforcement: how? External or internal monitoring?**
   Test 2: Public/private Cross-randomization
   - **Chiefs: external monitoring.** Activating ethnic peer enforcement.
     ↳ Public records should strengthen chief effects.
   - **Pastors: internal monitoring.** Activating divine judgment.
     ↳ Pastor effects should not require public records.

Public records identify who contributed

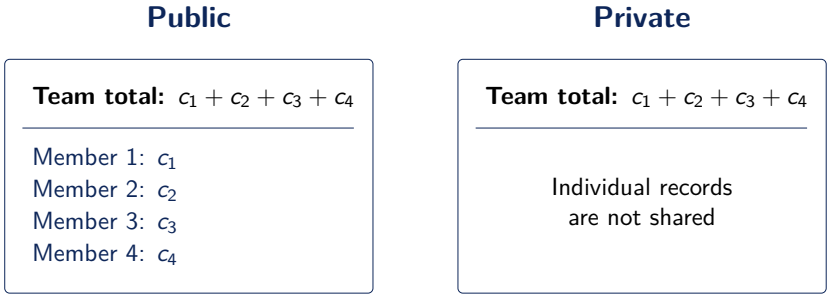


Information shared at the meeting

- **Work and meetings run in parallel:** teammates often cannot observe each other's contributions.

Implementation checks | Randomization

Public records identify who contributed



Information shared at the meeting

- Within each neighborhood and leadership–composition condition, **randomly assign one group to each reporting rule.**

Implementation checks | Randomization

Compare leader effects across public and private records

$$Y_i = \lambda_{s(i)} + \rho Y_{i0} + X_i'\beta + \sum_{k \in \mathcal{K}} \mathbf{1}\{k(i) = k\}(\tau_k^T T_{g(i)} + \tau_k^R R_{g(i)}) + \varepsilon_i.$$

$$\mathcal{K} = \{\text{Mixed}, \text{Homogeneous}\} \times \{\text{Public}, \text{Private}\}.$$

- $\lambda_{s(i)}$: stratum fixed effects (neighborhood $\times$ composition $\times$ reporting rule).
- X_i: the same 11 selected controls and current spillover exposure. LASSO selection
- Compare leader effects across reporting rules **within each composition**.

Hypotheses

	Public	Private
Chiefs Homogeneous groups		
Pastors Mixed groups		

Hypotheses

	Public	Private
Chiefs Homogeneous groups	Stronger effect	Weaker effect
Pastors Mixed groups		

- **Chiefs: external accountability.** Public records make shirking visible, encouraging participation of those who seek to avoid being seen as shirkers by the chief. (Bursztyn and Jensen, 2017, Baldwin, 2016)

Hypotheses

	Public	Private
Chiefs Homogeneous groups	Stronger effect	Weaker effect
Pastors Mixed groups	Stronger effect	Stronger effect

- **Chiefs: external accountability.** Public records make shirking visible, encouraging participation of those who seek to avoid being seen as shirkers by the chief. (Bursztyn and Jensen, 2017, Baldwin, 2016)
- **Pastors: internal accountability.** Intrinsic religious motivation needs no external audience.

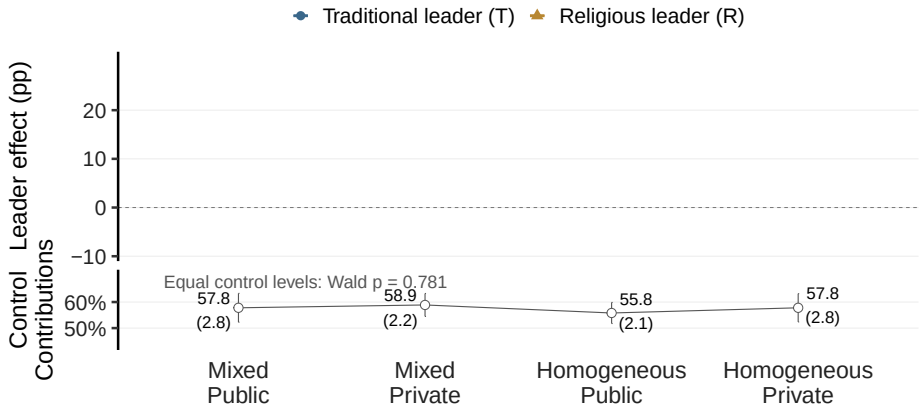
Hypotheses

	Public	Private
Chiefs Homogeneous groups	Stronger effect	Weaker effect
Pastors Mixed groups	Ambiguous	Stronger effect

- **Chiefs: external accountability.** Public records make shirking visible, encouraging participation of those who seek to avoid being seen as shirkers by the chief. (Bursztyn and Jensen, 2017, Baldwin, 2016)

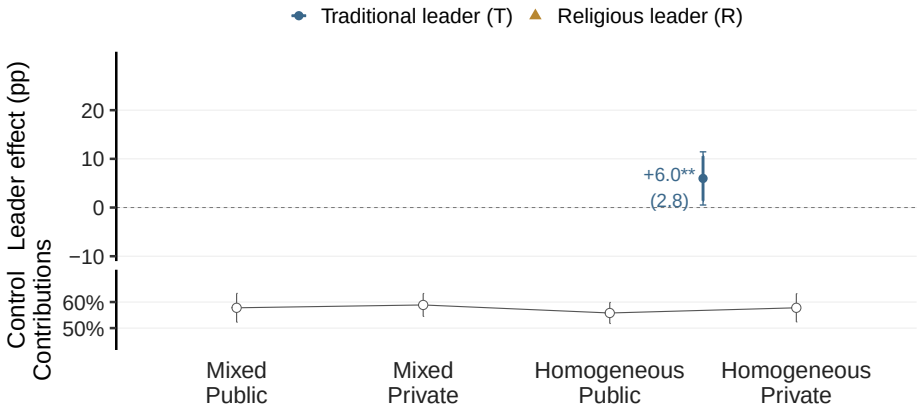
- **Pastors: internal accountability.** Intrinsic religious motivation needs no external audience. Publicity may make participation look like a search for approval, weakening that motivation. (Bénabou and Tirole, 2006)

Public accountability distinguishes enforcement channels



- **Control contributions:** levels are similar across the four environments (joint Wald $p = .781$).

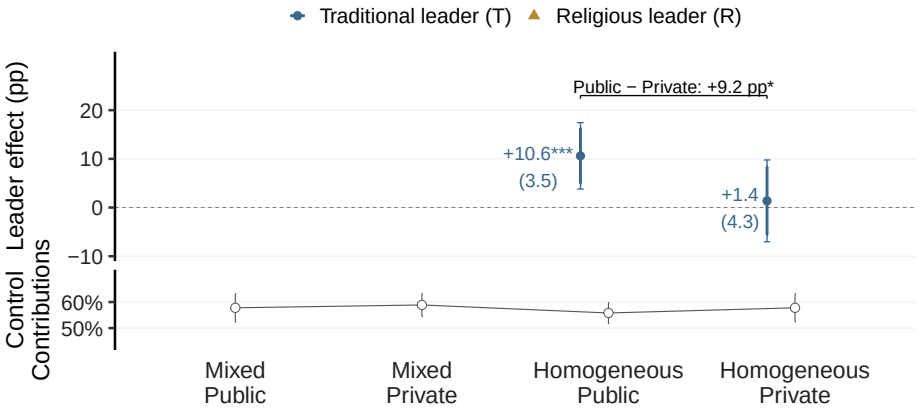
Public accountability distinguishes enforcement channels



- **Chiefs in homogeneous groups:** a 6.0 pp effect pooled across reporting rules. Does visibility matter?

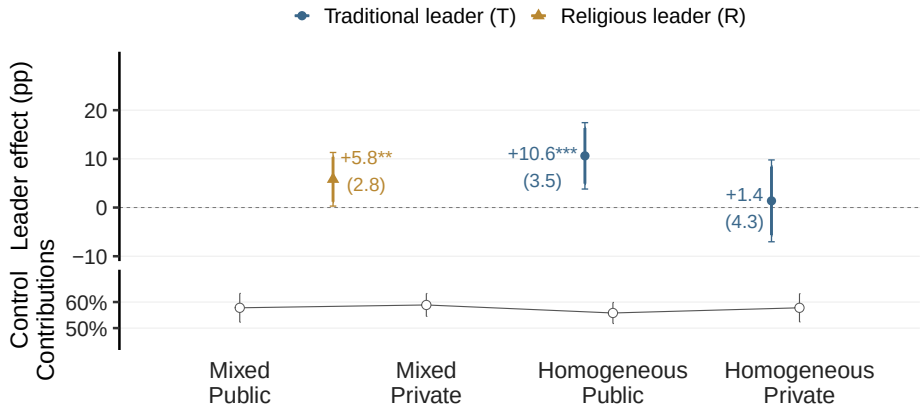
HTE specification ▸ Alternatives

Public accountability distinguishes enforcement channels



- **Chiefs in homogeneous groups:** effects concentrate under public records, consistent with external accountability.

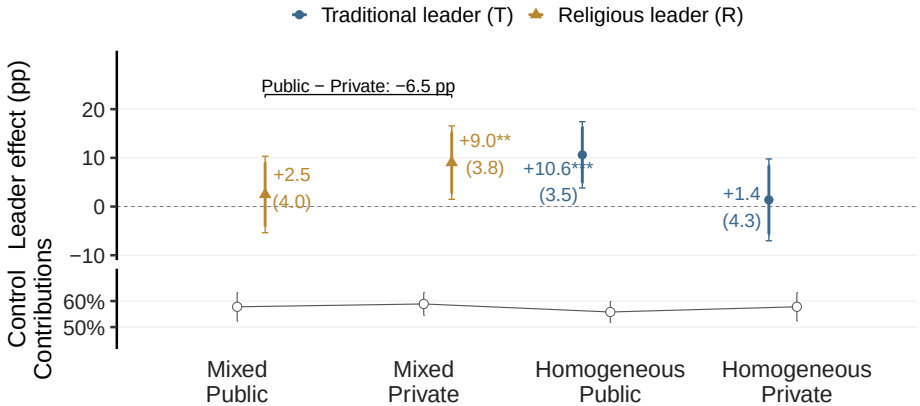
Public accountability distinguishes enforcement channels



- **Pastors in mixed groups:** a 5.8 pp effect pooled across reporting rules. Does visibility matter?

HTE specification ▸ Alternatives

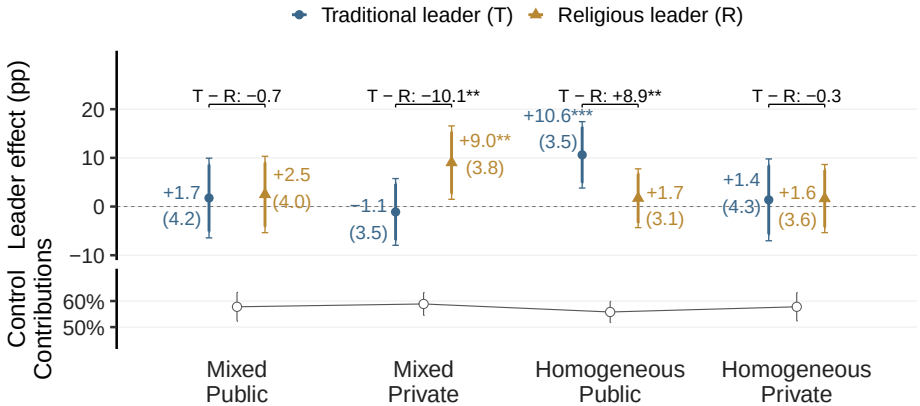
Public accountability distinguishes enforcement channels



- **Pastors in mixed groups:** effects are larger under private records, consistent with internal religious motivation.

HTE specification ▸ Alternatives

Public accountability distinguishes enforcement channels



- **Remaining environments:** neither reporting rule removes the group-context boundary.

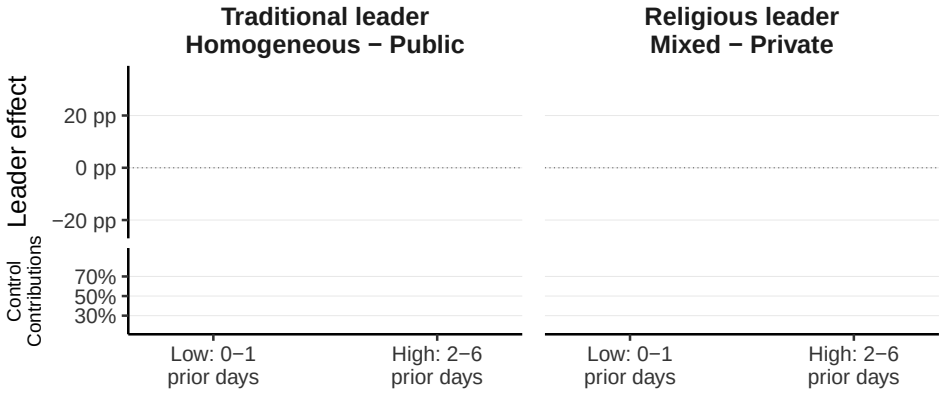
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   - **Chiefs: external monitoring.** Activating ethnic peer enforcement.
     ✓ Public records do strengthen chief effects.
   - **Pastors: internal monitoring.** Activating divine judgment.
     ✓ Pastor effects do not require public records.

Mechanisms

1. **Scope of authority: who can mobilize whom?**
   - **Chiefs: ethnic legitimacy.** Can mobilize via ethnic networks.
   - **Pastors: religious appeals.** Can mobilize across ethnic lines (conditional on the strength of ethnic identity).
2. **Source of enforcement: how? External or internal monitoring?**
   Test 3: Contribution Histories
   - **Chiefs: external monitoring.** Activating ethnic peer enforcement.
     ↳ Chiefs should mobilize shirkers (low prior contributors).
   - **Pastors: internal monitoring.** Activating divine judgment.
     ↳ Pastors should mobilize both low and high prior contributors.

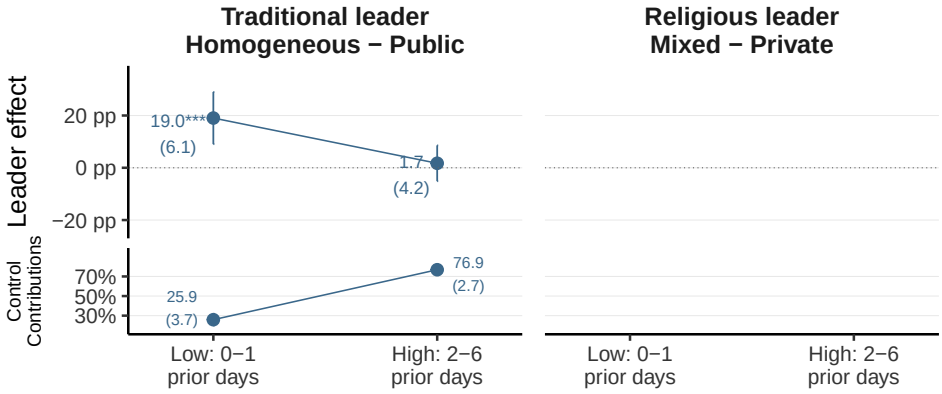
Chiefs mobilize shirkers



* p<.10, ** p<.05, *** p<.01. Control Contributions, tags: estimate (SE), 90% confidence intervals.

- **Social judgment:** Public records expose weak contributors. If traditional leaders enforce through reputation, their effect should concentrate on these residents.

Chiefs mobilize shirkers

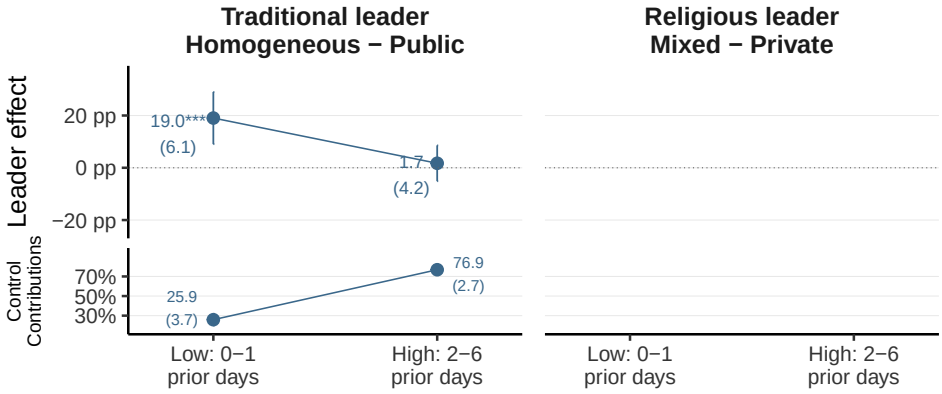


* p<.10, ** p<.05, *** p<.01. Control Contributions, tags: estimate (SE), 90% confidence intervals.

- **Public records:** traditional-leader effects are $+19.0$ pp for low contributors and $+1.7$ pp for high contributors. The -17.3 pp contrast has $p = .030$.

HTE specification ▸ Other environments ▸ All history estimates ▸ 15 min: Policy + contributions

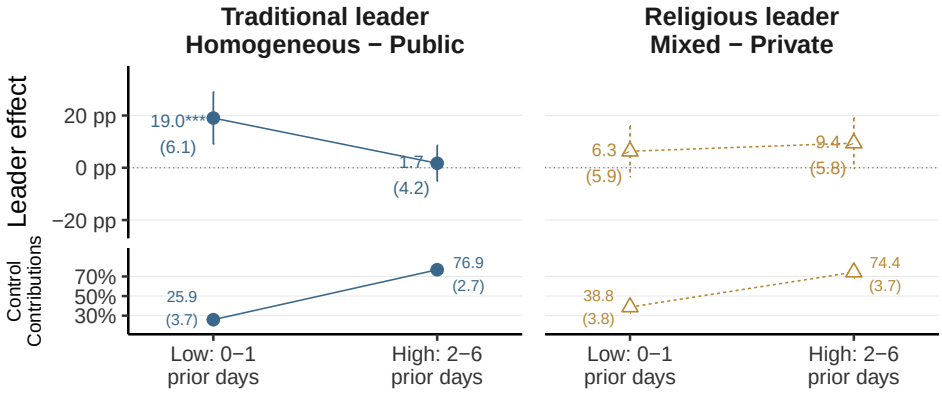
Pastors mobilize everyone



* p<.10, ** p<.05, *** p<.01. Control Contributions, tags: estimate (SE), 90% confidence intervals.

- **Internal obligation:** under Private records, compare pastor effects by prior contributions (0–1 vs. 2–6 days). Religious appeals should motivate both groups.

Pastors mobilize everyone



* p<.10, ** p<.05, *** p<.01. Control Contributions, tags: estimate (SE), 90% confidence intervals.

- **Religious leaders under Private records:** effects are +6.3 pp for low contributors and +9.4 pp for high contributors, with no clear difference between them.

HTE specification ▸ Other environments ▸ All history estimates ▸ 15 min: Policy + contributions

Mechanisms

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Outline

1. Setting
2. Experiment
3. **Results**
   - Main results
   - Mechanisms
   - **Alternative explanations**
4. Conclusion

Alternative explanations and robustness

1. **Leader endogenous behavior.** Speech varies little across experimental environments. Leaders do not follow up more where mobilization succeeds.

   Leader speech | Follow-up effort | Arriving together

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   Same population

▸ Mechanisms ▸ Composition results ▸ Disclosure results

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▸ Mechanisms ▸ Composition results ▸ Disclosure results

Alternative explanations and robustness

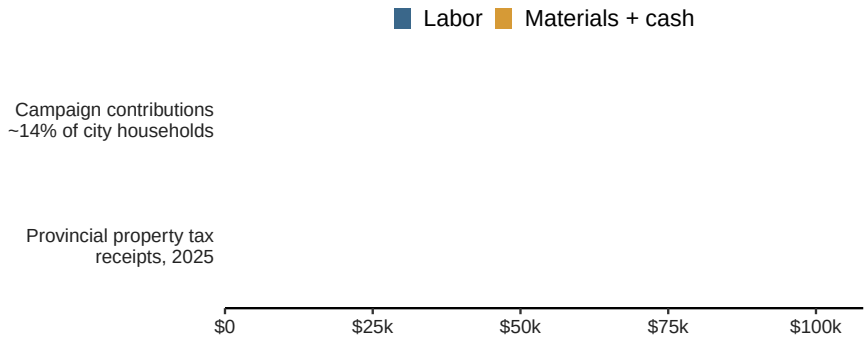
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4. **Leadership by example.** Leaders' prior participation does not explain the pattern.
   Leadership by example
5. **Robustness.** Results survive flexible baseline adjustment, alternative windows, outcomes, inference, spillovers and controls.
   Balance | Trends / day one | Baseline slopes | Spillovers
   Outcome windows | Measurement | Scope / inference | Composition | Disclosure

▸ Mechanisms ▸ Composition results ▸ Disclosure results

Outline

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2. Experiment
3. Results
4. **Conclusion**
   - **Policy**
   - Taking stock
   - Concluding thoughts

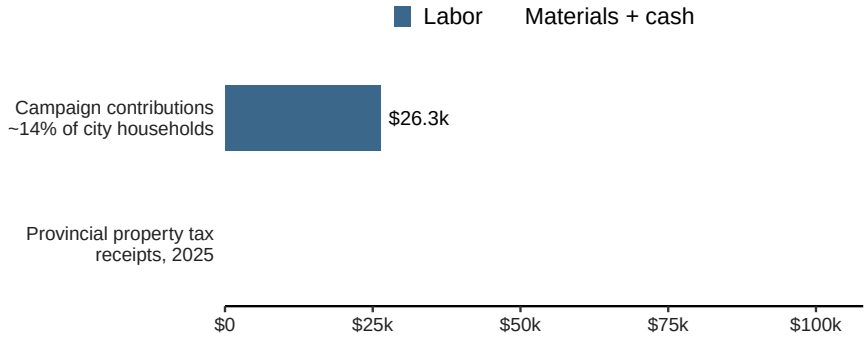
Taking stock of community contributions



- The campaign recorded **70,250 hours of labor**: about **8,800 eight-hour workdays** across 28 neighborhoods.

Wage benchmark | Valuation details | Fiscal source

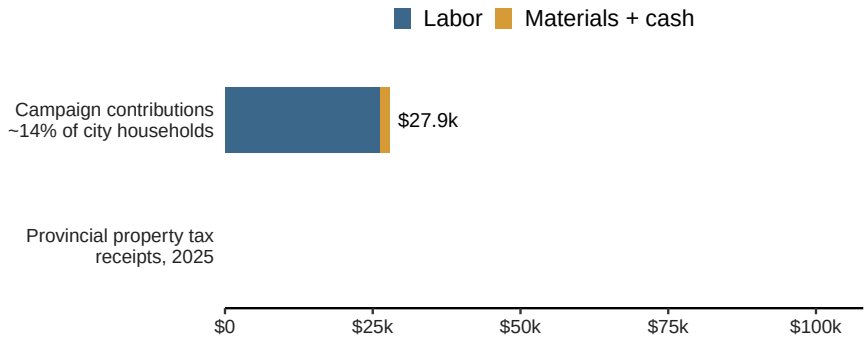
Taking stock of community contributions



- **Value labor at a World Bank program rate in Kananga**: $3/day. Assuming an eight-hour workday: $0.375/hour. At 70,250 hours: **$26,344**.

Wage benchmark Valuation details Fiscal source

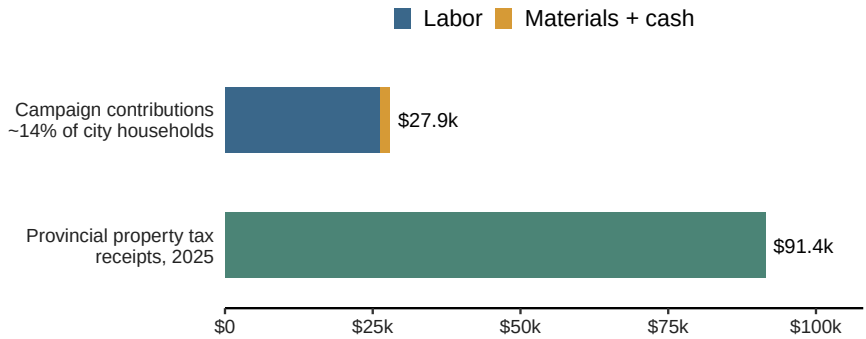
Taking stock of community contributions



- Labor, materials and cash amount to **$27,900 in recorded contributions**.

Wage benchmark Valuation details Fiscal source

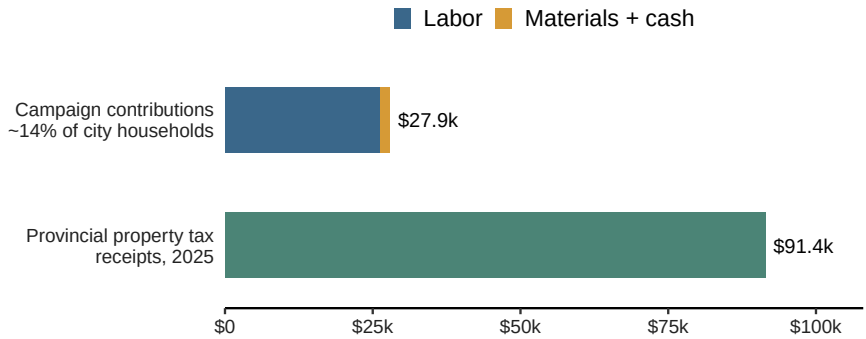
Taking stock of community contributions



- Contributions equal **30.5% of annual provincial property-tax receipts**.

Wage benchmark | Valuation details | Fiscal source

Taking stock of community contributions



- Contributions equal **30.5% of annual provincial property-tax receipts**.
- Illustrates importance of informal labor contributions to public finances in developing countries. (Olken and Singhal, 2011)

Wage benchmark | Valuation details | Fiscal source

Matching leaders and information could mobilize more labor

- **Benchmark:** the 1,040 enrolled residents recorded **47,700 hours** over the full campaign period.
- **Match leaders and reporting rules to group composition.** Hours regression
  - Homogeneous groups: chief + Public records, **18% more hours**.
  - Mixed groups: pastor + Private records, **14% more hours**.
- **Projection:** assume these six-day proportional effects persist across all days of the campaign.

Calculation and uncertainty Other counterfactuals

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- **Projection:** assume these six-day proportional effects persist across all days of the campaign.
  - **7,300 additional hours** relative to a modeled no-leader total of **46,000 hours** under the same reporting rules.
  - About **75 worksite workloads**: 42 from chief/Public, 33 from pastor/Private.

Calculation and uncertainty Other counterfactuals

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Taking stock: answers to the research questions

1. Can religious leaders promote cooperation in ethnically divided societies?

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$\Rightarrow$ **Collective action depends on matching leaders to audiences and environments.**

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Concluding thoughts

- **Local leaders fill gaps in public-good provision and can complement state efforts.** What other roles do they already play that states can build on to strengthen capacity?

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- **Pastors shape beliefs about poverty and responsibility for helping others.**
  - → Does support for redistribution depend on how reforms fit religious worldviews? *Is Pentecostalism Eroding Support for Redistribution in Africa? Experimental Evidence from a Progressive Tax Reform*. (Boyer et al., work in progress)

Appendix summary

Thank you!

gabrielgranato@berkeley.edu

Appendix


2025/10/18 08:3
Kananga, République démocratique du Congo

Appendix

1. Context and implementation
Chief profile / Pastor profile / Regional examples
Chiefs and ethnicity / Territories and conflict / Recruitment
Timeline / Windows / Meetings / Composition / Disclosure
Assignment balance / Randomization / Sample
Coverage / Delivery / Fieldwork interruption
HTE specifications / Networks: proximity / cohesion

2. Main results and alternative explanations
LASSO procedure / Selected controls
Specification comparison: composition / environments
Contrasts and inference: composition / environments
Spillovers / No-leader levels / Pastor origins / Same population / Prior leader effort
Windows: Pooled disclosure / Mixed / Homogeneous
Baseline imbalance / Trends and day one / Baseline slopes
Contact specification / Contact model / Timing model
Policy hours regression
Campaign accounting / Fiscal / Project costs / Income
Policy projection / Counterfactuals / Cash and goods

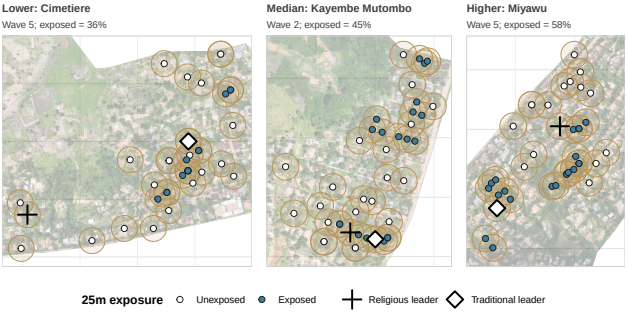
3. Outcome definitions (both leaders)
Pooled disclosure:
Rate / days / Hours / effort / Asinh hours / cash
Cash / In-kind / Asinh material
Four environments:
Rate / days / Hours / effort / Asinh hours / cash
Cash / In-kind / Asinh material

4. Scope of authority
Favorable evaluations / Sanctioning authority
Majority origin / Baseline identity
Personal faith / Social religiosity / Denomination
Leader traits: local roots / conviction / dual office

5. Appeals and enforcement
Leader speech: pooled disclosure / four environments
Resident speech: attendance / divine / material / time
Chief-reference exchange
Own histories: other records / traditional / religious
Peer histories: traditional / religious

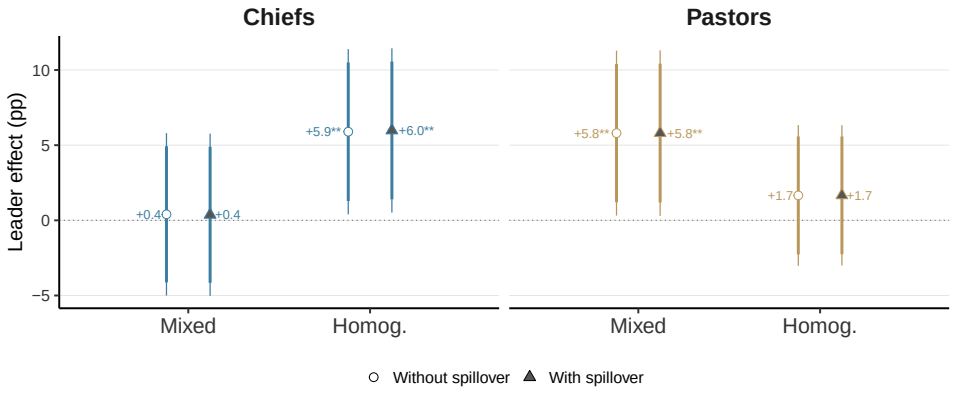
Could nearby groups affect participation?

- **Exposure:** another group's study household within 25 metres, measured at the end of the wave's rollout. Own-group neighbors are excluded.
- **483 of 1,040 residents exposed (46.4%).** The radius was selected for balanced exposure support, without using outcomes.



Alternative explanations | Composition results | Four environments | Appendix

Composition effects with and without spillover adjustment

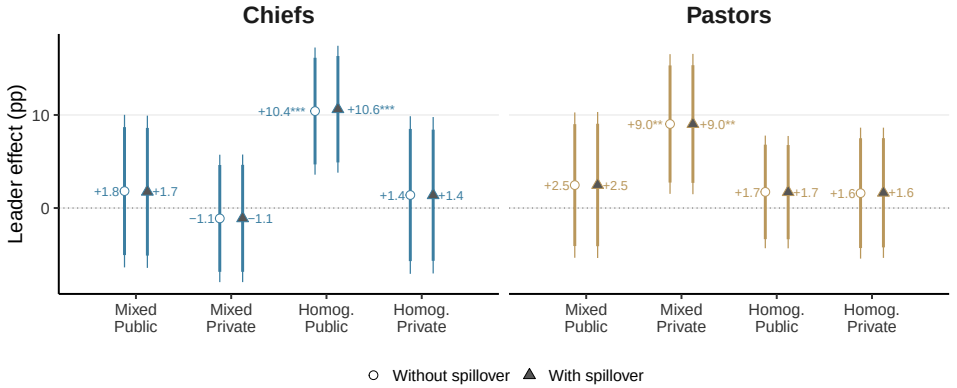


95% / 90% group−clustered intervals. * p<.10, ** p<.05, *** p<.01.

Same 11 LASSO-selected controls, baseline Salongo contribution and design fixed effects. Only spillover adjustment changes.

Alternative explanations | Maps / definition | Other specification | Appendix

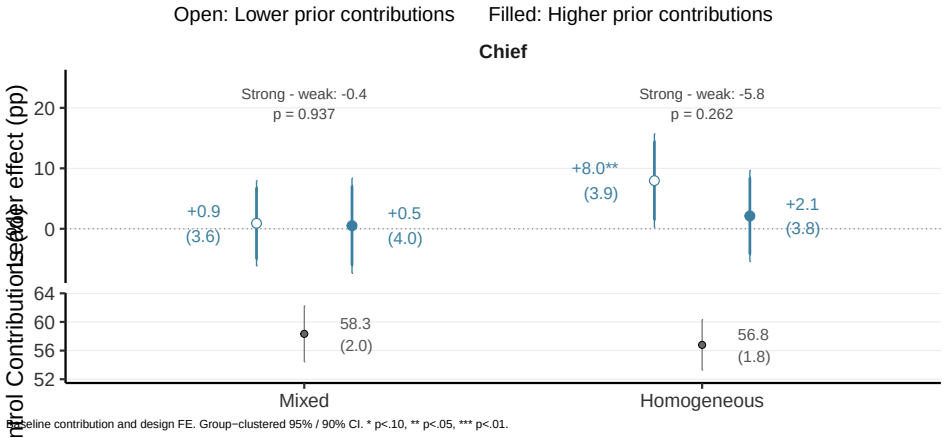
Environment effects with and without spillover adjustment



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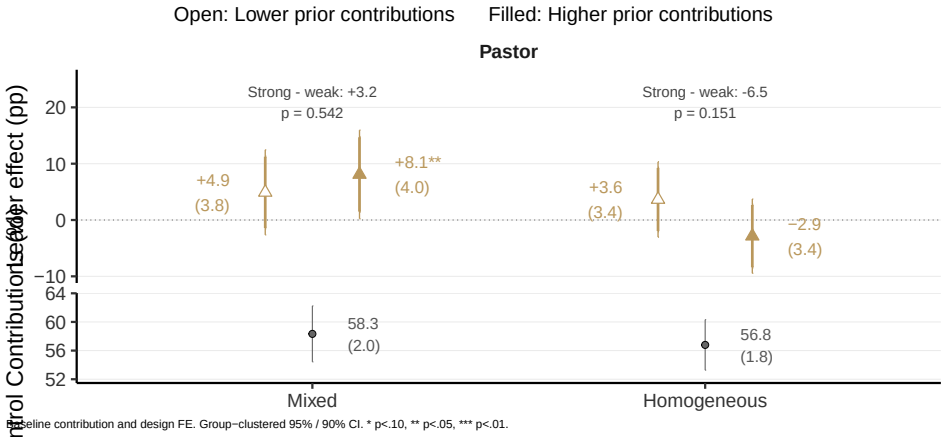
Same 11 LASSO-selected controls, baseline Salongo contribution and design fixed effects. Only spillover adjustment changes.

Leadership by example: chiefs (disclosure pooled)



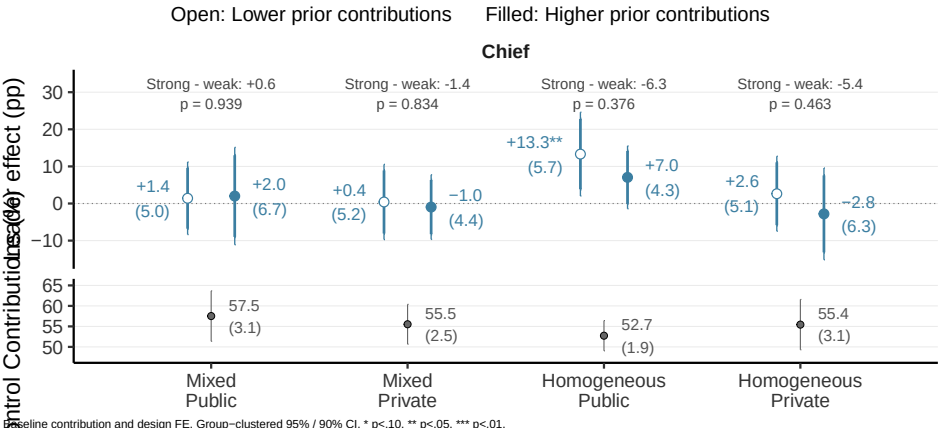
Baseline contribution and design FE. Group−clustered 95% / 90% CI. * p<.10, ** p<.05, *** p<.01.

Leadership by example: pastors (disclosure pooled)



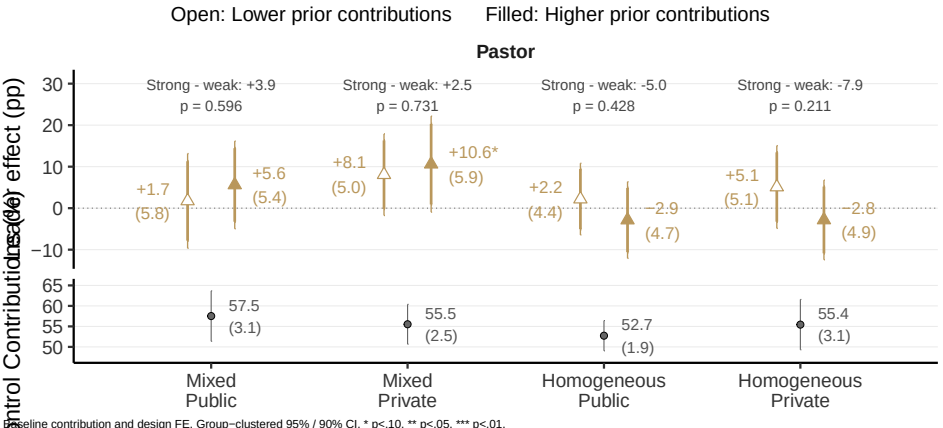
Baseline contribution and design FE. Group−clustered 95% / 90% CI. * p<.10, ** p<.05, *** p<.01.

Leadership by example: chiefs (four environments)



Baseline contribution and design FE. Group−clustered 95% / 90% CI. * p<.10, ** p<.05, *** p<.01.

Leadership by example: pastors (four environments)



Leader speech: composition specification

Estimate separately for chiefs and pastors:

$$S_{gn} = \alpha_n + \beta \, \text{Homogeneous}_g + \varepsilon_{gn}.$$

- **One observation per speaking leader-meeting.** 90 chief meetings and 92 pastor meetings.
- S_{gn}: share of speech devoted to a topic, or total speaking minutes. Pool public and private disclosure.
- α_n: leader/neighborhood fixed effects. β: homogeneous minus mixed, comparing the same leader across compositions.
- Plot regression-adjusted levels using the same leader weights within each leader type. Standard errors cluster by leader/neighborhood.

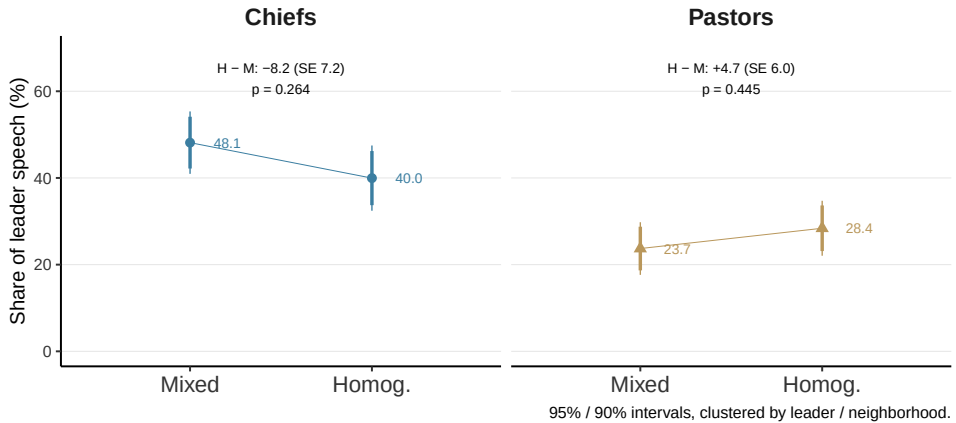
Leader speech: four-environment specification

Estimate separately for chiefs and pastors:

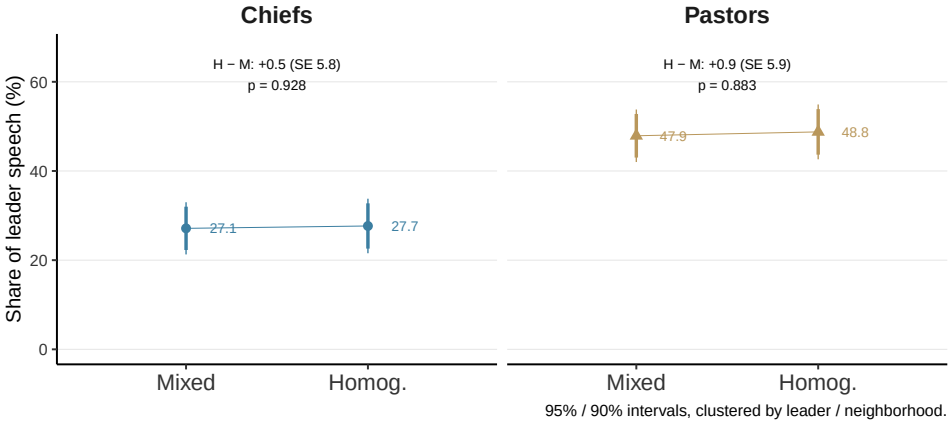
$$S_{gn} = \alpha_n + \beta H_g + \gamma P_g + \delta H_g P_g + \varepsilon_{gn}.$$

- **Same 90 chief and 92 pastor meetings.** H_g: homogeneous group. P_g: public disclosure.
- S_{gn}: topic share or total speaking minutes. α_n: leader/neighborhood fixed effects.
- Plot all four regression-adjusted levels, using common leader weights within each type.
- Public minus private: γ in mixed groups, $\gamma + \delta$ in homogeneous groups. Standard errors cluster by leader/neighborhood.

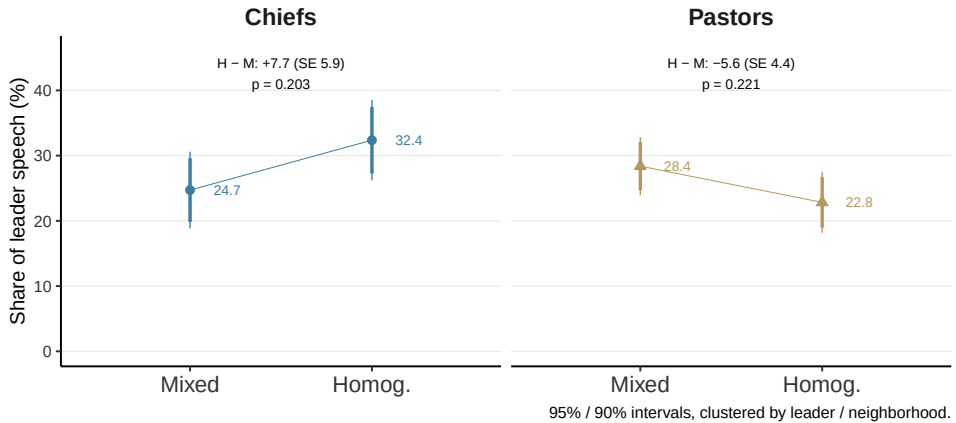
Attendance and performance appeals (disclosure pooled)



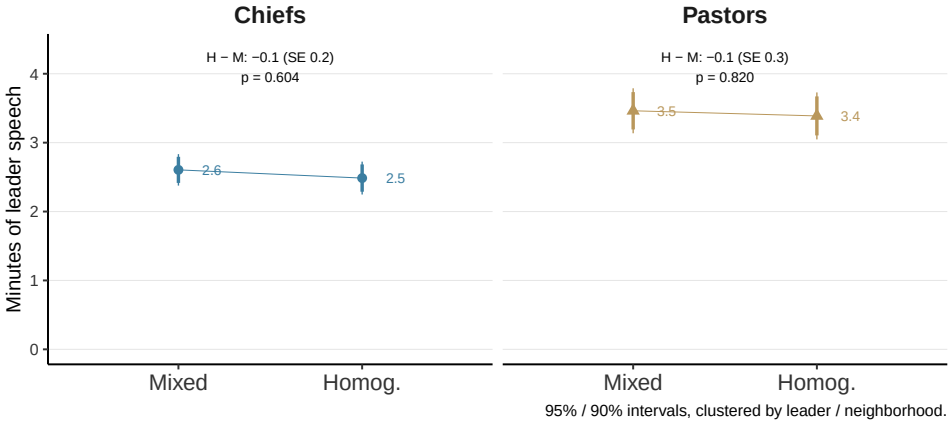
Divine reward and solidarity (disclosure pooled)



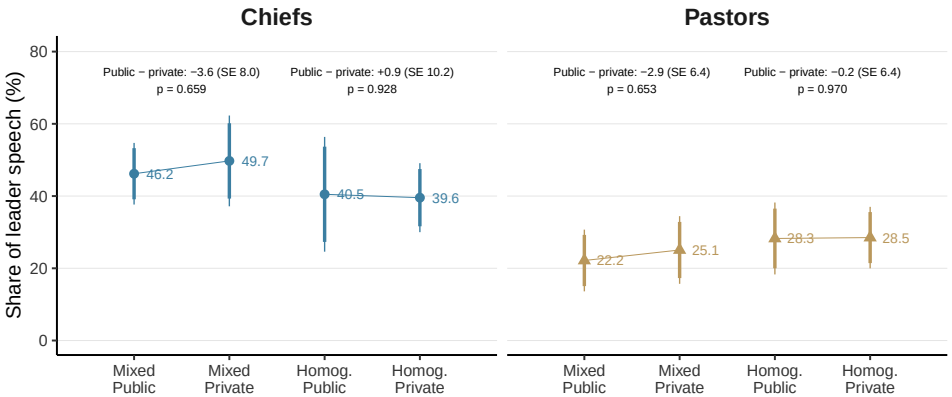
Requests for material support (disclosure pooled)



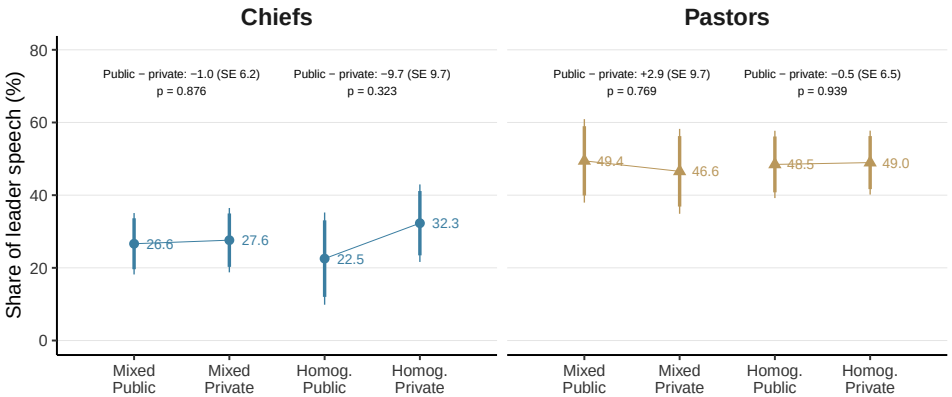
Total leader speaking time (disclosure pooled)



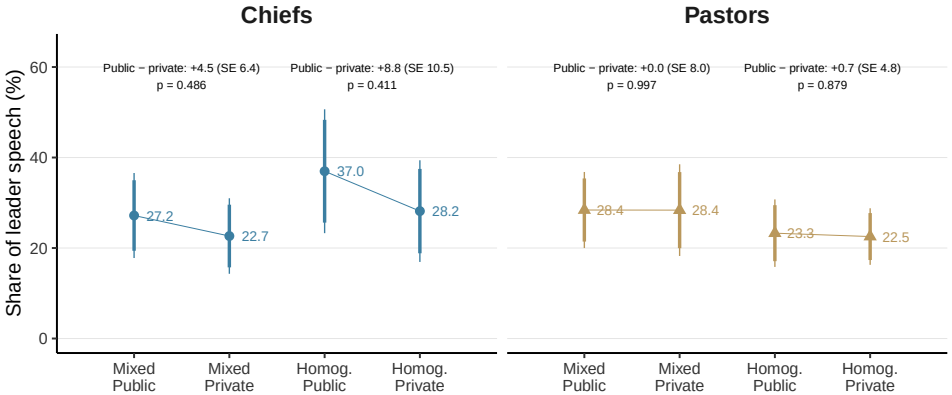
Attendance and performance appeals (four environments)



Divine reward and solidarity (four environments)

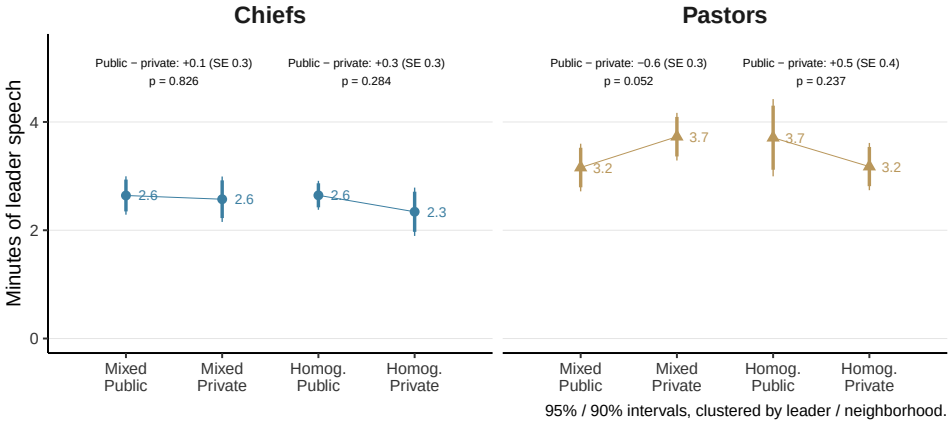


Requests for material support (four environments)



Alternative explanations | Next outcome | Other specification | Specification | Appendix

Total leader speaking time (four environments)



How the participation controls are selected

1. **Candidate pool:** 1,158 correctly coded baseline columns: survey items and constructs, category and threshold indicators, and missingness indicators. Constant and duplicate columns removed. Missing values mean-imputed and columns standardized before CV.

2. **Select once:** Gaussian LASSO ($\alpha = 1$), five fixed folds holding out whole groups. Choose the literal CV minimum, $\lambda_{\text{min}} = 0.02093$, by mean squared prediction error.

3. **Always included, never penalized:** leader-by-composition assignments, baseline Salongo contribution, neighborhood $\times$ composition $\times$ disclosure FE, and 25 m spillover exposure.

4. **Refit OLS:** retain the 11 selected controls. Freeze this set across the main participation and disclosure models. Heterogeneity figures omit selected controls, with both specifications in the appendix. Cluster SEs by group. Selected controls

Selection is exploratory following earlier specification searches. Reported tests are nominal: they do not adjust for selection or earlier searches.

The 11 selected baseline controls

Demographics and service provision

- Education, female, age (3).
- NGO named as provider of poor relief and schools (2).
- Chief's salongo performance: response category 4 (1).

Attitudes and beliefs

- Residents fulfill salongo responsibilities: category 2 (1).
- Understanding others' emotions: category 3 (1).
- Government security performance and avenue-chief performance: category ≥ 2 (2).
- Ancestors reward good deeds promptly in this life: category ≥ 2 (1).

Categories refer to questionnaire response codes. These are selected adjustment variables, not treatment-effect moderators. Missingness indicators were candidates. None were selected.

Back to LASSO procedure

Appendix

Civic work and recognition are used across the region

Kinshasa: Kin Bopeto



Prizes for the cleanest quartier in each commune.

Togo: cleanliness contests



Neighborhood streets evaluated by a jury.

- **Rwanda's Umuganda**: regular community work for infrastructure, environmental protection, and cleanliness.
- **Cameroon's Villes propres**: a city-cleanliness competition originating in a neighborhood contest.

Back

Appendix

Why do urban chiefs have an ethnic basis?

1. **An urban office rests on local social standing.**
   - Avenue chiefs are respected residents nominated by elders and ratified by the state. Their influence depends on local knowledge and recognition. (Balan et al., 2022)
2. **Origin ties remain relevant after migration to the city.**
   - Origin-based associations organize mutual support and links to customary authorities. Ethnic origins also shape expectations of political representation. (Gobbers, 2016. Hoffmann et al., 2022)
3. **A shared Lulua identity does not erase these distinctions.**
   - Around Kananga, kin groups and chiefly histories underpin geographically distinct identities within the Lulua umbrella. (Ngalamulume, 1995)

$\Rightarrow$ The chief's administrative reach can exceed the origin networks that sustain his standing.

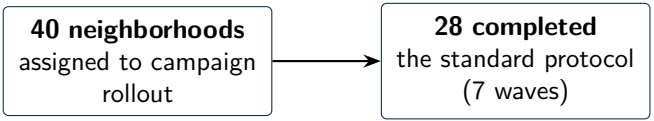
Territories distinguish origins. Conflict reshuffled neighbors



- **Territories are provincial administrative divisions:** Demba, Dibaya, Dimbelenge, Kazumba, and Luiza. We also distinguish Kananga city.
- **Origin territory proxies kin networks and chiefly histories within the Lulua umbrella**, rather than current residence. (Ngalamulume, 1995)
- **Kamuina Nsapu began in 2016 over chiefly succession in Dibaya.** Displacement brought different origin communities into the same urban neighborhoods. (CRG, 2018. Mercy Corps, 2019)

$\Rightarrow$ Shared public-good needs need not imply shared origin networks.

Sampling crosses local origins within each neighborhood

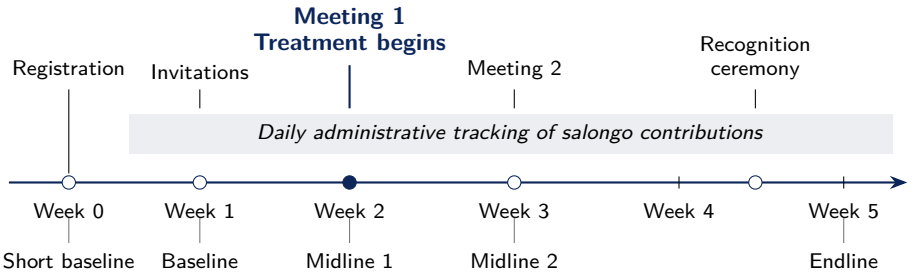


- Household visits register adults willing to join and elicit leader nominations.
  - The most-mentioned willing chief and pastor are selected before team assignment.
- Sampling oversamples the neighborhood majority to make same-origin teams feasible.
- **Main comparisons are between teams within intervention neighborhoods.**

Sample accounting Randomization

Back Appendix

Two meetings coordinate several weeks of daily work



Surveys around the campaign and the two meetings

- At Meeting 1, residents first encounter their assigned team and disclosure condition.
- The same team meets again one week later. Work continues after recognition.

Window definitions

Back

Appendix

Meetings coordinate contributions but do not supply the work





- Four people review recent team contributions, coordinate work, and encourage one another.
- Leaders join as team members, with **no script or special mobilization instructions**.
- Daily work takes place at different sites and times: meetings create a shared audience.

Treatment delivery

Back

Appendix

Composition varies homogeneous and mixed teams



Illustration: A is the neighborhood majority's origin, the chief belongs to it, letters represent origins, not specific people.

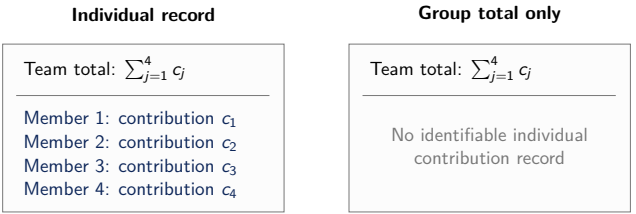
- Homogeneous composition uses the **neighborhood majority in every leader arm**.
  - The pastor need not share that origin.
- Mixed-team formation draws a new ethnic origin whenever one remains available. Nearly 95% attained the intended diversity.

$\Rightarrow$ 22.2% of initial roster slots were replaced before revelation, preserving the intended treatment cells.

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Appendix

Disclosure makes individual contributions visible to teammates



Schematic of the information disclosed, c_j denotes member j's recorded contribution.

- Both conditions receive the same team total at the meeting.
- Individual disclosure adds an identifiable record for each member.

$\Rightarrow$ Disclosure can strengthen social judgment. People already know their own behavior.

Treatment delivery

Back

Appendix

Assignment balances groups within each rollout wave

- 40 neighborhoods assigned to the intervention rollout.
  - Main leader effects compare groups **within intervention neighborhoods**.
- Planned rollout: ten waves of four neighborhoods. Seven waves completed under the standard protocol.
- Within neighborhoods: eligibility pools, resident selection, and assignment to the twelve treatment cells.
- Cumulative rerandomization uses prespecified covariate-balance criteria.
  - 100 candidate allocations in wave 1. 48 in waves 2–7.
  - Replacement lists preserve the intended leader/composition/disclosure cells.

The analysis follows 1,040 distinct residents

- 28 neighborhoods $\times$ 12 groups $=$ **336 groups**.
- Three ordinary residents per group $=$ 1,008 residents.
- During waves 1–2, no-leader groups contained four distinct ordinary residents.
  - 8 neighborhoods $\times$ 4 no-leader groups $=$ 32 additional residents.
- From wave 3, a repeated placebo member filled the fourth position in no-leader groups.
  - Repeated fourth members, like repeated chiefs and pastors, are excluded from the main outcome sample.
- **1,008 + 32 = 1,040 non-repeated residents.**

Pretreatment balance

The assigned meetings and disclosure protocols were delivered

- Residents first learned their team composition and disclosure assignment at Meeting 1.
- 95% attended the first meeting. Assigned-leader attendance was 99.6%.
- The individual-record announcement and subsequent record protocol were verified in at least 99.4% of cases.
- Leaders received the common meeting instructions, without a script or special leader incentive.

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Appendix

The Ebola emergency interrupted the planned rollout

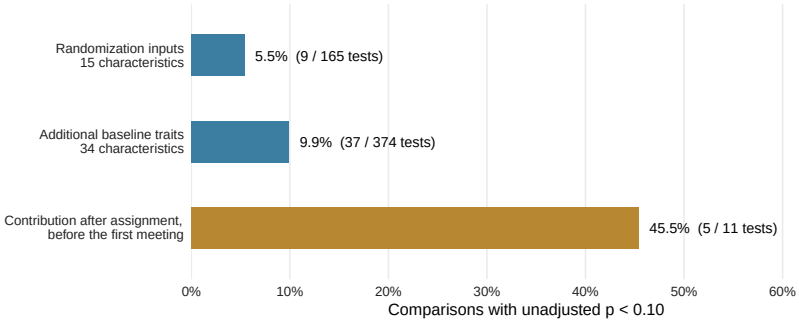
- **Planned rollout:** ten waves of four neighborhoods, covering 40 neighborhoods.
- **Fieldwork stopped on May 30, 2026.** The Ebola emergency disrupted Wave 8 and prevented the remaining rollout. Wave 8 was only 30.4% complete when the outbreak was declared.
- **Main sample:** seven completed waves: 28 neighborhoods, 336 teams, and 1,040 residents. The paper's appendix reports checks adding disrupted Wave 8 observations.

Source: paper appendix, "Fieldwork stoppage."

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Appendix

Pre-assignment characteristics are broadly balanced



Most pre-assignment comparisons show no detectable difference. Administrative participation before Meeting 1 is an exception.

Paper's assignment-balance table: 11 contrasts per characteristic. Neighborhood FE, group-clustered SEs. Counts of unadjusted tests are descriptive, not a joint balance test.

Participation imbalance

Back to design

Appendix

Groups largely retain their assigned composition

Leader	Composition	Disclosure	Assigned residents	Replacement share	Mean territories	Final majority-origin share	Composition fidelity
No leader	Mixed	Private	92	21.7	3.04	22.8	96.4
		Public	92	18.5	3.07	22.8	100.0
	Homogeneous	Private	92	17.4	1.07	97.8	92.9
		Public	92	22.8	1.07	97.8	92.6
Traditional	Mixed	Private	84	26.2	2.61	23.8	96.4
		Public	84	27.4	2.82	28.6	100.0
	Homogeneous	Private	84	21.4	1.07	96.4	92.9
		Public	84	25.0	1.11	95.2	89.3
Religious	Mixed	Private	84	29.8	2.79	26.2	96.4
		Public	84	21.4	2.82	26.2	100.0
	Homogeneous	Private	84	15.5	1.07	97.6	92.9
		Public	84	20.2	1.18	94.0	85.7
Total			1,040	22.2	1.98	60.7	94.6

Back to composition

Appendix

Residents, leaders, and records reached the meetings

Leader	Composition	Disclosure	Assigned residents	Resident attendance	Realized group size	Full-group share	Leader attendance	M1 announcement verified	M2 record verified
No leader	Mixed	Private	92	95.7	3.82	82.1	–	96.4	100.0
		Public	92	95.7	3.75	75.0	–	100.0	100.0
	Homogeneous	Private	92	97.8	3.82	85.7	–	100.0	100.0
		Public	92	91.3	3.64	64.3	–	100.0	100.0
Traditional	Mixed	Private	84	94.0	3.82	85.7	100.0	100.0	100.0
		Public	84	96.4	3.86	85.7	96.4	96.4	100.0
	Homogeneous	Private	84	92.9	3.79	78.6	100.0	100.0	100.0
		Public	84	94.0	3.82	82.1	100.0	100.0	100.0
Religious	Mixed	Private	84	95.2	3.86	89.3	100.0	100.0	100.0
		Public	84	96.4	3.89	89.3	100.0	100.0	100.0
	Homogeneous	Private	84	96.4	3.89	89.3	100.0	100.0	100.0
		Public	84	94.0	3.82	82.1	100.0	100.0	100.0
Total			1,040	95.0	3.82	82.4	99.6	99.4	100.0

Back to assignment Disclosure Appendix

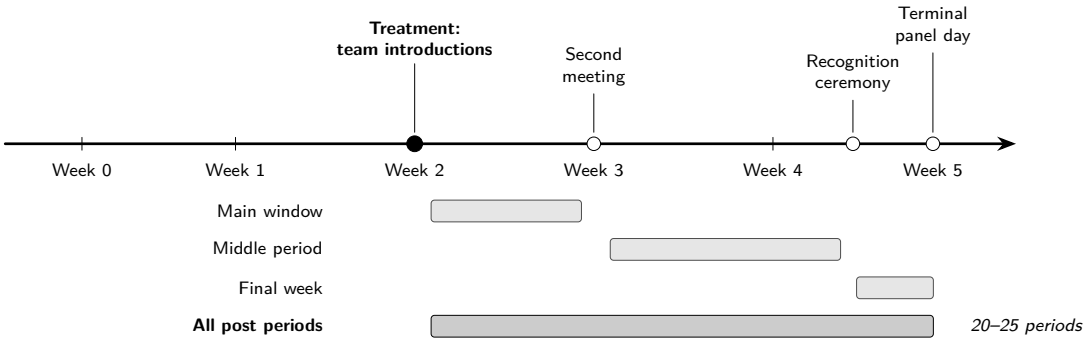
Administrative participation is observed for every resident

Leader	Composition	Disclosure	Assigned residents	Baseline survey	Meeting 1 usable	Meeting 2 usable	Endline survey	Audio recorded	Administrative outcome
No leader	Mixed	Private	92	100.0	95.7	93.5	88.0	89.3	100.0
		Public	92	98.9	95.7	90.2	95.7	57.1	100.0
	Homogeneous	Private	92	100.0	97.8	94.6	93.5	78.6	100.0
		Public	92	98.9	91.3	92.4	97.8	64.3	100.0
Traditional	Mixed	Private	84	100.0	94.0	92.9	94.0	92.9	100.0
		Public	84	98.8	96.4	90.5	92.9	60.7	100.0
	Homogeneous	Private	84	98.8	92.9	85.7	94.0	89.3	100.0
		Public	84	98.8	94.0	92.9	97.6	60.7	100.0
Religious	Mixed	Private	84	100.0	95.2	89.3	85.7	89.3	100.0
		Public	84	100.0	96.4	94.0	95.2	64.3	100.0
	Homogeneous	Private	84	100.0	96.4	90.5	96.4	100.0	100.0
		Public	84	98.8	94.0	86.9	86.9	50.0	100.0
Total			1,040	99.4	95.0	91.2	93.2	74.7	100.0

Back to measurement

Appendix

Outcome windows span meetings and recognition



Chiefs: office, networks, and standing

1. Selection and institutional role
Resident-nominated avenue notables. Organize salongo, mediate disputes, and link households to city authorities.

2. Demographic and economic profile
Relative to ordinary residents: 22 years older, 1.4 more years of schooling, 12.5 more years on the avenue. Overwhelmingly men. Higher income.

3. Social networks and state links
Same-ethnicity shares: +16 pp among confidants, +13 pp in material-support networks. Knowing the quartier chief: +48 pp.

4. Civic and religious involvement
Plan 10 additional salongo hours. 29% also report a religious role.

5. Personal traits
Higher aggregate Big Five score, especially conscientiousness. No clear difference in emotional intelligence or intrinsic religiosity.

6. Residents' assessments
Known and recognized as authorities across origins. Trust, moral standing, motivation, and hard work are rated more favorably mainly by majority-origin residents.

7. Expected contributions
Residents across origins expect more Salongo contribution, with particularly strong expectations of labor effort.

An urban local office with customary roots. Not necessarily a hereditary rural sovereign.

Profile comparisons use ordinary residents in the same locality as the reference. Descriptive differences, not treatment effects. Source: manuscript leader profiles and resident evaluations.

Back Appendix

Pastors: congregations, networks, and standing

1. Selection and institutional role
Resident-nominated religious leaders, mostly outside mission churches. Of 28: 12 Pentecostal, 7 Kimbanguist, 4 Neo-Apostolic. 2 Protestant/Presbyterian, 1 traditional religion, 2 other/unreported. No Catholic priests.

2. Demographic and economic profile
Relative to ordinary residents: 13 years older, 1.8 more years of schooling, 12.9 more years on the avenue. Overwhelmingly men. No clear income difference.

3. Social networks and state links
Over 90% of congregations mix ethnic groups. Coethnic share: 48.2%, versus 48.8% without sorting. Knowing the quartier chief: +30 pp.

4. Civic and religious involvement
More church attendance and giving. Plan 6.8 additional salongo hours. About 40% also hold a traditional role.

5. Personal traits
Similar standard leadership traits. More positive and less negative religious coping. Intrinsic religiosity is 0.11 points higher, with weaker evidence.

6. Residents' assessments
Better known, motivating, morally exemplary, trustworthy, and hardworking among both coethnics and residents of other origins.

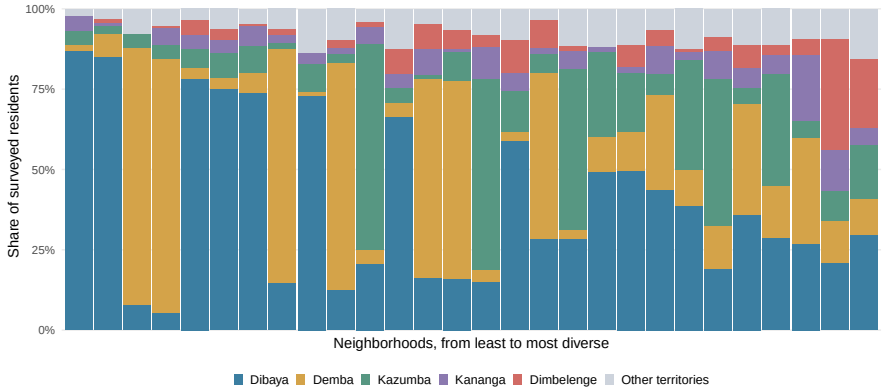
7. Expected contributions
Residents expect participation and financial support. Expected labor effort is weaker than for chiefs.

Profile differences use ordinary residents in the same locality as the reference. Denominations describe the 28 assigned leaders. Recruitment was open to all denominations.

Back

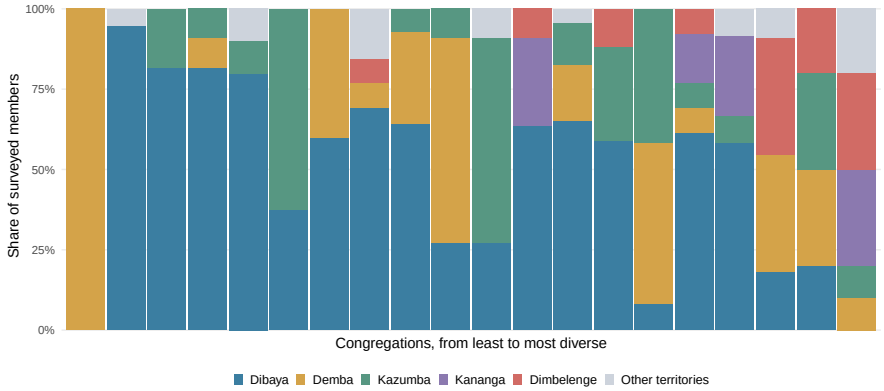
Appendix

Neighborhoods span ethnic origins



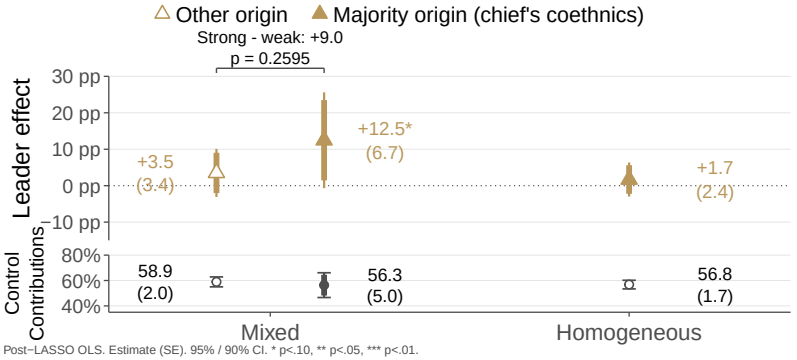
Back to chiefs
Recruitment profile

Church congregations span ethnic origins



Back to pastors
Recruitment profile

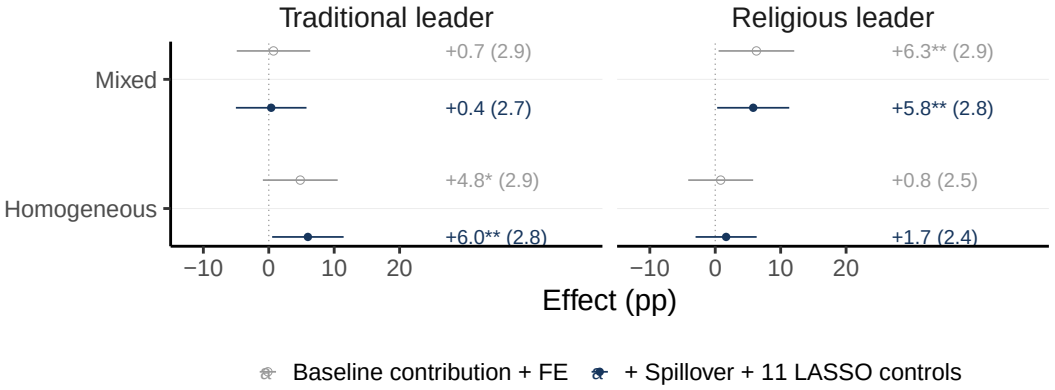
Pastors reach majority residents in mixed groups



- Among the chief's coethnics, pastor effects are **12.5 pp in mixed groups versus 1.7 pp in homogeneous groups**. The 10.8 pp contrast is imprecise ($p = .129$).
- Consistent with **group context shaping religious appeals**, rather than individual majority status.

Identity and religiosity
◂ Mixed-group results
Back to alternative explanations
Appendix

Composition effects: baseline versus LASSO adjustment

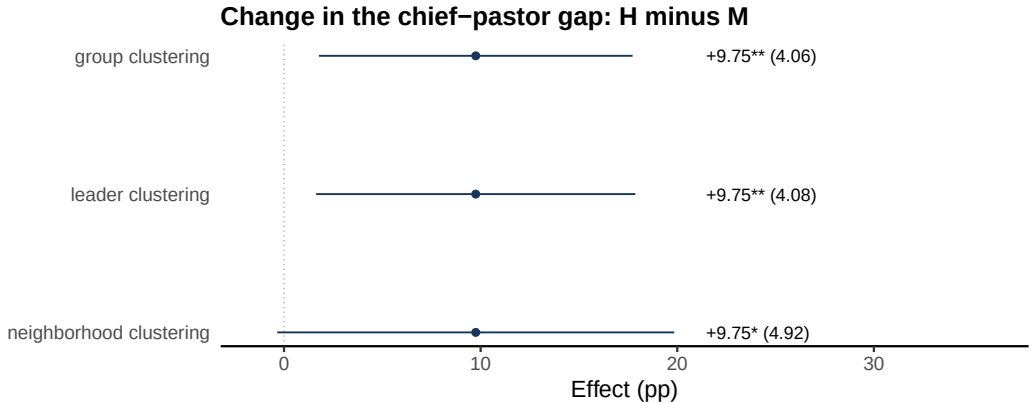


95% CI, tags: effect (SE). * p<.10, ** p<.05, *** p<.01.
Observed no−leader means: Mixed 58.3%, Homogeneous 56.8%.

Both models retain baseline Salongo contribution and design fixed effects. The current specification adds 25 m spillover exposure and the frozen 11 LASSO-selected controls. No-leader means are observed cell means. Group-clustered inference.

Scope checks Next Appendix

Composition contrasts and alternative inference

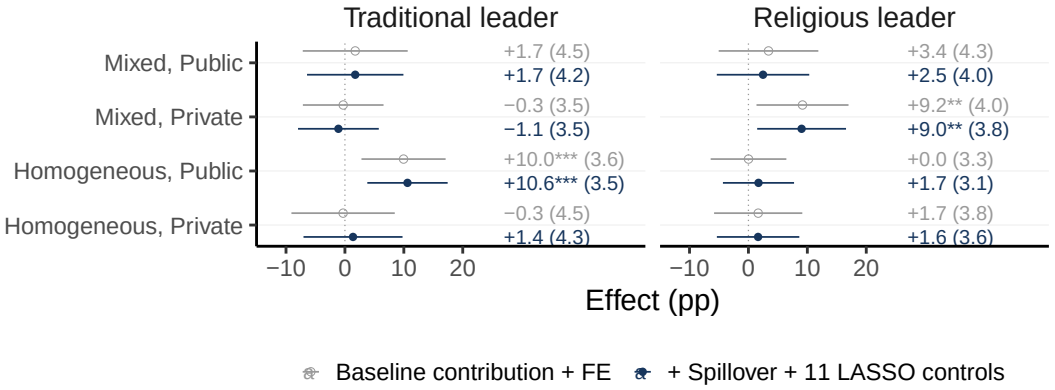


Main LASSO model: baseline, current exposure, 11 selected controls and NQC FE.
95% intervals. Updated randomization inference unavailable pending the assignment/exposure/nuisance replay protocol.

Baseline Salongo contribution and neighborhood × composition × disclosure FE. Group-clustered SEs. No-leader references follow the matching source specification.

Scope checks Previous Appendix

Environment effects: baseline versus LASSO adjustment

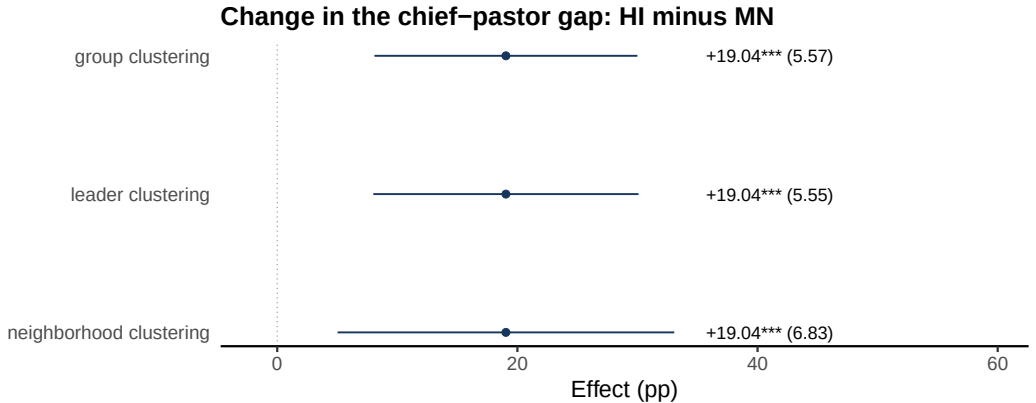


95% CI, tags: effect (SE). * p<.10, ** p<.05, *** p<.01.
Observed no−leader means: Mixed Public 57.8%, Private 58.9%, Homogeneous Public 55.8%, Private 57.8%.

Both models retain baseline Salongo contribution and design fixed effects. The current specification adds 25 m spillover exposure and the frozen 11 LASSO-selected controls. No-leader means are observed cell means. Group-clustered inference.

Enforcement checks Next Appendix

Environment contrasts and alternative inference

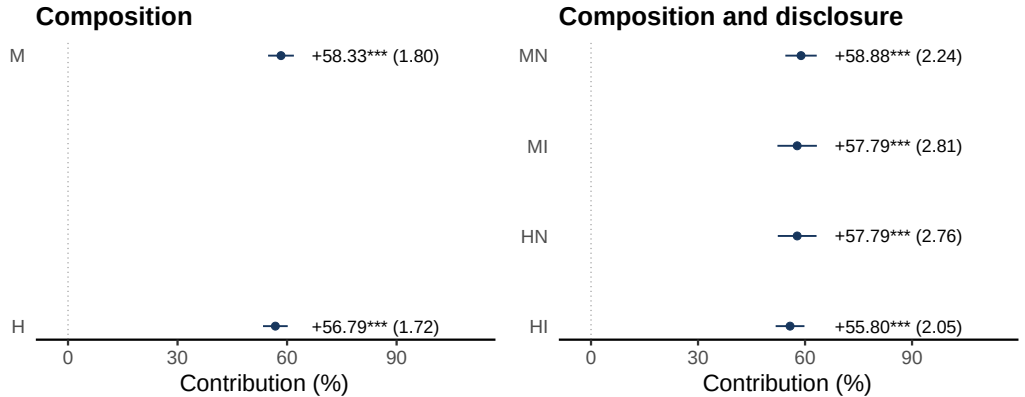


Main LASSO model: baseline, current exposure, 11 selected controls and NQC FE.
95% intervals. Updated randomization inference unavailable pending the assignment/exposure/nuisance replay protocol.

Baseline Salongo contribution and neighborhood × composition × disclosure FE. Group-clustered SEs. No-leader references follow the matching source specification.

Enforcement checks Previous Appendix

No-leader participation across experimental conditions



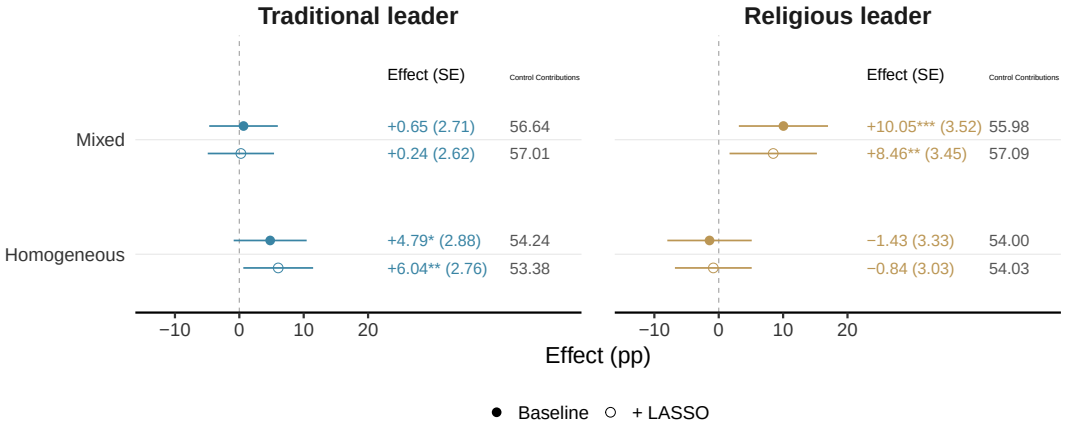
Observed−covariate no−leader profile from the matching LASSO model. Composition difference p=.536, environment omnibus p=.781.
, baseline, current exposure, selected controls and NQC FE. Group-clustered 95% intervals.

Baseline Salongo contribution and neighborhood × composition × disclosure FE. Group-clustered SEs. No-leader references follow the matching source specification.

Enforcement checks

Appendix

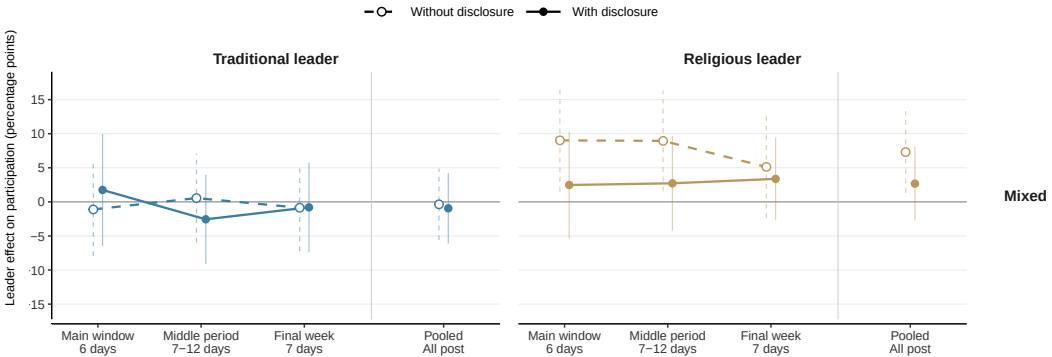
Restricting pastors to the neighborhood majority



Composition-pooled effects. Baseline Salongo contribution and design-stratum FE. Group-clustered SEs. Pastor-arm residents retained in 17 matching neighborhoods. Other arms unchanged. No-leader references use each leader's retained support, with controls at sample means. Added controls: 25 m spillover exposure and the frozen 11 LASSO controls.

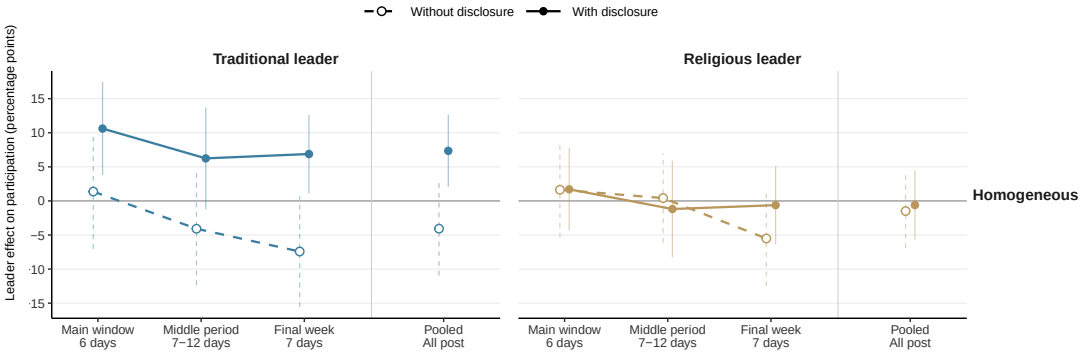
Back Scope checks Next Appendix

Mixed groups: leader effects across outcome windows

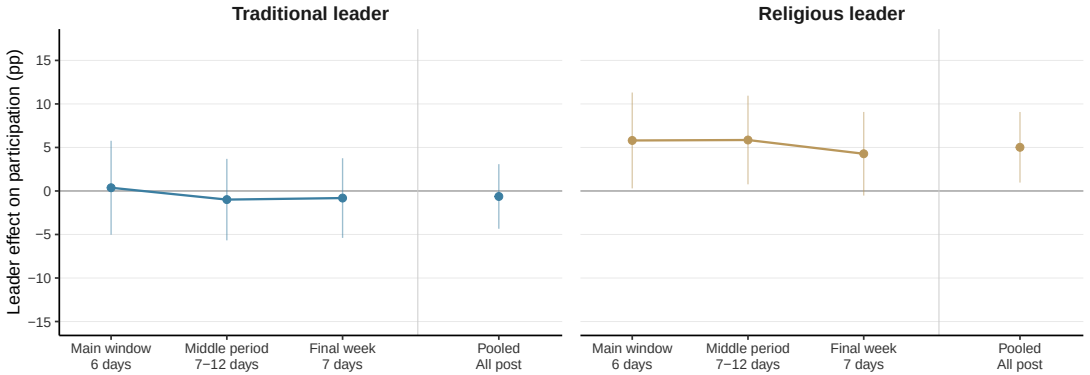


Enforcement checks | Definitions | Next | Pooled disclosure | Appendix

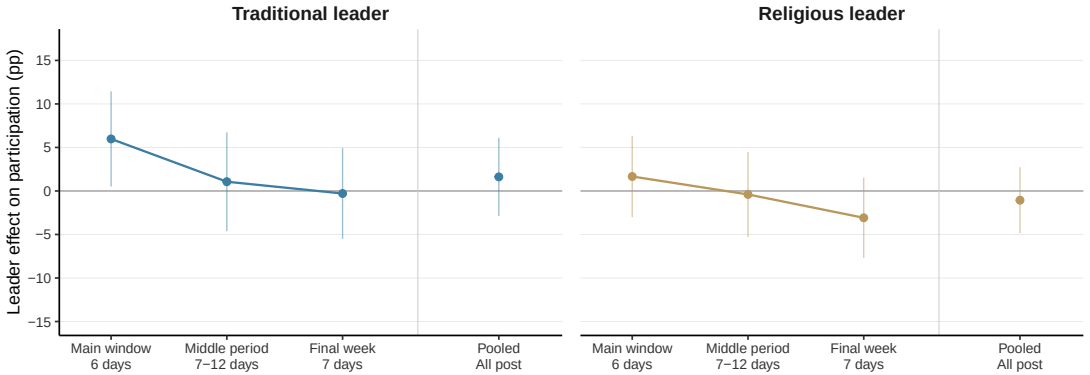
Homogeneous groups: leader effects across outcome windows



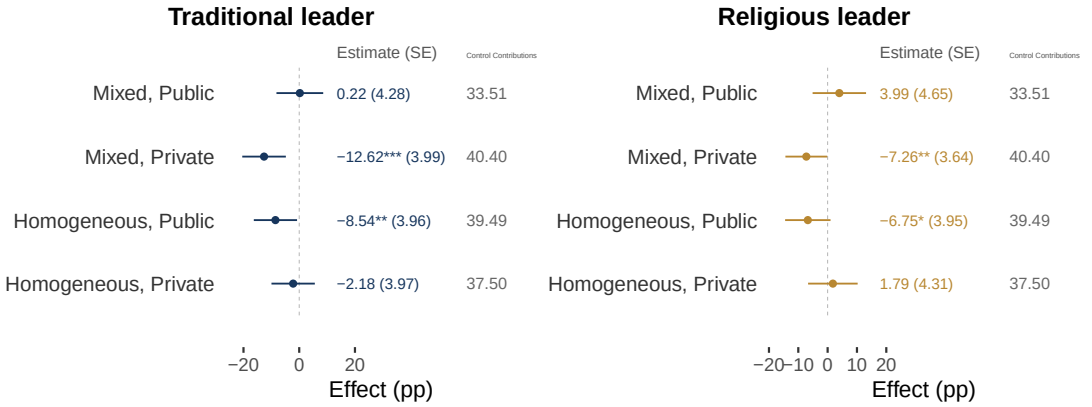
Mixed groups: effects pooled across disclosure



Homogeneous groups: effects pooled across disclosure



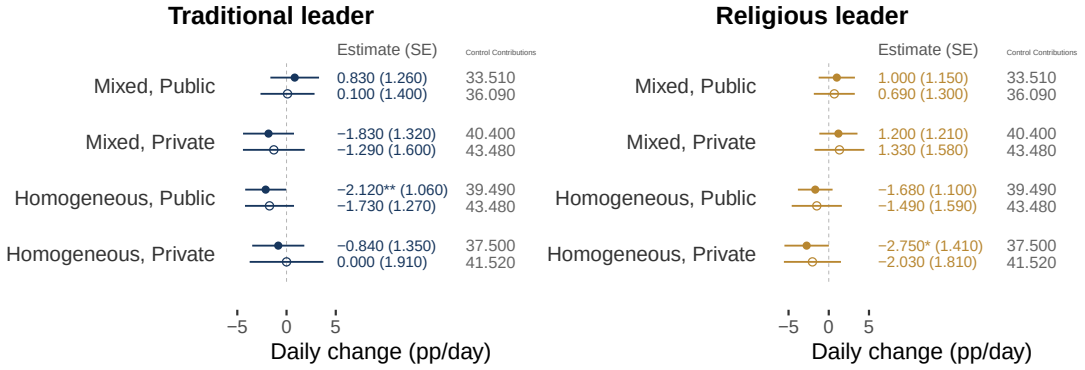
Participation already differs before the first meeting



Bars: 95% CI. * p<.10, ** p<.05, *** p<.01 (source tests).

Six common days after assignment, before Meeting 1. Differences compare residents within neighborhood. Group-clustered SEs. Stars use unadjusted p-values: * $p < .10$, ** $p < .05$, *** $p < .01$.

Pre-meeting trends, including the day-one exclusion

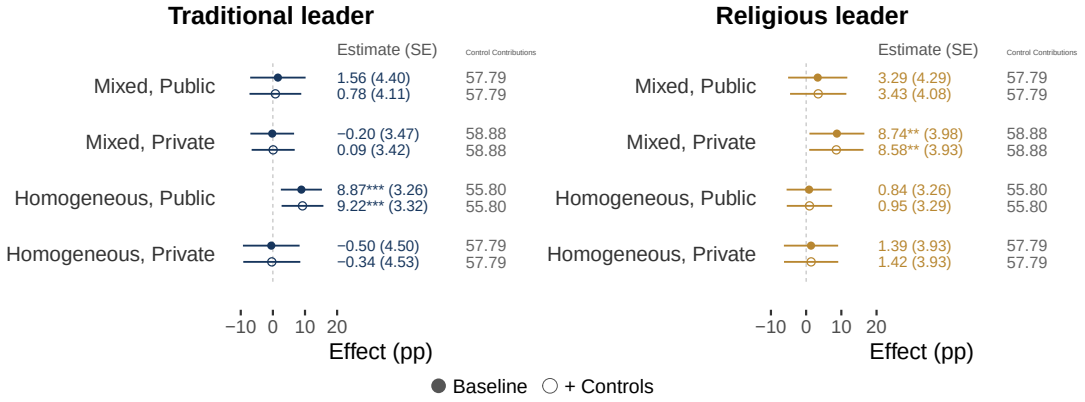


Bars: 95% CI. * p<.10, ** p<.05, *** p<.01 (source tests).

Respondent and neighborhood × disclosure × composition × day FE. Group-clustered SEs. No-leader references use the matching six- or five-day window. * $p < .10$, ** $p < .05$, *** $p < .01$.

Enforcement checks Previous Next Appendix

Separate baseline slopes in every assignment cell



Bars: 95% CI. * p<.10, ** p<.05, *** p<.01 (source tests).

Baseline Salongo contribution and neighborhood × composition × disclosure FE. Group-clustered SEs. No-leader references follow the matching source specification. Separate baseline-participation slope in each of the twelve assignment cells.

Enforcement checks Previous Appendix

Leader contact: composition specification

Run the same regression separately in chief groups and pastor groups:

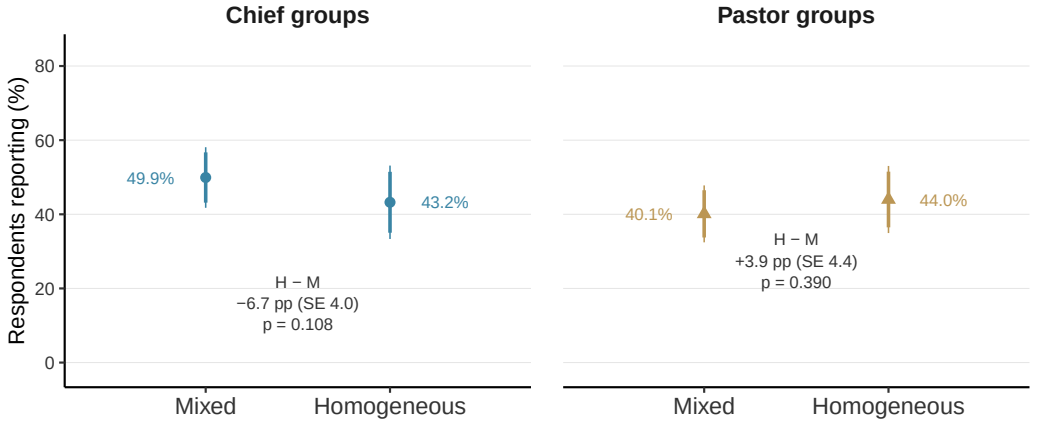
$$Y_{ig} = \alpha + \beta \,\mathrm{Homogeneous}_g + \rho X_i + \varepsilon_{ig}.$$

- **Chief groups:** 281 respondents reporting about their assigned chief.
- **Pastor groups:** 291 respondents reporting about their assigned pastor.
- Y_{ig} is respondent i's report about the assigned leader in group g. $\mathrm{Homogeneous}_g = 1$ for homogeneous groups and 0 for mixed groups.
- β**: homogeneous minus mixed**, adjusted for X_i: mean-centered baseline Salongo contribution. Each regression has its own coefficients. Standard errors cluster by neighborhood.

Plot both group levels at the same baseline Salongo contribution mean.

Composition levels | Disclosure specification | Alternative explanations | Appendix

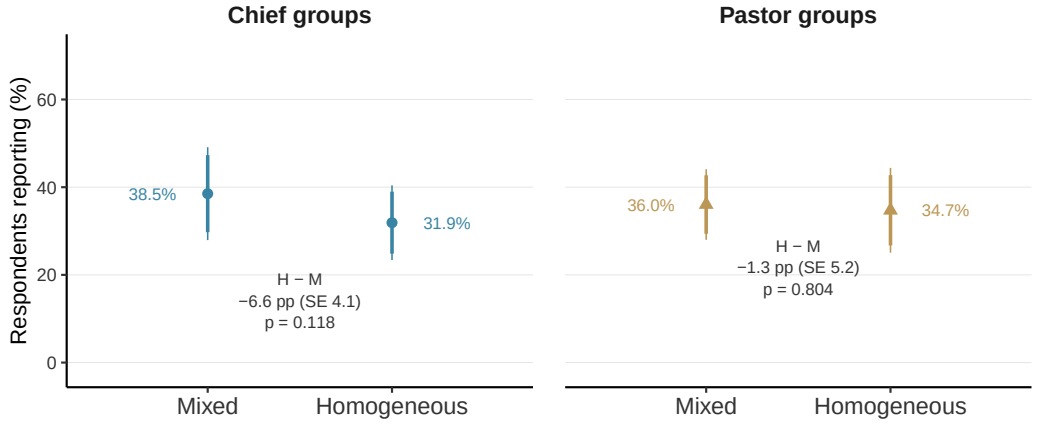
Any contact with the assigned leader



Adjusted levels. One report per respondent about the assigned leader in leader groups only. Baseline Salongo contribution adjustment. 90% / 95% neighborhood-clustered intervals. Separate chief-group and pastor-group regressions: homogeneous indicator + baseline Salongo contribution.

Specification Previous Next Alternative explanations

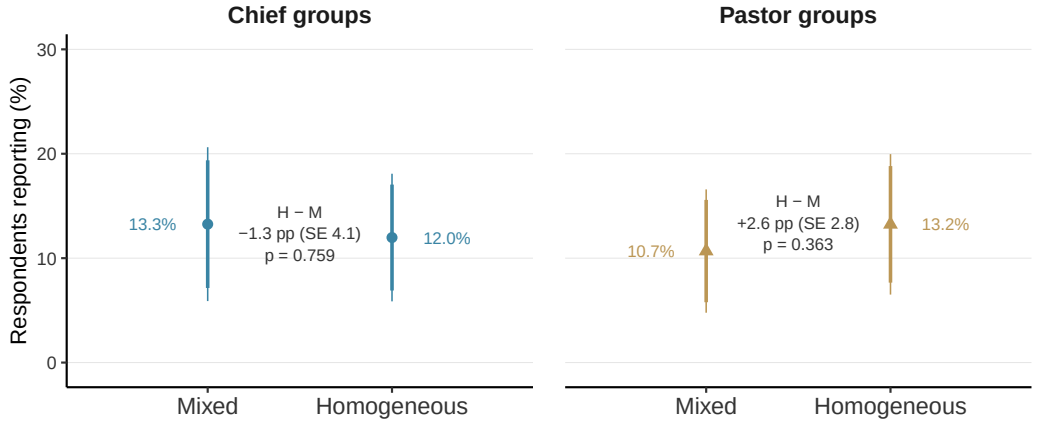
Discussed salongo with the assigned leader



Adjusted levels. One report per respondent about the assigned leader in leader groups only. Baseline Salongo contribution adjustment. 90% / 95% neighborhood-clustered intervals. Separate chief-group and pastor-group regressions: homogeneous indicator + baseline Salongo contribution.

Specification Previous Next Alternative explanations Question details

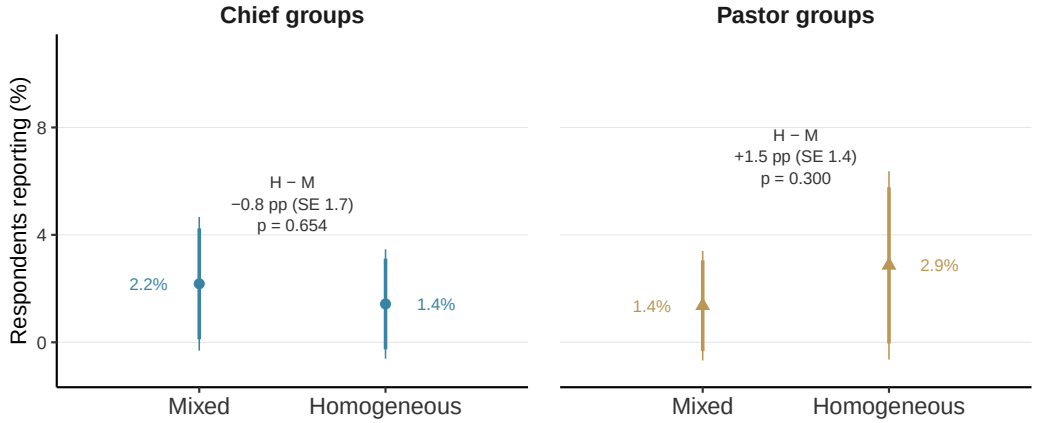
Assigned leader asked respondent to participate



Adjusted levels. One report per respondent about the assigned leader in leader groups only. Baseline Salongo contribution adjustment. 90% / 95% neighborhood-clustered intervals. Separate chief-group and pastor-group regressions: homogeneous indicator + baseline Salongo contribution.

Specification Previous Next Alternative explanations Question details

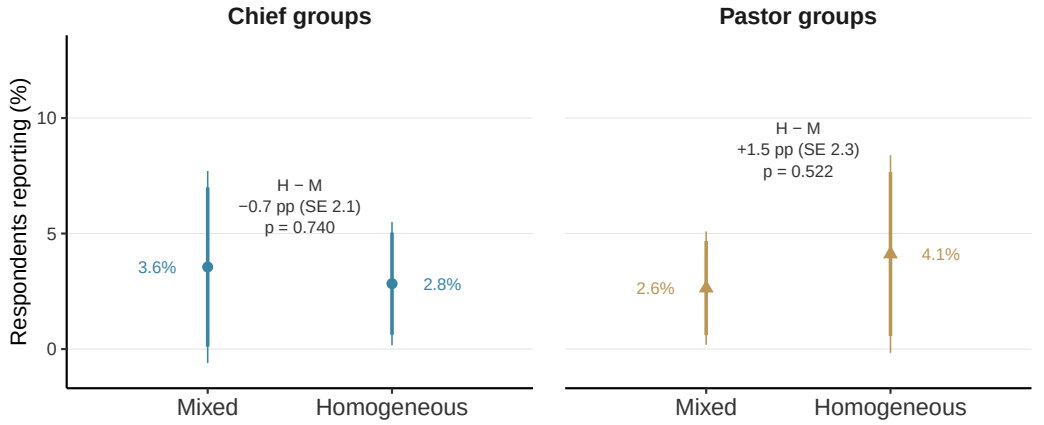
Assigned leader mentioned social consequences



Adjusted levels. One report per respondent about the assigned leader in leader groups only. Baseline Salongo contribution adjustment. 90% / 95% neighborhood-clustered intervals. Separate chief-group and pastor-group regressions: homogeneous indicator + baseline Salongo contribution.

Specification Previous Next Alternative explanations Question details

Assigned leader mentioned divine consequences



Adjusted levels. One report per respondent about the assigned leader in leader groups only. Baseline Salongo contribution adjustment. 90% / 95% neighborhood-clustered intervals. Separate chief-group and pastor-group regressions: homogeneous indicator + baseline Salongo contribution.

Leader contact: disclosure specification

$$Y_{ig} = \alpha_L + \beta_L H_g + \gamma_L P_g + \delta_L H_g P_g + \rho_L X_i + \varepsilon_{ig}.$$

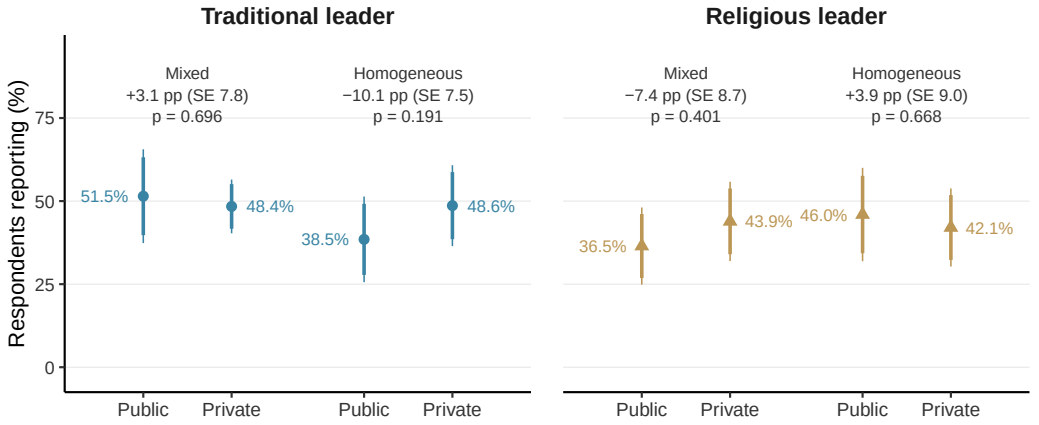
- **One respondent, one report about the assigned leader.** Leader groups only: 281 chief-group and 291 pastor-group respondents.
- Estimate separately for $L =$ chief and $L =$ pastor. H_g: homogeneous, P_g: public disclosure. X_i: mean-centered baseline Salongo contribution.
- Levels use the same baseline mean. Tests compare those plotted levels. Standard errors cluster by neighborhood (shared leader).

Public − Private: γ_L in Mixed groups,
$\gamma_L + \delta_L$ in Homogeneous groups.

Disclosure levels Composition specification

Alternative explanations Appendix

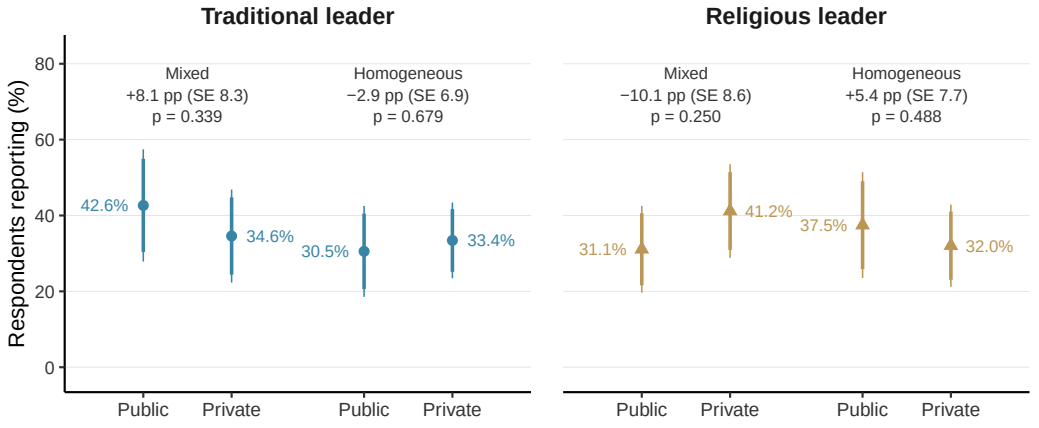
Any contact with the assigned leader



Adjusted levels. One report per respondent about the assigned leader in leader groups only. Baseline Salongo contribution adjustment. 90% / 95% neighborhood-clustered intervals. Within-composition Public minus Private comparisons.

Specification Previous Next Alternative explanations Question details

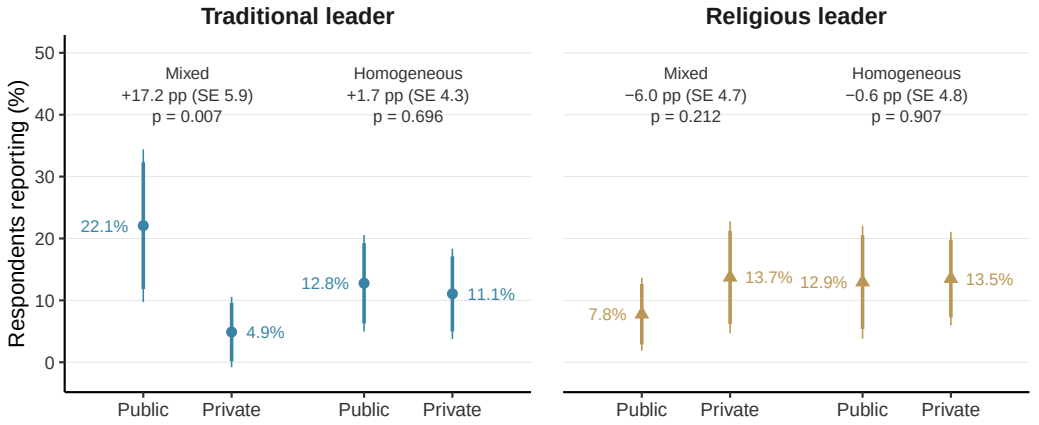
Discussed salongo with the assigned leader



Adjusted levels. One report per respondent about the assigned leader in leader groups only. Baseline Salongo contribution adjustment. 90% / 95% neighborhood-clustered intervals. Within-composition Public minus Private comparisons.

Specification Previous Next Alternative explanations Question details

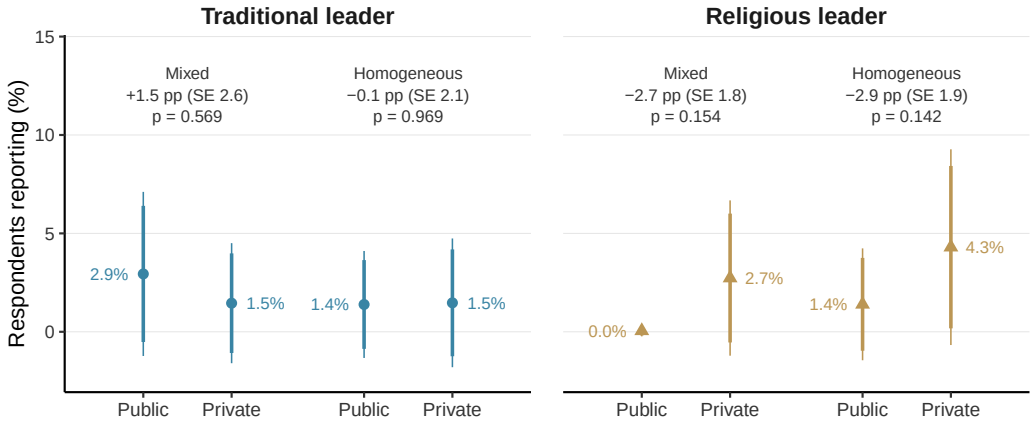
Assigned leader asked respondent to participate



Adjusted levels. One report per respondent about the assigned leader in leader groups only. Baseline Salongo contribution adjustment. 90% / 95% neighborhood-clustered intervals. Within-composition Public minus Private comparisons.

Specification Previous Next Alternative explanations Question details

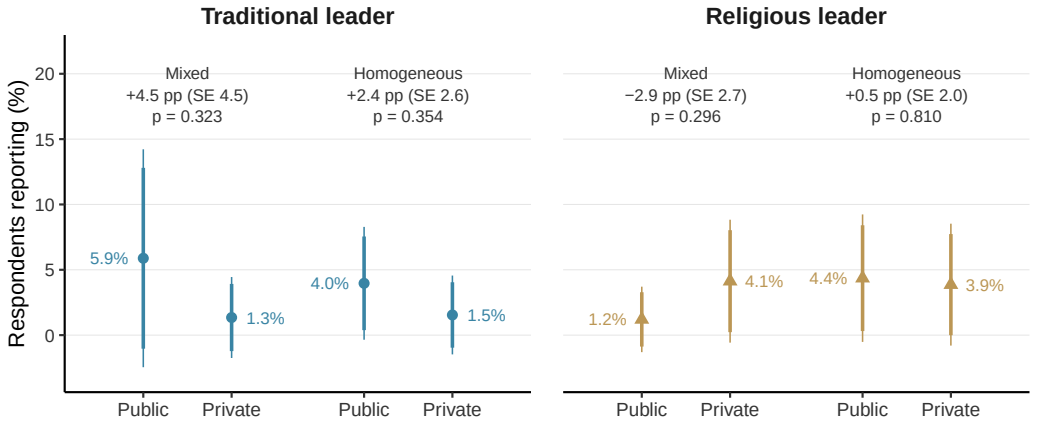
Assigned leader mentioned social consequences



Adjusted levels. One report per respondent about the assigned leader in leader groups only. Baseline Salongo contribution adjustment. 90% / 95% neighborhood-clustered intervals. Within-composition Public minus Private comparisons.

Specification Previous Next Alternative explanations Question details

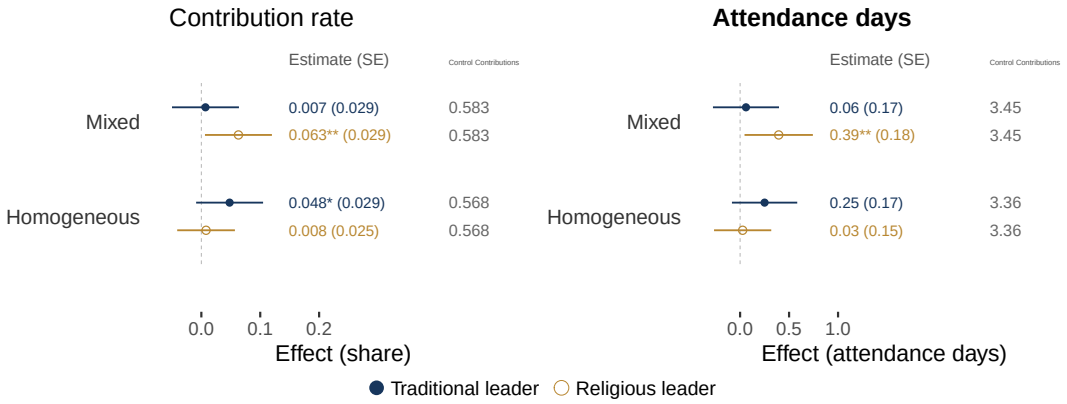
Assigned leader mentioned divine consequences



Adjusted levels. One report per respondent about the assigned leader in leader groups only. Baseline Salongo contribution adjustment. 90% / 95% neighborhood-clustered intervals. Within-composition Public minus Private comparisons.

Specification Previous Next Alternative explanations Question details

Participation rate and attendance days: by composition

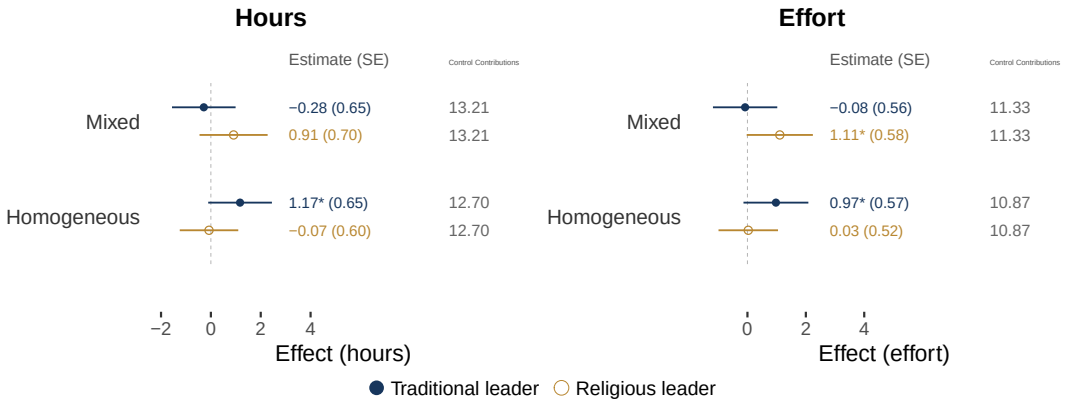


Bars: 95% CI. * p<.10, ** p<.05, *** p<.01 (source tests).

Matched pretreatment outcome and design FE. Group-clustered SEs. Six-day main window. No-leader means use the same outcome and environment. Amounts in CDF. Asinh retains zero contributions. Rate is a share. Days, hours, and effort are window totals.

Scope checks | Controls | More outcomes | Inference | Appendix

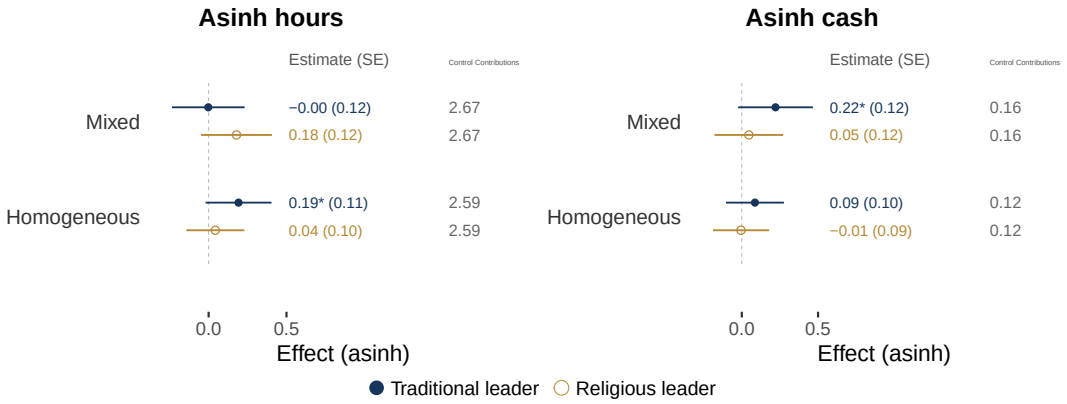
Hours and effort: by composition



Bars: 95% CI. * p<.10, ** p<.05, *** p<.01 (source tests).

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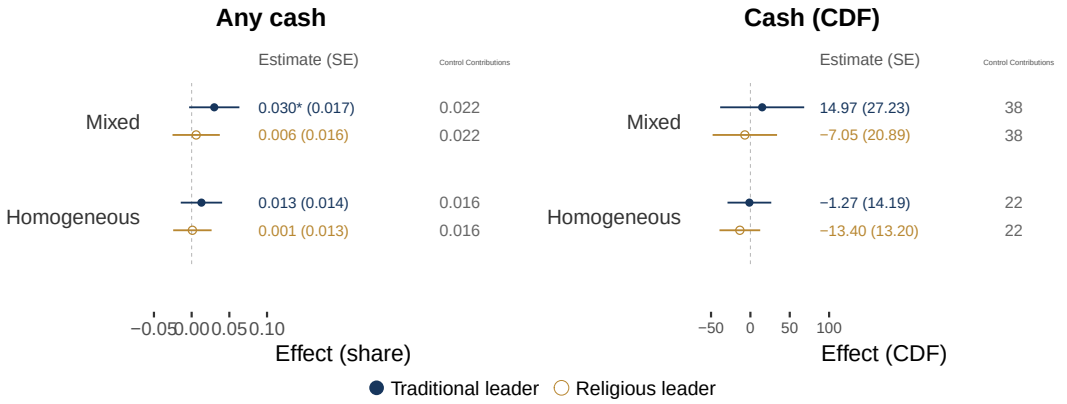
Transformed hours and cash: by composition



Bars: 95% CI. * p<.10, ** p<.05, *** p<.01 (source tests).

Matched pretreatment outcome and design FE. Group-clustered SEs. Six-day main window. No-leader means use the same outcome and environment. Amounts in CDF. Asinh retains zero contributions. Rate is a share. Days, hours, and effort are window totals.

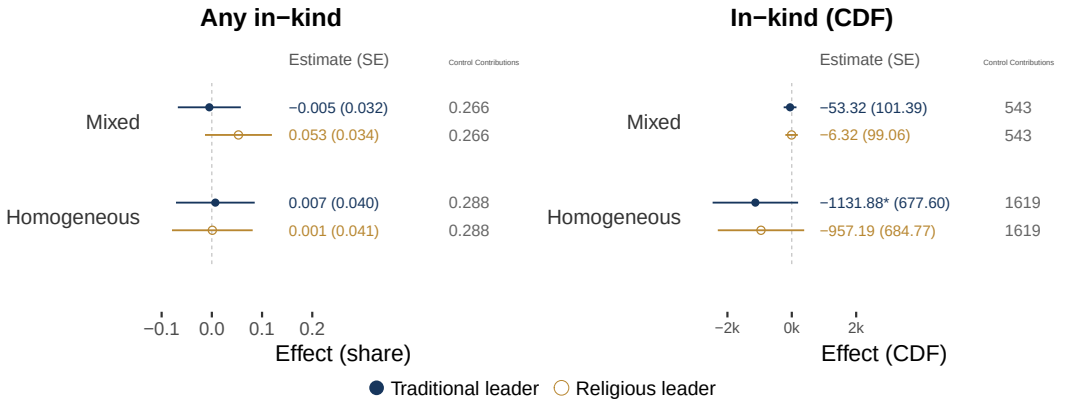
Cash contributions: by composition



Bars: 95% CI. * p<.10, ** p<.05, *** p<.01 (source tests).

Matched pretreatment outcome and design FE. Group-clustered SEs. Six-day main window. No-leader means use the same outcome and environment. Amounts in CDF. Asinh retains zero contributions. Rate is a share. Days, hours, and effort are window totals.

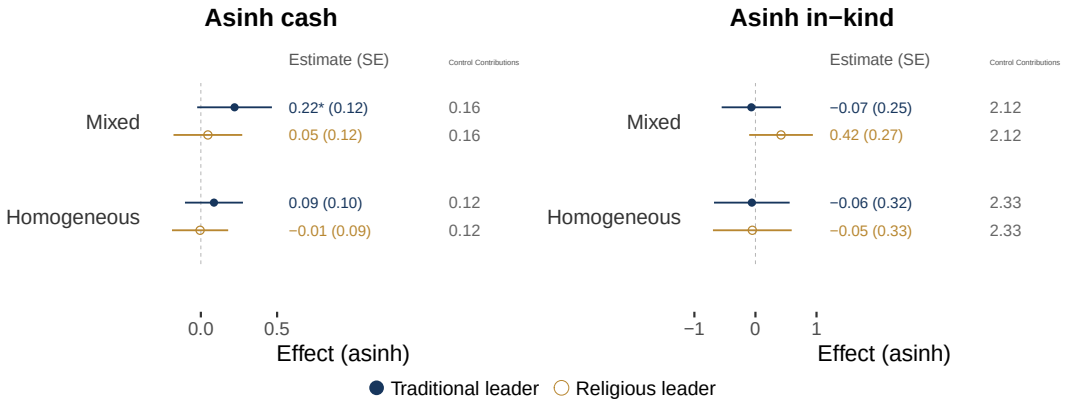
In-kind contributions: by composition



Bars: 95% CI. * p<.10, ** p<.05, *** p<.01 (source tests).

Matched pretreatment outcome and design FE. Group-clustered SEs. Six-day main window. No-leader means use the same outcome and environment. Amounts in CDF. Asinh retains zero contributions. Rate is a share. Days, hours, and effort are window totals.

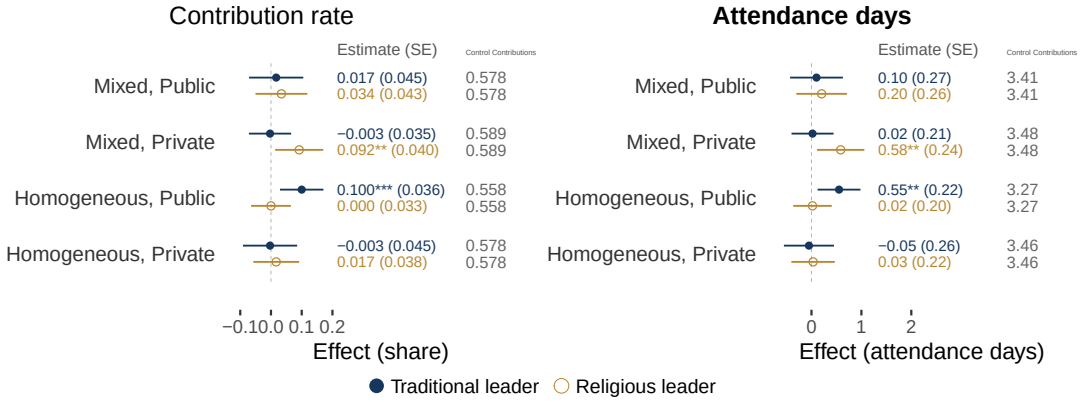
Transformed material contributions: by composition



Bars: 95% CI. * p<.10, ** p<.05, *** p<.01 (source tests).

Matched pretreatment outcome and design FE. Group-clustered SEs. Six-day main window. No-leader means use the same outcome and environment. Amounts in CDF. Asinh retains zero contributions. Rate is a share. Days, hours, and effort are window totals.

Participation rate and attendance days: by environment

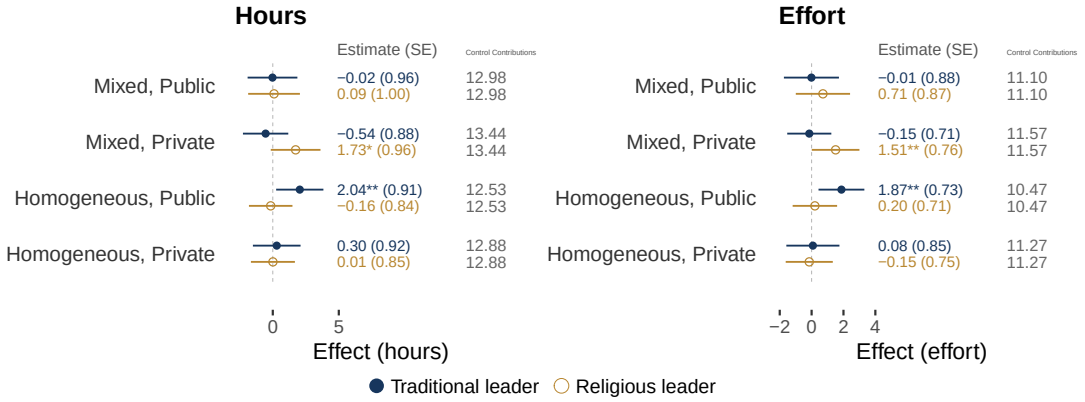


Bars: 95% CI. * p<.10, ** p<.05, *** p<.01 (source tests).

Matched pretreatment outcome and design FE. Group-clustered SEs. Six-day main window. No-leader means use the same outcome and environment. Amounts in CDF. Asinh retains zero contributions. Rate is a share. Days, hours, and effort are window totals.

Enforcement checks | Controls | More outcomes | Inference | Windows | Appendix

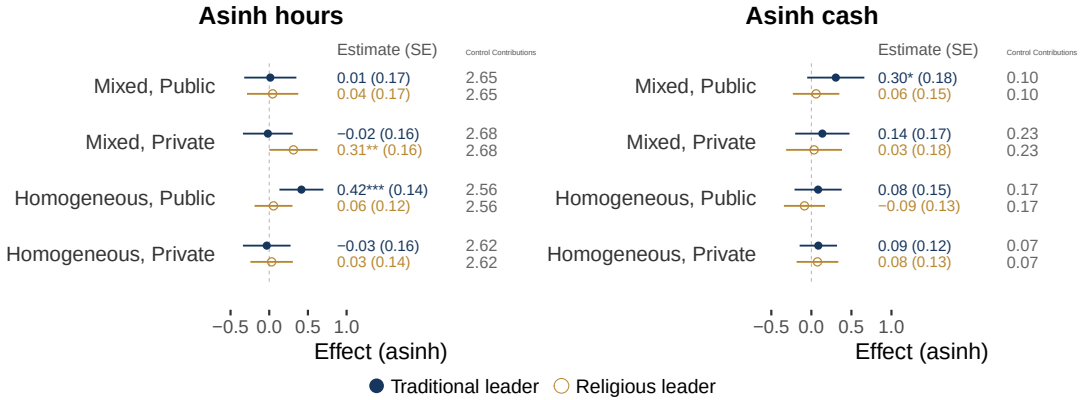
Hours and effort: by environment



Bars: 95% CI. * p<.10, ** p<.05, *** p<.01 (source tests).

Matched pretreatment outcome and design FE. Group-clustered SEs. Six-day main window. No-leader means use the same outcome and environment. Amounts in CDF. Asinh retains zero contributions. Rate is a share. Days, hours, and effort are window totals.

Transformed hours and cash: by environment

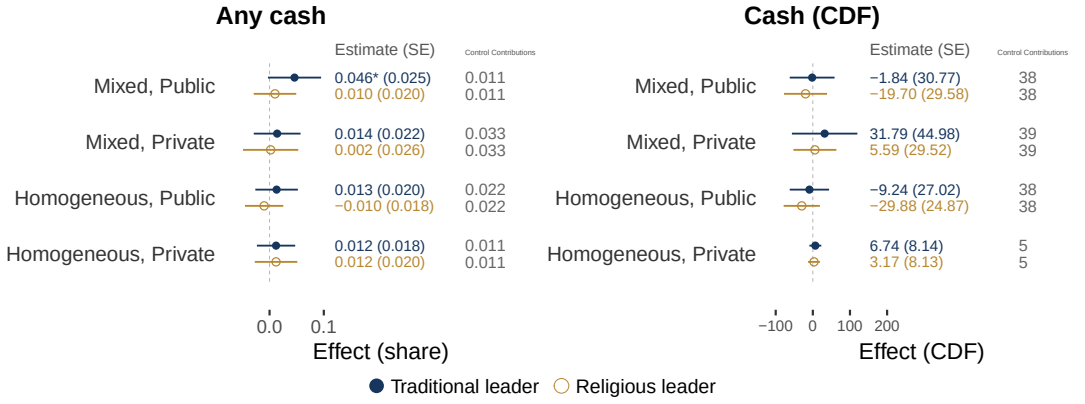


Bars: 95% CI. * p<.10, ** p<.05, *** p<.01 (source tests).

Matched pretreatment outcome and design FE. Group-clustered SEs. Six-day main window. No-leader means use the same outcome and environment. Amounts in CDF. Asinh retains zero contributions. Rate is a share. Days, hours, and effort are window totals.

Enforcement checks Previous Next Appendix

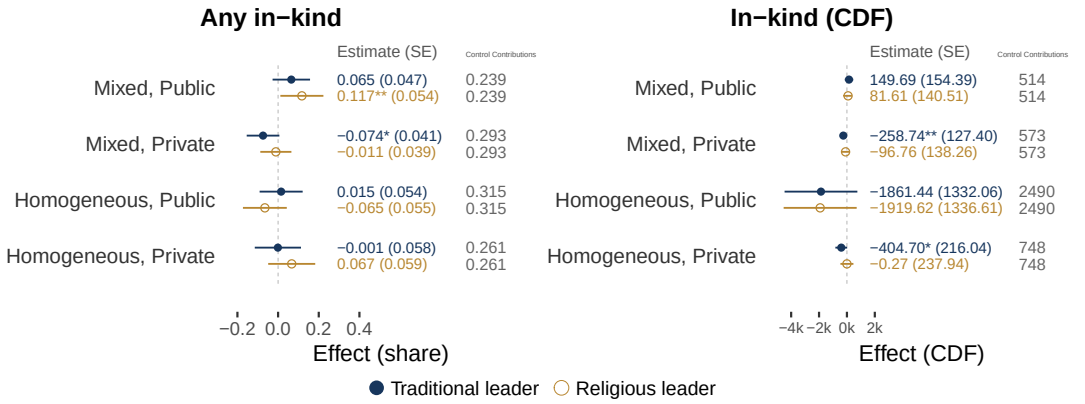
Cash contributions: by environment



Bars: 95% CI. * p<.10, ** p<.05, *** p<.01 (source tests).

Matched pretreatment outcome and design FE. Group-clustered SEs. Six-day main window. No-leader means use the same outcome and environment. Amounts in CDF. Asinh retains zero contributions. Rate is a share. Days, hours, and effort are window totals.

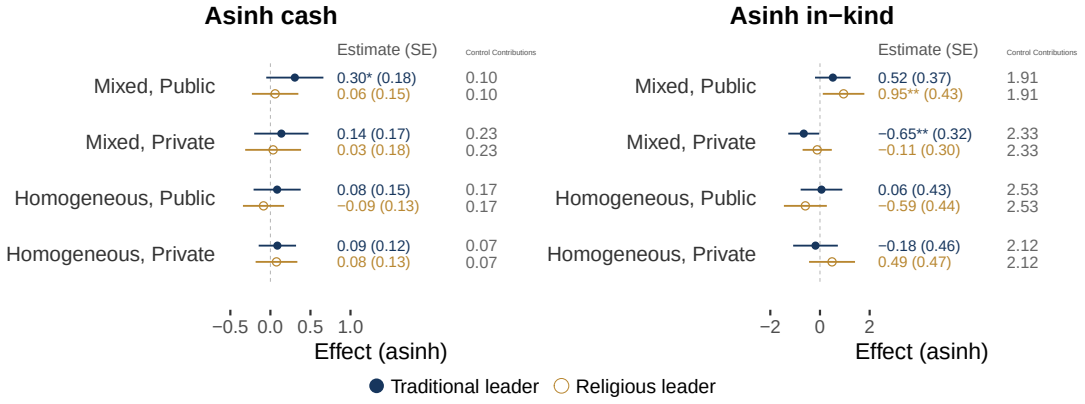
In-kind contributions: by environment



Bars: 95% CI. * p<.10, ** p<.05, *** p<.01 (source tests).

Matched pretreatment outcome and design FE. Group-clustered SEs. Six-day main window. No-leader means use the same outcome and environment. Amounts in CDF. Asinh retains zero contributions. Rate is a share. Days, hours, and effort are window totals.

Transformed material contributions: by environment

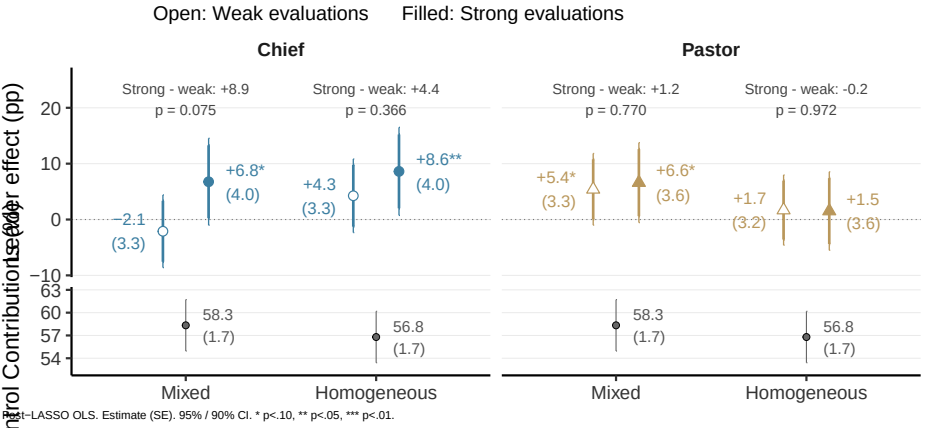


Bars: 95% CI. * p<.10, ** p<.05, *** p<.01 (source tests).

Matched pretreatment outcome and design FE. Group-clustered SEs. Six-day main window. No-leader means use the same outcome and environment. Amounts in CDF. Asinh retains zero contributions. Rate is a share. Days, hours, and effort are window totals.

Enforcement checks Previous Next Appendix

Perceived sanctioning authority

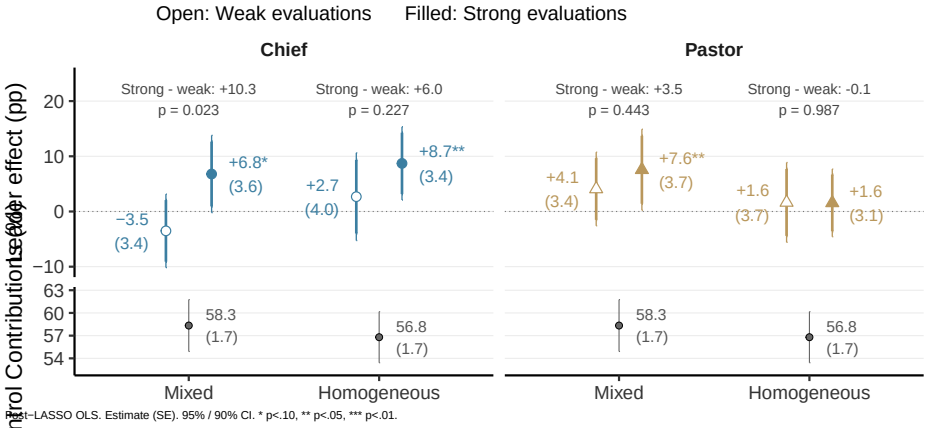


Post–LASSO OLS. Estimate (SE). 95% / 90% CI. * p<.10, ** p<.05, *** p<.01.

Perceived authority of the assigned leader, split at the median.

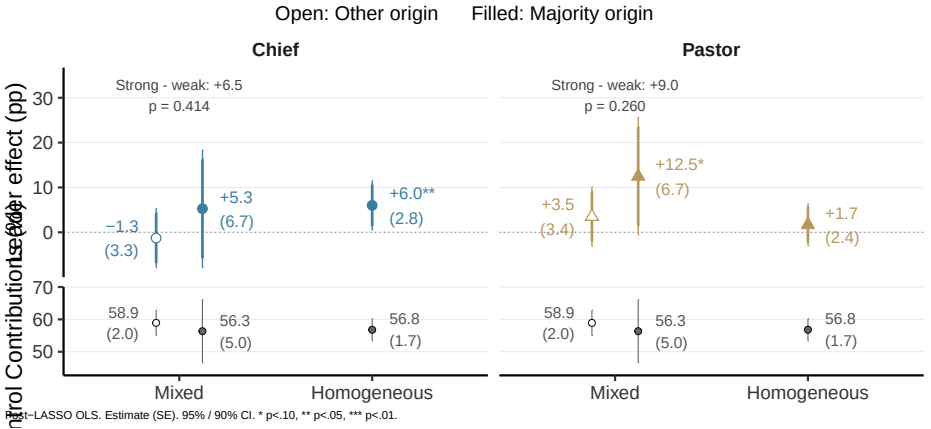
Chief evidence Previous Next Appendix

Favorable evaluations of the assigned leader



Trust, hard work, moral example and inspiration, split at the median.

Majority origin and group context

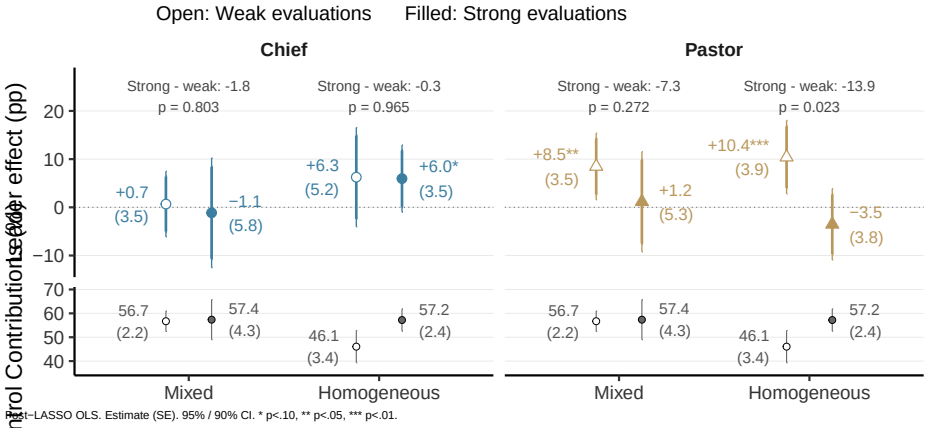


Post–LASSO OLS. Estimate (SE). 95% / 90% CI. * p<.10, ** p<.05, *** p<.01.

The chief shares the majority origin. Split mixed groups by residents' origin.

Chief evidence | Pastor evidence | Previous | Appendix

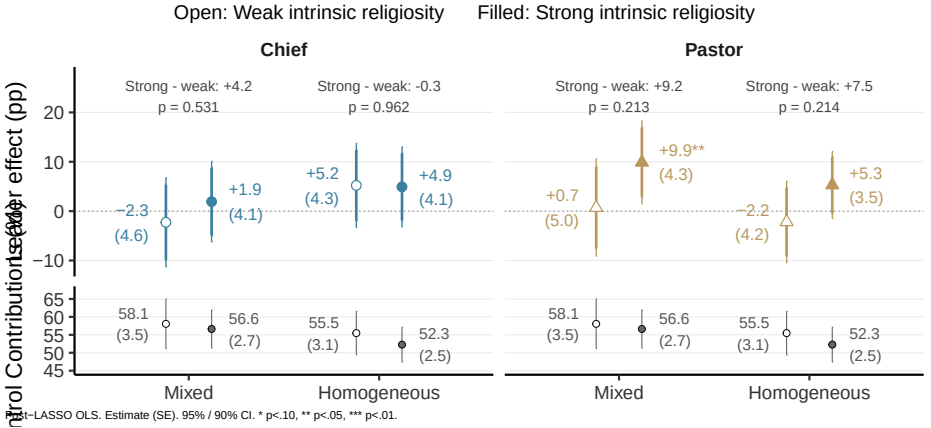
Resident networks: supplementary heterogeneity



Post–LASSO OLS. Estimate (SE). 95% / 90% CI. * p<.10, ** p<.05, *** p<.01.

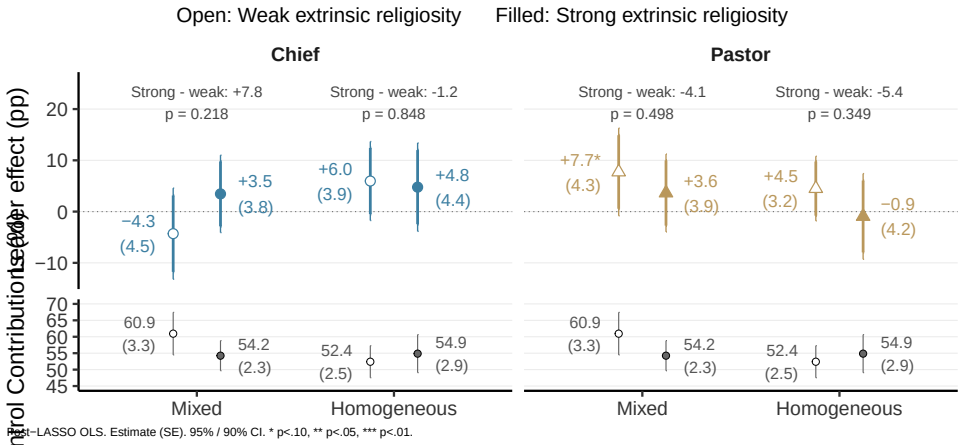
Share of confidants from members' own ethnic groups, split at the median.

Intrinsic religiosity



Faith as a guide to life: intrinsic score at/below versus above 3.2.

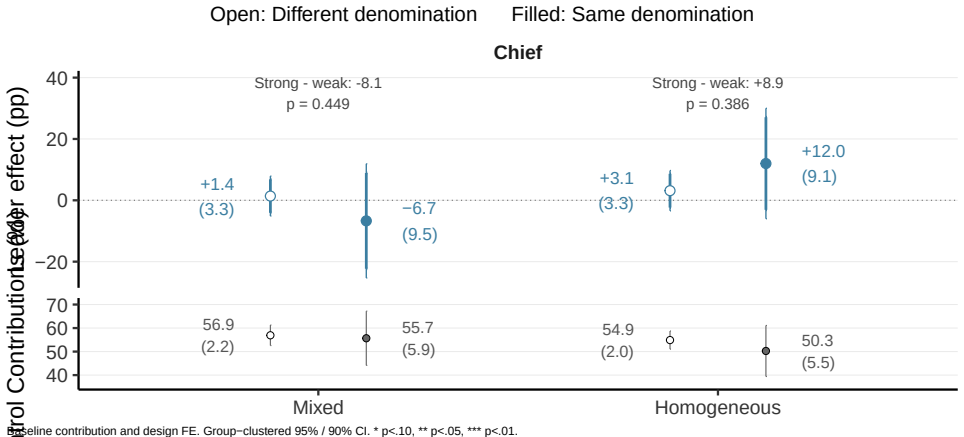
Extrinsic religiosity



Post-LASSO OLS. Baseline Salongo contribution, spillover exposure, selected controls and stratum fixed effects.

▸ Religiosity measure ▸ Religiosity results Chief evidence Pastor evidence Previous Next Appendix Question details

Shared denomination: traditional leader

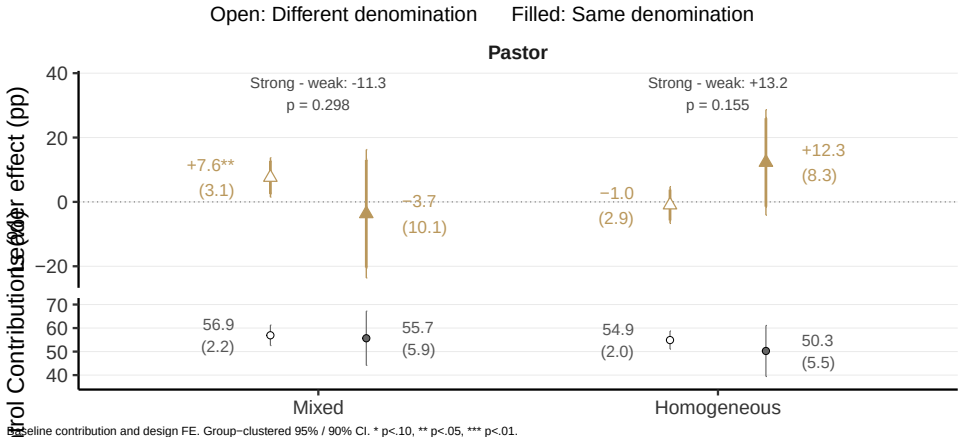


Baseline contribution and design-stratum fixed effects. Current paper denomination model; disclosure pooled within composition. Group-clustered 95% / 90% CI. Fitted no-leader references.

Religious leader HTE specifications

Chief evidence Pastor evidence Previous Appendix

Shared denomination: religious leader



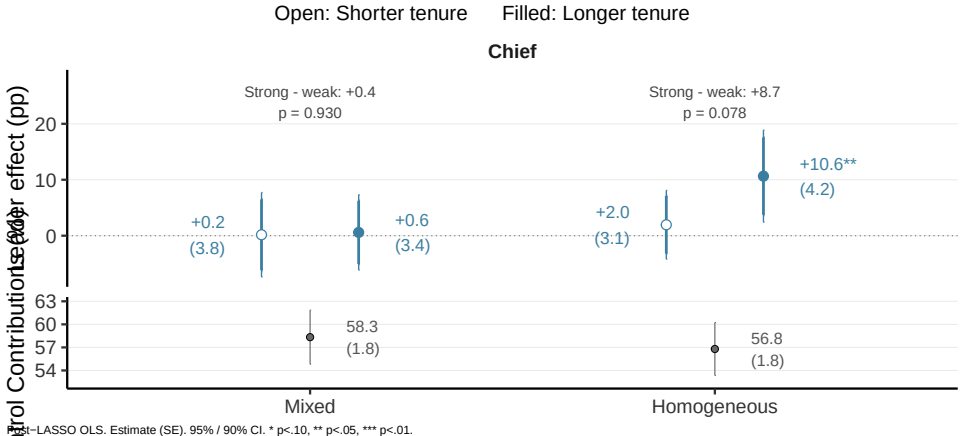
Baseline contribution and design-stratum fixed effects. Current paper denomination model; disclosure pooled within composition. Group-clustered 95% / 90% CI. Fitted no-leader references.

Traditional leader HTE specifications

Traditional leader HTE specifications

Appendix

Leaders' local roots: traditional leader

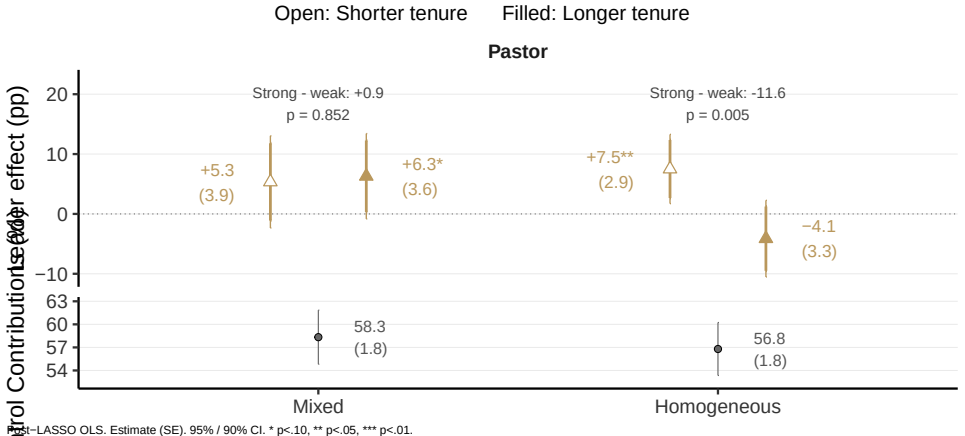


Post-LASSO OLS. Estimate (SE). 95% / 90% CI. * p<.10, ** p<.05, *** p<.01.

Current paper preferred model: baseline contribution, 25 m spillover, frozen 11 controls and design-stratum FE. Group-clustered 95% / 90% CI. Fitted no-leader references.

Religious leader
HTE specifications
Chief evidence
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Leaders' local roots: religious leader



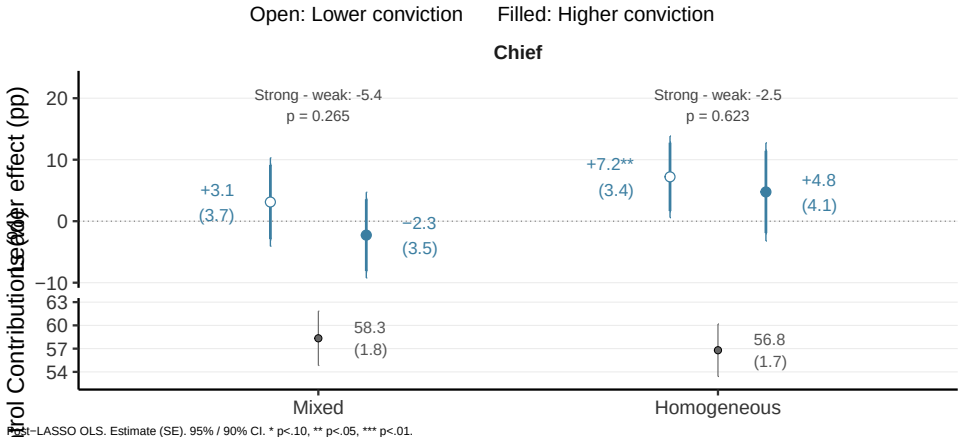
Current paper preferred model: baseline contribution, 25 m spillover, frozen 11 controls and design-stratum FE. Group-clustered 95% / 90% CI. Fitted no-leader references.

Traditional leader HTE specifications

Traditional leader HTE specifications

Appendix

Belief in divine enforcement: traditional leader

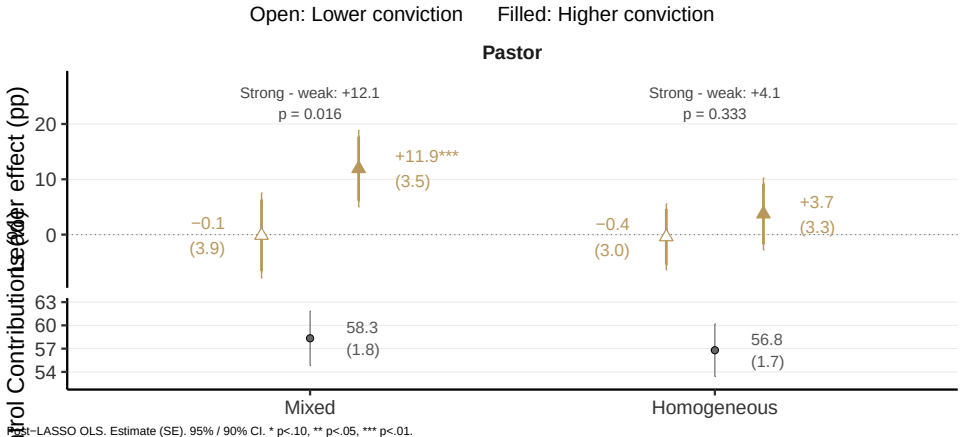


Current paper preferred model: baseline contribution, 25 m spillover, frozen 11 controls and design-stratum FE. Group-clustered 95% / 90% CI. Fitted no-leader references.

Religious leader HTE specifications

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Belief in divine enforcement: religious leader

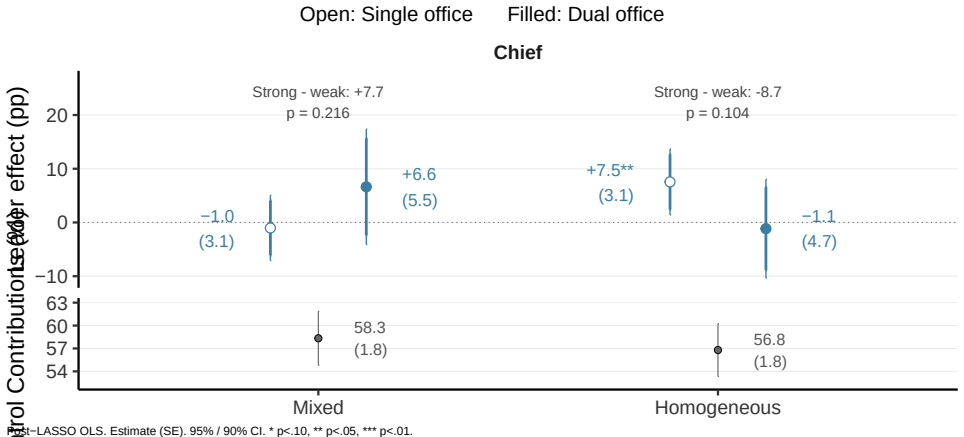


Post-LASSO OLS. Estimate (SE). 95% / 90% CI. * p<.10, ** p<.05, *** p<.01.

Current paper preferred model: baseline contribution, 25 m spillover, frozen 11 controls and design-stratum FE. Group-clustered 95% / 90% CI. Fitted no-leader references.

Traditional leader HTE specifications

Holding office in both institutions: traditional leader



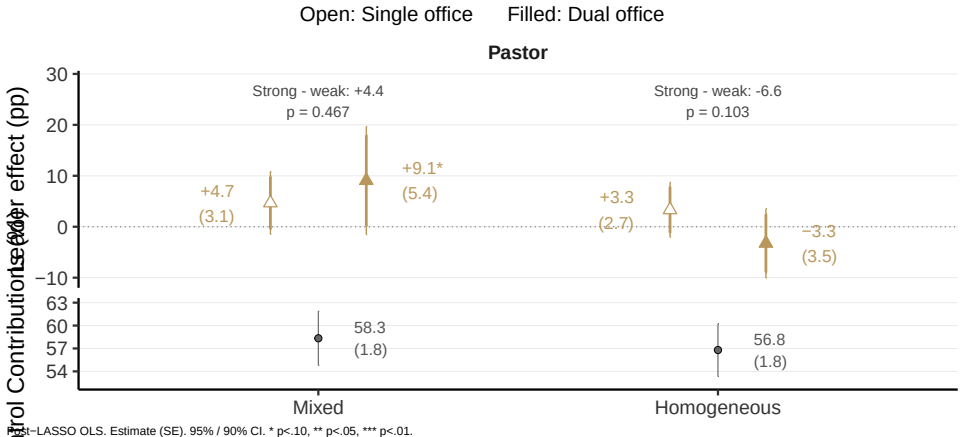
Post−LASSO OLS. Estimate (SE). 95% / 90% CI. * p<.10, ** p<.05, *** p<.01.

Current paper preferred model: baseline contribution, 25 m spillover, frozen 11 controls and design-stratum FE. Group-clustered 95% / 90% CI. Fitted no-leader references.

Religious leader HTE specifications

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Holding office in both institutions: religious leader



Post-LASSO OLS. Estimate (SE). 95% / 90% CI. * p<.10, ** p<.05, *** p<.01.

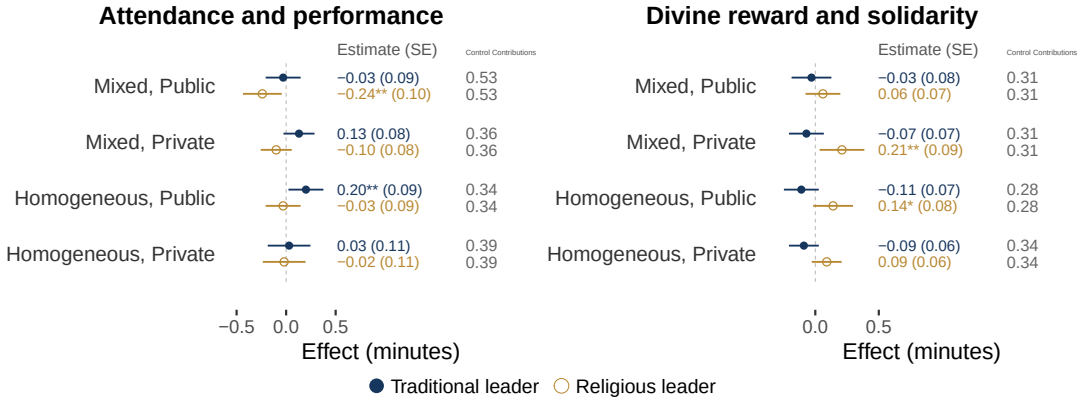
Current paper preferred model: baseline contribution, 25 m spillover, frozen 11 controls and design-stratum FE. Group-clustered 95% / 90% CI. Fitted no-leader references.

Traditional leader HTE specifications

Traditional leader HTE specifications

Appendix

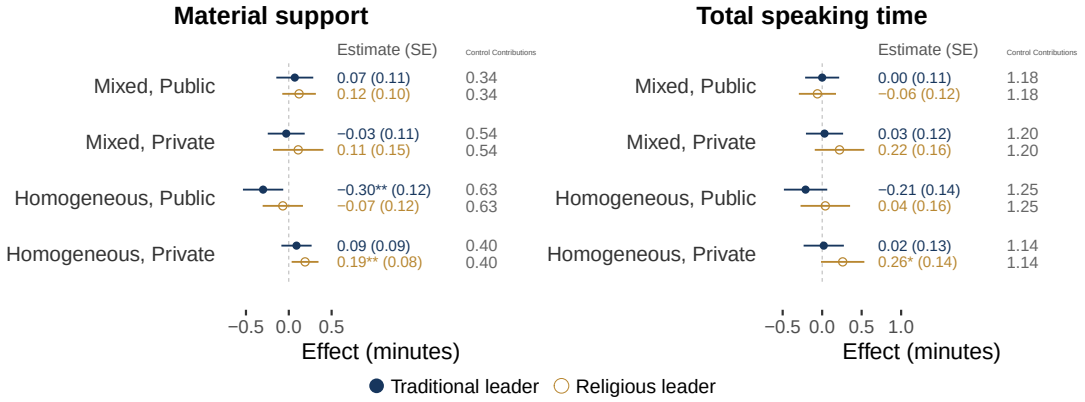
Residents' attendance and divine appeals



Bars: 95% CI. * p<.10, ** p<.05, *** p<.01 (source tests).

Ordinary participants' minutes of speech. Silent participants count as zero. Assignment effects relative to no leader. Design FE and group-clustered SEs. The no-leader mean is shown beside each corresponding effect.

Residents' material appeals and speaking time

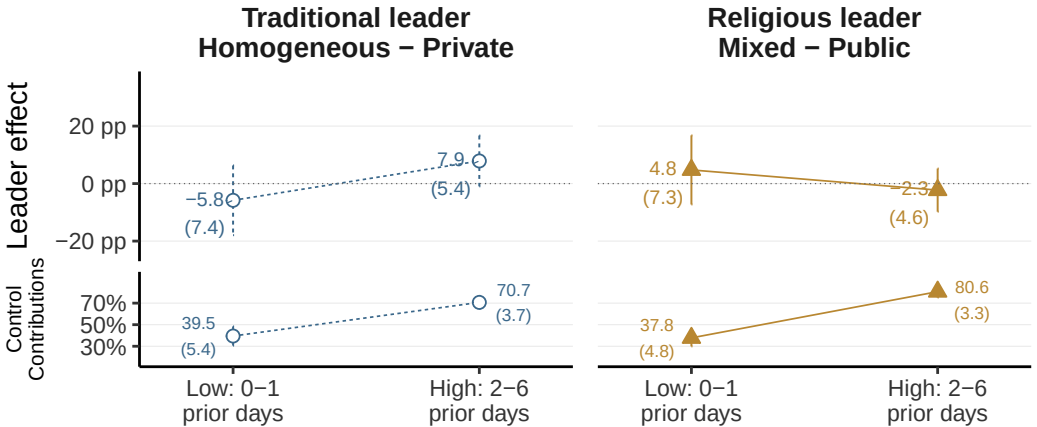


Bars: 95% CI. * p<.10, ** p<.05, *** p<.01 (source tests).

Ordinary participants' minutes of speech. Silent participants count as zero. Assignment effects relative to no leader. Design FE and group-clustered SEs. The no-leader mean is shown beside each corresponding effect.

Speech Previous Appendix

Participation histories under the other reporting rule

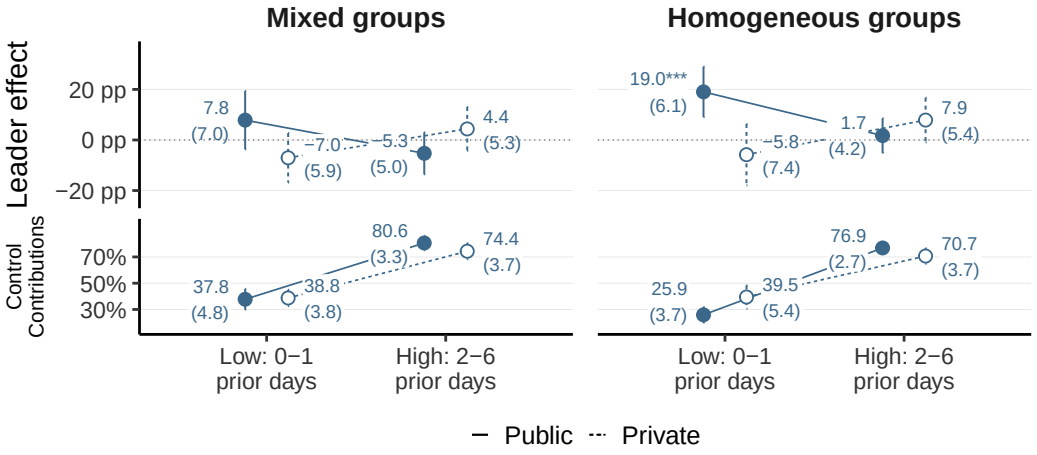


* p<.10, ** p<.05, *** p<.01. Control Contributions, tags: estimate (SE), 90% confidence intervals.

Post-LASSO OLS with the frozen 11 controls and spillover. The same outcome, sample split, fixed effects, baseline adjustment, and group-clustered uncertainty as on the main slide. Effects and matched no-leader levels are shown together.

Back to histories | All traditional | All religious | Appendix

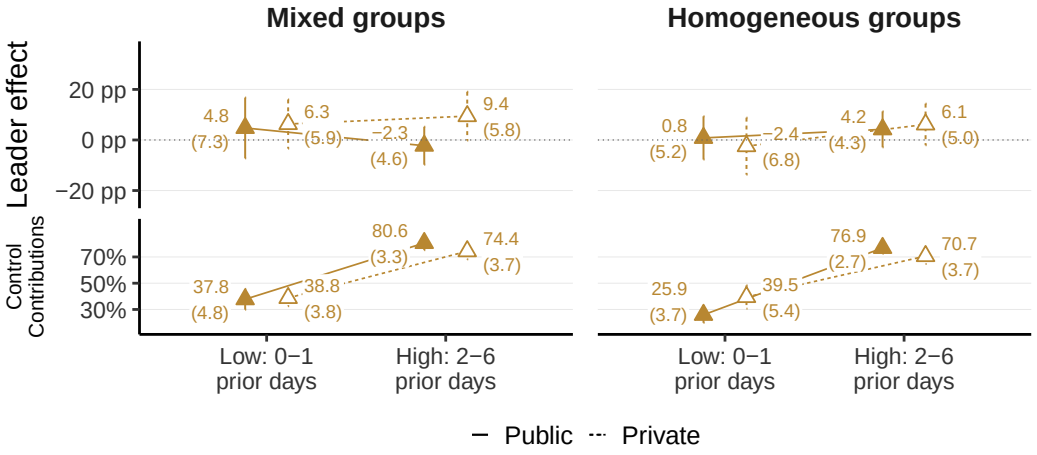
Own pretreatment participation: traditional leader



Post-LASSO OLS with the frozen 11 controls and spillover. Baseline Salongo contribution and neighborhood × composition × disclosure FE. Group-clustered SEs. No-leader predictions use the residents supporting each subgroup effect. Own history: 0–1 versus 2–6 pretreatment participation days. Connected Public/Private series follow the paper figure. 90% CI. References average actual no-leader residents within each environment and history category.

Histories Next Appendix

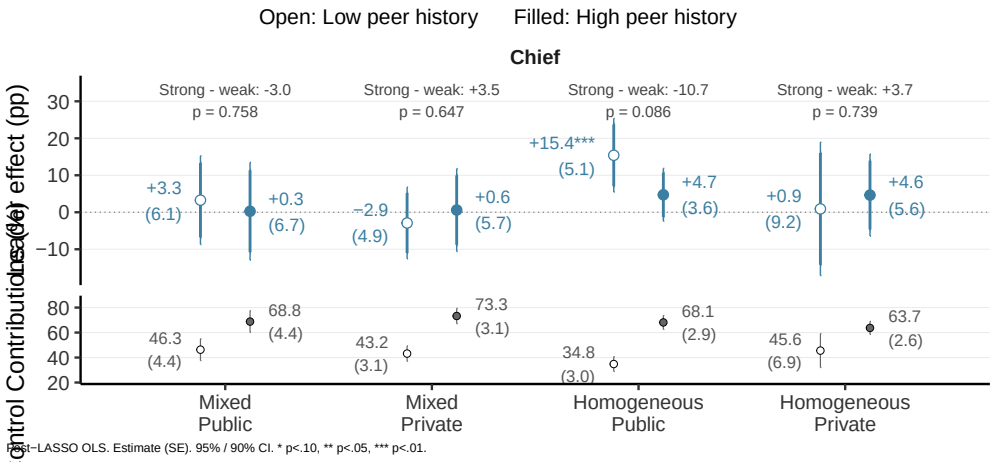
Own pretreatment participation: religious leader



Post-LASSO OLS with the frozen 11 controls and spillover. Baseline Salongo contribution and neighborhood × composition × disclosure FE. Group-clustered SEs. No-leader predictions use the residents supporting each subgroup effect. Own history: 0–1 versus 2–6 pretreatment participation days. Connected Public/Private series follow the paper figure. 90% CI. References average actual no-leader residents within each environment and history category.

Histories Previous Next Appendix

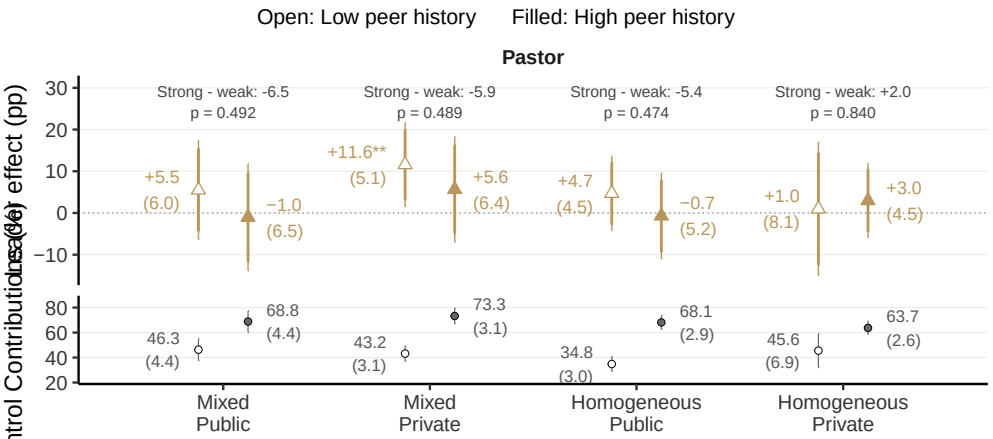
Peers' pretreatment participation: traditional leader



Post–LASSO OLS. Estimate (SE). 95% / 90% CI. * p<.10, ** p<.05, *** p<.01.

Current paper preferred model: baseline contribution, 25 m spillover, frozen 11 controls and design-stratum FE. Group-clustered 95% / 90% CI. Fitted no-leader references.

Peers' pretreatment participation: religious leader



Post–LASSO OLS. Estimate (SE). 95% / 90% CI. * p<.10, ** p<.05, *** p<.01.

Current paper preferred model: baseline contribution, 25 m spillover, frozen 11 controls and design-stratum FE. Group-clustered 95% / 90% CI. Fitted no-leader references.

A divine appeal does not displace the chief as a reference

With homogeneous groups, the chief remains a reference

Pastor, retelling the parable of the talents: "Enter the kingdom of your Lord. Thus I urge you to commit yourselves to the work."

Resident 1: "God in heaven will reward us. No person will give us a wage."

Resident 2: "Let us do this work that makes our neighborhood clean. It will benefit the chief we have in our neighborhood. [...] God will help us."

Who responds more to each leader?

$$Y_i = \lambda_{s(i)} + \rho Y_{i0} + X_i'\beta + \sum_{c \in \{H,M\}} \mathbf{1}\{c(i) = c\}\left[\eta_c Z_i + (\tau_c^T + \gamma_c^T Z_i) T_{g(i)} + (\tau_c^R + \gamma_c^R Z_i) R_{g(i)}\right] + \varepsilon_i.$$

- $Z_i \in \{0, 1\}$: the heterogeneous effects dimension (e.g., coethnic or not). c: composition.
- $Z_i = 0$: τ_c^L $\quad$ $Z_i = 1$: $\tau_c^L + \gamma_c^L$ $\quad$ **Difference:** γ_c^L.

Baseline Salongo contribution, stratum fixed effects, the frozen 11 LASSO controls and current spillover exposure.

▸ Specification details Specification comparisons

◂ Chiefs HTE ◂ Pastors HTE ◂ Source HTE

Pretreatment timing and flexible baseline adjustment

1. **The balance problem:** some pre-assignment characteristics are imbalanced. Participation between assignment and Meeting 1 differs across treatment cells.
2. **Anticipation:** estimate pre-meeting differences day by day over the six pretreatment days.
3. **Mean reversion:** augment the main ANCOVA with a separate baseline slope in every assignment cell:

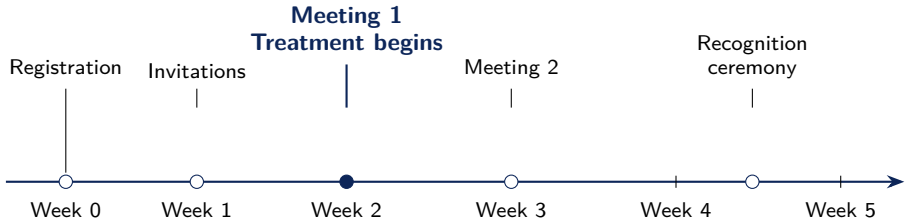
$$Y_i = \lambda_{n(i),q(i),c(i)} + \sum_{a=1}^{12} \rho_a Y_{i0} \mathbf{1}\{A_i = a\} + \sum_{L,e} \tau_e^L \mathbf{1}\{L_i = L, e_i = e\} + \varepsilon_i.$$

4. Neither the daily evidence nor flexible adjustment explains the post-meeting reversal.

Enforcement checks

Appendix

Measurement before and after treatment

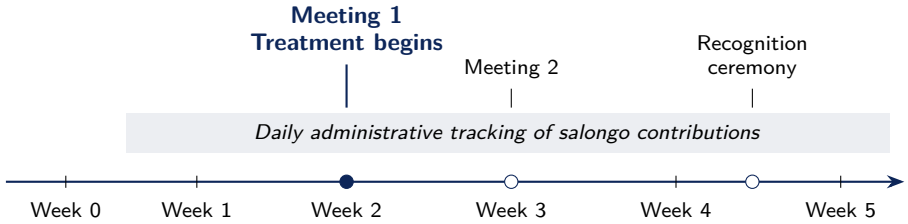


- **Registration and invitations:** residents register for the city campaign. Selected residents are invited to neighborhood improvement teams.

Survey and fieldwork timing | Outcome windows | Data coverage

▸ Back: Measures

Measurement before and after treatment

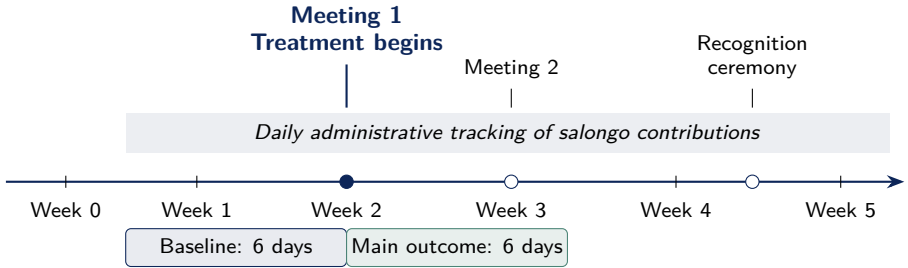


- **Daily contribution records:** attendance and hours from digital arrival and departure records, plus effort, cash, and in-kind support. Tracking starts before treatment.

Survey and fieldwork timing | Outcome windows | Data coverage

▸ Back: Measures

Measurement before and after treatment

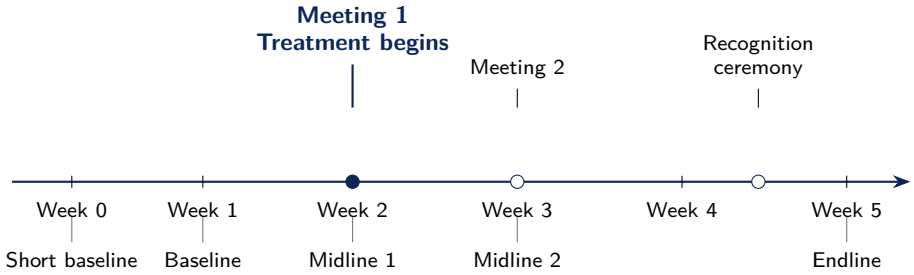


- **Main outcome:** share of six days attended after the first meeting, controlling for participation during the six-day baseline window.

Survey and fieldwork timing | Outcome windows | Data coverage

▸ Back: Measures

Measurement before and after treatment



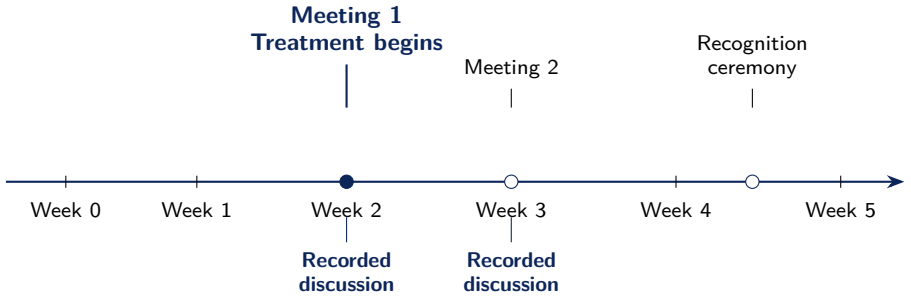
Surveys around the campaign and the two meetings

- **Surveys:** baseline before treatment, midlines before each meeting's discussion, and endline after recognition. Measure social ties, leader evaluations, beliefs, and expectations.

Survey and fieldwork timing | Outcome windows | Data coverage

▸ Back: Measures

Measurement before and after treatment

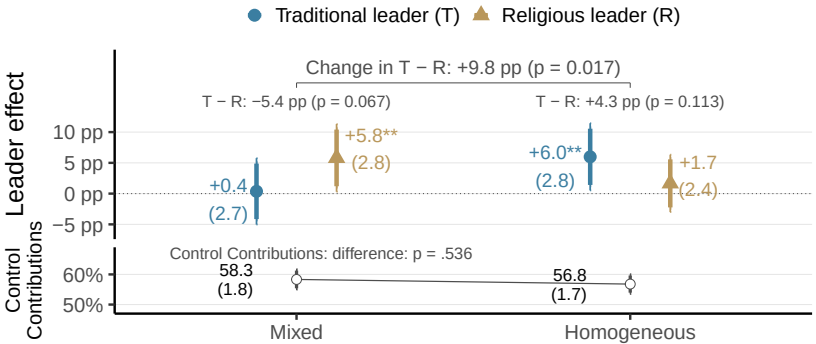


- **Recorded discussions at both meetings:** transcripts capture leaders' unscripted appeals and residents' responses.

Survey and fieldwork timing | Outcome windows | Data coverage

▸ Back: Measures

All composition contrasts



Post−LASSO OLS. N = 1,040 residents, 336 groups. Labels: effect (SE). * p<.10, ** p<.05, *** p<.01.

Same leader, homogeneous minus mixed:

Chiefs: $+5.6$ pp ($p = 0.152$) **Pastors:** -4.1 pp ($p = 0.261$)

Back: Results

Other chief ratings: exploratory heterogeneity

- Favorable evaluations predict larger chief effects in mixed groups (10.3 pp).

  Evaluation results

- Recognized sanctioning authority predicts a suggestive 8.9 pp mixed-group contrast.

  Authority results

These additional correlations do not isolate a causal mechanism.

Back: Chief results

Baseline identity: survey question

Verbatim English questionnaire prompt

"Now I would like to ask you to think about what parts of your identity are most important to you. Some people might identify more with being Congolese or with being from your territory of origin or with being a Christian or with being Kasaien. There is no right answer. Please rank each of the following aspects of identity in order of importance to you personnally."

Six options to rank

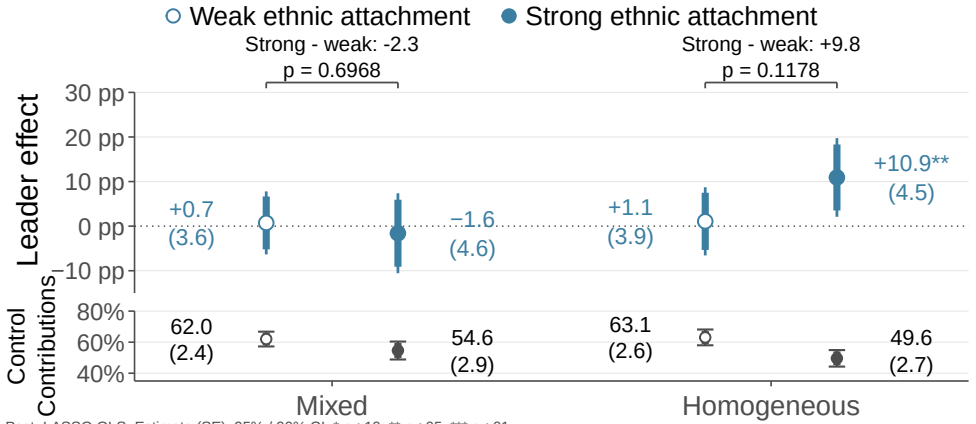
- Being Congolese
- Being Kasaien
- Being Christian
- Being from [**ethnicity**]
- Being part of [denomination]
- Being part of [**umbrella ethnicity**]

Randomized display order at each rank. Respondents choose first through sixth; previously selected options are removed.

Brackets denote the respondent's own group: ethnicity = territory of origin; umbrella ethnicity = tribe in the questionnaire.

Chief HTE | Pastor HTE | Test logic

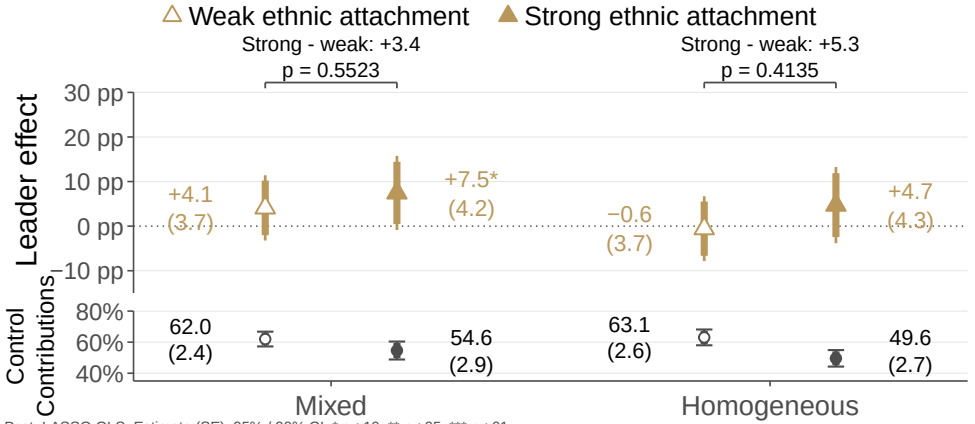
Chiefs: individual ethnic identity



Post−LASSO OLS. Estimate (SE). 95% / 90% CI. * p<.10, ** p<.05, *** p<.01.

Individual high: baseline ethnic-identity score above 3, ties are low. Exploratory tests, see Definitions for inference and multiplicity.

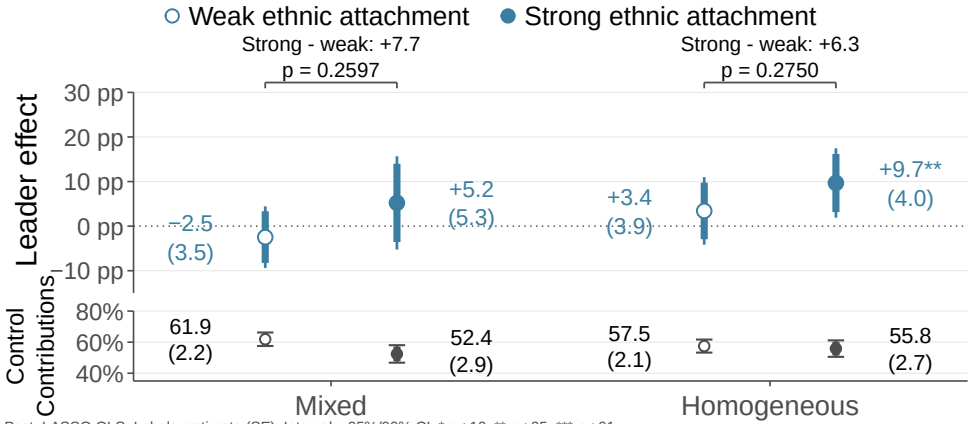
Pastors: individual ethnic identity



Post−LASSO OLS. Estimate (SE). 95% / 90% CI. * p<.10, ** p<.05, *** p<.01.

Individual high: baseline ethnic-identity score above 3, ties are low. Exploratory tests, see Definitions for inference and multiplicity.

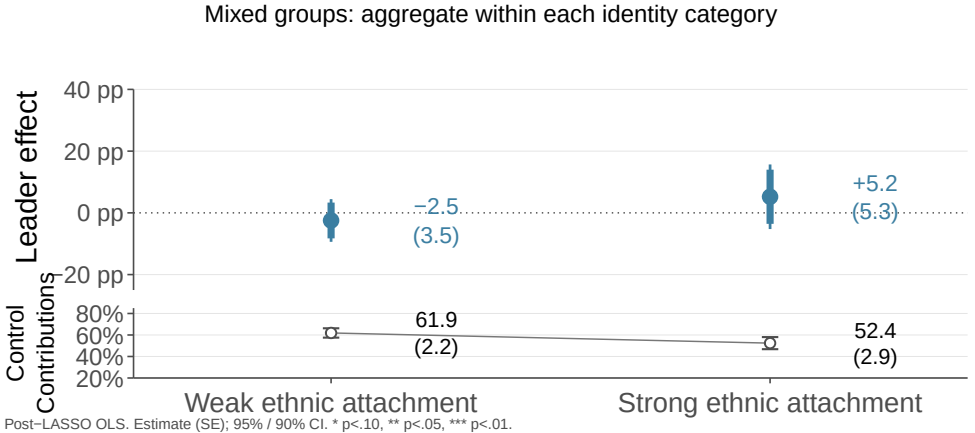
Chiefs: group ethnic identity



Post−LASSO OLS. Labels: estimate (SE). Intervals: 95%/90% CI. * p<.10, ** p<.05, *** p<.01.

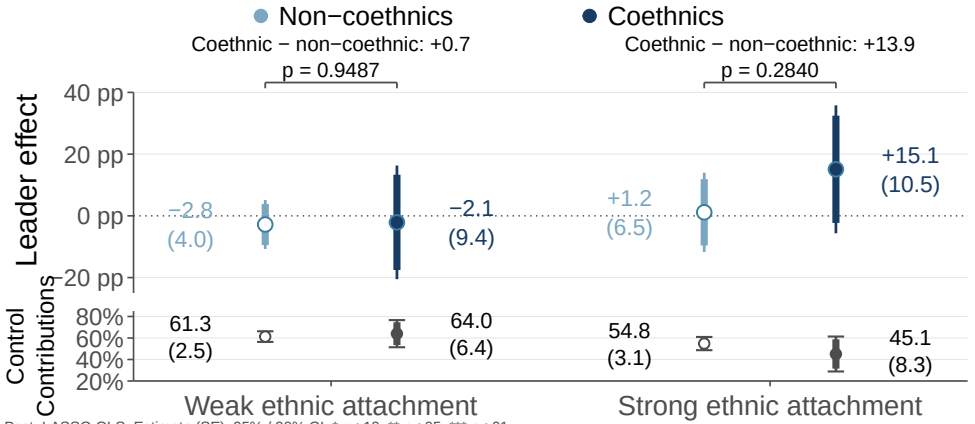
Group high: mean ethnic-identity score above 10/3 among ordinary members with available scores, ties are low. Exploratory tests, see Definitions for inference and multiplicity.

Chiefs: ethnic identity and coethnicity in mixed groups



Mixed groups only. Coethnicity refers to the neighborhood chief. Group identity uses the same fixed median split; the decomposition retains the current paper's sample and controls.

Chiefs: ethnic identity and coethnicity in mixed groups

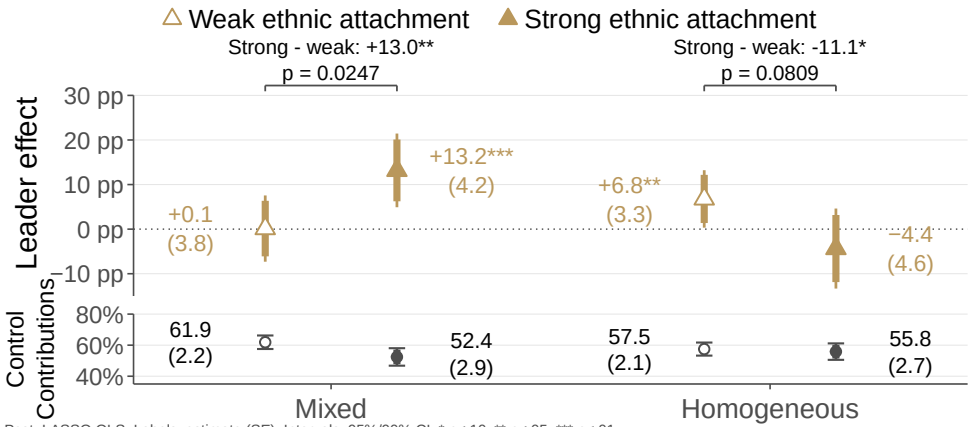


Post−LASSO OLS. Estimate (SE); 95% / 90% CI. * p<.10, ** p<.05, *** p<.01.

Mixed groups only. Coethnicity refers to the neighborhood chief. Group identity uses the same fixed median split; the decomposition retains the current paper's sample and controls.

Main results | Pastors | Individual split | Definitions | Appendix

Pastors: group ethnic identity



Post−LASSO OLS. Labels: estimate (SE). Intervals: 95%/90% CI. * p<.10, ** p<.05, *** p<.01.

Group high: mean ethnic-identity score above 10/3 among ordinary members with available scores, ties are low. Group estimates and fitted no-leader levels match the current paper. Exploratory, see Definitions.

Baseline identity: definitions and exploratory inference

- **Group model: 1,040 residents in 336 groups.** Baseline Salongo contribution, frozen 11 LASSO controls, spillover and NQC fixed effects, group-clustered SEs. Four leader–composition arms pool disclosure.
- **Individual high:** ethnic-identity score above the median (> 3). **Group high:** mean score $> 10/3$, ties low.
- **Group mean:** mean available score among ordinary members, including own rank where applicable; three scores in 330 groups and two in six.
- **Separate regressions:** each split has treatment interactions and composition-specific main effects. No extra own-identity adjustment in the group model.
- **Exploratory:** displayed group tests are unadjusted. Individual estimates use all 1,034 residents with valid rankings, without a complete-teammate restriction. No separately identified peer mechanism.

Individual results Group results

Heterogeneity specifications

Preferred heterogeneous effects estimates retain baseline Salongo contribution, 25 m spillover exposure, design fixed effects and the frozen 11 LASSO-selected controls. The links below show the current paper estimates. Denomination and leadership-by-example retain their separate baseline-adjusted paper specifications.

Leader and resident characteristics

- Leader authority
- Leader evaluations
- Majority origin
- Resident networks
- Intrinsic religiosity
- Extrinsic religiosity
- Shared denomination

Leader supply and history

- Leader tenure
- Leader conviction
- Dual office
- Leader prior effort
- Chief: peer history
- Pastor: peer history
- Chief: own history
- Pastor: own history

All current estimates and source audits are saved with the local deck build record.

▸ Back: HTE specification

Heterogeneity: specification details

- $s(i)$ indexes neighborhood $\times$ composition $\times$ disclosure strata. $\lambda_{s(i)}$ are fixed effects.
- $T_{g(i)}$ and $R_{g(i)}$ indicate the assigned chief and pastor. No leader is the omitted category.
- Y_{i0} is baseline Salongo contribution. X_i contains 25 m spillover exposure and the frozen 11 LASSO-selected baseline controls.
- Z_i indicates the higher category. The $\eta_c Z_i$ terms allow moderator-specific no-leader levels within each composition.
- Assigned-leader ratings enter only through leader interactions: omit $\eta_c Z_i$ for those specifications.
- Samples and moderator definitions follow the current paper. Standard errors are clustered by group.

Back: HTE specification

Supplementary network sensitivity checks

Homogeneous groups: pastor effect above minus at/below the 0.8 network-share cutoff.

Added proxy interactions	Difference (pp)	SE	p
Reference	−14.59	6.26	.020
Residential proximity	−14.72	6.44	.023
Baseline community cohesion	−18.44	6.61	.006
Both baseline proxies	**−18.48**	**6.78**	**.007**

Proximity: mean $-\log(1 + \text{residential distance in meters})$ across the three resident pairs.
Community cohesion: residents' baseline avenue safety, trust and closeness index, not direct team ties.

Post-assignment familiarity sensitivity: −15.69 (SE 6.61), $p = .018$, 1,033 residents, 334 groups.
All three surveys before Meeting 1: −23.29 (SE 6.12), same-sample reference −21.56, $N = 990$.

Supplementary exploratory regressions; not a current-paper exhibit. Main rows: 1,040 residents, 336 groups. Baseline participation, 25 m spillover and neighborhood × composition × disclosure FE, group-clustered t inference. No selected controls. Each proxy has composition-specific levels and four leader-by-composition interactions. Cohesion coverage controls included. Network: original ordinary-resident confidant share including self, high > .8, no leaders or placebos in proxy construction. Nominal p-values. Baseline timing is not uniformly pre-randomization.

Campaign accounting and GPS coverage

Recorded contribution	Amount	Coverage
Attendance-clock hours	70,250	Waves 1–7
Hours excluding no-effort spells	69,003	Waves 1–7
Recorded ID–day appearances	20,046	28 neighborhoods
Distinct recorded IDs	1,664	Not a unique-person count
Mapped worksite histories	596	Waves 1–5
Usable moved-on labor histories	566	Mean: 96.7 hours

- A site groups different-day GPS records with maximum pairwise distance of 5 m. Separately recorded same-day sites stay separate. All revisits are summed.
- 573 histories were followed by work elsewhere, and 23 reached the last mapped date. Moving on does not establish completion.
- The experimental sample contains 1,040 ordinary residents: 13,127 hours in the six-day window and 47,714 over the full recorded waves.

Back Appendix

From six-day effects to full-period workloads

$$\Delta H_c = \underbrace{\widehat{\tau}_c}_{\text{six-day hours/person}} \times \underbrace{\frac{\overline{H}^0_{c,\text{full}}}{\overline{H}^0_{c,6}}}_{\text{no-leader time profile}} \times \underbrace{520}_{\text{residents/composition}} .$$

Combination	$\widehat{\tau}_c$ (SE)	Multiplier	Extra hours	Workloads
Chief / homogeneous / public	2.10 (0.89)	3.74	4,083	42.2
Pastor / mixed / private	1.70 (0.94)	3.58	3,168	32.8
Total			**7,251**	**75.0**

- Workloads: extra hours / 96.7 observed hours per usable worksite history.
- Conditional nominal 95% interval: **2,424–12,078 hours** (25–125 workloads). Uses joint coefficient covariance. Excludes policy selection, persistence and GPS uncertainty.
- Hours model: 11 frozen participation controls, matched baseline, spillover, stratum FE, group-clustered SEs.

Back Appendix

The counterfactual determines the policy gain

Comparison for all 1,040 residents	Extra hours	Gain	Workloads
No leaders, same selected information rules	7,251	15.8%	75.0
No leaders, original information mix	5,528	11.6%	57.2
Original randomized leader/information mix	4,296	8.8%	44.4

- Each row compares the same selected policy with a different baseline. All full-period totals are projections.
- The 11 participation controls are used for hours. Leader/reporting assignments and time multipliers are held fixed.
- Hold enrolled residents fixed. No citywide extrapolation, additional leader labor, capital complementarity or displacement is modeled.

Back Appendix

Fiscal benchmark: reported 2025 property-tax receipts

Valuation and comparison	Amount
Labor value + recorded cash and in-kind	62,752,363 CDF
Same total value at 2,250 CDF/$	$27,889.94
DGRKAC 2025 property tax (*impôt foncier*)	205,682,923 CDF
Same property-tax receipts at 2,250 CDF/$	$91,414.63
Total contribution value / annual property-tax receipts	**30.5%**

- **Source:** Kasaï-Central DGRKAC, *Rapport des recettes réalisées par service et par acte générateur*, 2025, dated January 2026. Property-tax row agrees with the companion workbook (N2, USD conversion N3).
- **Comparison:** valued labor plus cash and in-kind inputs versus reported tax receipts. The USD wage uses the same 2,250 CDF/USD conversion. Campaign: Aug. 2025–Apr. 2026, taxes: calendar 2025.
- **Scope:** source totals disagree, so the comparison uses only the verified property-tax row. It is not total provincial tax revenue.

Back STEP wage Income sensitivity Appendix

Survey income: conversion and sensitivity

Income per adult household member: mean 76,918 CDF/month ($34.19), from 1,015 valid income–size pairs among 1,040 enrolled residents.

Alternative hourly proxy: $76,918/(20 \times 8) = 480.74$ CDF ($0.214).

Per-adult income and work-time normalization	Labor value	Share of property tax
Mean, 20 days × 8 hours	$15,000	16.4%
Trimmed mean (2.5–97.5%), 20 days	$11,761	12.9%
Median, 20 days	$8,781	9.6%
Mean, 30 days × 8 hours	$10,007	10.9%

- Divide each household's earnings by its adult count (mean: 4.10), including elders and excluding children. Average the ratios. This is not an observed wage.
- Untrimmed mean, retaining 15 zero incomes. Exclude six missing surveys, eight income nonresponses and 11 records with zero adults.
- The 160-hour month follows the fieldwork convention. Applying this common rate to all campaign labor is an assumption.

Back Fiscal source Appendix

Contribution accounting: labor, cash and in-kind

Same campaign coverage: 28 neighborhoods, waves 1–7, Aug. 2025–Apr. 2026.

Component	CDF	USD
Labor: STEP wage benchmark	59,273,696	26,344
In-kind goods: recorded quantity × unit price	3,324,417	1,478
Cash: recorded contributions	154,250	69
Gross contribution value	**62,752,363**	**27,889.94**

- **Cash:** 197 entries in 23 neighborhoods, including redistribution to participants. Subsequent purchases are not added again.
- **In-kind:** 1,936 entries in 28 neighborhoods: sacks, other items, food, coffee and water. Values use recorded prices and the existing panel's price correction.
- **Labor accounts for 94.5% of the gross total.** Values combine imputed labor and recorded cash/goods. They do not measure net resources after transfers or output benefits.

Back STEP wage Income sensitivity Appendix

Hours regression behind the policy projection

Outcome: hours over the six days after Meeting 1.

Environment	Chief	Pastor
Mixed, private	-0.67 (0.85)	$+1.70^{*}$ (0.94)
Mixed, public	$+0.03$ (0.91)	$+0.00$ (0.93)
Homogeneous, private	$+0.66$ (0.91)	-0.03 (0.81)
Homogeneous, public	$+2.10^{**}$ (0.89)	$+0.13$ (0.82)

Policy gains: chief/Public $2.10/11.65 = 18\%$, pastor/Private $1.70/12.51 = 14\%$.

Matched baseline hours, stratum fixed effects, spillover and the frozen 11 controls. $N = 1{,}040$, 336 groups.

Group-clustered SEs in parentheses, $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Back to policy Projection calculation Appendix

Project spending: waves 1–7, including research

Recorded field-project spending	USD
Personnel, including preparation	45,710.83
Daily field expenses	18,670.72
Other field expenses	4,666.56
Total (allocated)	**69,048.11**

- Includes research, surveys and recorded preparation costs through wave-7 closeout on April 25, 2026.
- April payroll: \$5,287 × 25/30 days. Excludes explicit wave-8 costs and cash transfers.

Elie's finance reports: April 26 and June 6, 2026, December 25, 2025.

Back to graph Implementation only Appendix

Implementation costs: five enumerators, materials and prizes

Allocated implementation costs	USD
Five-enumerator salary allocation	16,092.61
Work materials (sacks, shovels, machetes)	1,707.80
Prizes and certificates	680.66
Total	**18,481.07**

- Monthly payroll $\times$ 5 / enumerator count (12–15), including July preparation and 25/30 of April.
- Materials and prizes use individual ledger entries. Excludes transport, phones, survey payments and other field costs.

One prize-ceremony payment includes ceremony-chief payments. Source: Elie's payroll and expense reports.

Back to graph All field costs Appendix

Labor valuation: Kananga's STEP II public-works wage

- **Kananga, 2021: $3 per day.** World Bank-supported STEP II recruited 6,750 workers for 100 days of road maintenance. (ACP, 22 January 2021, reporting FSRDC)
- **Assume an eight-hour workday: $0.375/hour.** Eight hours is documented in the earlier eastern-DRC STEP evaluation, not independently for Kananga.
- **$0.375/hour $\times$ 70,250 hours = $26,344.** With cash and materials: **$27,890**, or **30.5%** of reported provincial property-tax receipts.
- Historical nominal wage benchmark, without inflation adjustment.

Sources: ACP (2021), Kananga STEP II Brandily et al. (2024), p. 5.

Back Accounting Income sensitivity

Arriving together: composition specification

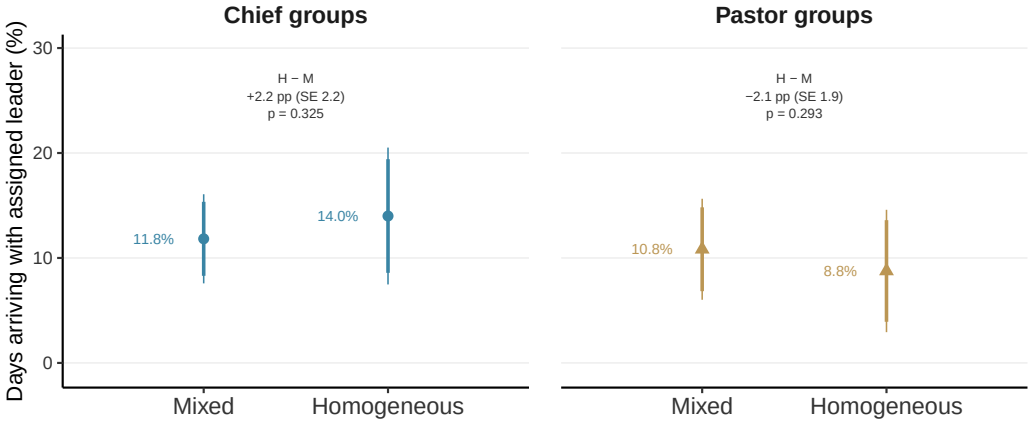
Run separately in chief groups and pastor groups:

$$Y_{ig} = \alpha + \beta H_g + \rho(Y_{i0} - \overline{Y}_0) + \varepsilon_{ig}.$$

- Y_{ig}: share of six post-meeting days arriving in the same chain as the assigned leader; both contribute labor. Nonattendance is zero.
- Consecutive check-ins at most **two minutes apart** form a chain. Build chains using all attendees, including non-roster arrivals. Include the first chain.
- Leader groups only: 336 chief-group and 336 pastor-group residents. $H_g = 1$ for homogeneous groups. Adjust for the same outcome over six pre-meeting days.
- β: homogeneous minus mixed. Pool disclosure without extra regressors. Neighborhood-clustered standard errors; levels at the common baseline mean.

Composition levels | Disclosure specification | Alternative explanations | Levels | Other specification | Appendix

Arriving with the assigned leader: composition



Two-minute links; all chains included. Separate leader-group regressions: homogeneous indicator + baseline coarrival. Six post-meeting days; nonattendance = 0. 90% / 95% neighborhood-clustered intervals.

Alternative explanations | Specification | Other specification | Appendix

Arriving together: disclosure specification

$$Y_{ig} = \alpha_L + \beta_L H_g + \gamma_L P_g + \delta_L H_g P_g + \rho_L (Y_{i0} - \overline{Y}_0) + \varepsilon_{ig}.$$

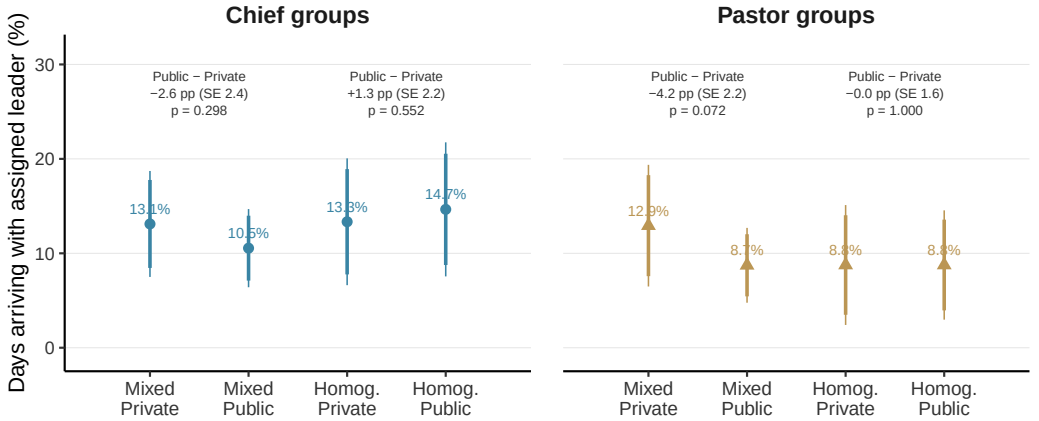
- Same two-minute chain outcome, including the first chain. Nonattendance is zero. One observation per resident, in leader groups only.
- Estimate separately for chiefs and pastors (336 residents each). H_g: homogeneous, P_g: public disclosure. Adjust for baseline own-leader coarrival.
- Levels use the same baseline mean. Tests compare plotted levels. Standard errors cluster by neighborhood.

Public − Private: γ_L in Mixed groups,
$\gamma_L + \delta_L$ in Homogeneous groups.

Disclosure levels | Composition specification

Alternative explanations | Levels | Other specification | Appendix

Arriving with the assigned leader: disclosure



Two-minute links; all chains included. Separate leader-group regressions: composition × disclosure + baseline coarrival. Six post-meeting days; nonattendance = 0. 90% / 95% neighborhood-clustered intervals.

Alternative explanations | Specification | Other specification | Appendix

Survey questions: wording and response options

- Ethnic identity ranking
- Allport: prompt and scale
- Intrinsic
- Extrinsic
- Leader evaluations
- Authority and trust
- Divine enforcement
- Leadership roles
- Leader contact
- Reported consequences
- Social networks
- Network composition

Appendix

Religiosity: questionnaire and response scale

Verbatim English prompt

"Here are a number of statements that may or may not apply to you. There are no right or wrong answers, so just answer honestly, considering how you compare to most people."

Verbatim response labels

1. Disagree strongly
2. Disagree a little
3. Neither agree nor disagree
4. Agree a little
5. Agree strongly

8888: JE NE SAIS PAS (don't know).

The current paper scores five intrinsic and four extrinsic items. Responses outside 1–5 are excluded from scoring.

Five intrinsic items | Four extrinsic items

Back to results | Survey questions

Intrinsic religiosity: five survey items

- "I enjoy reading about my religion."
- "It doesn't much matter what I believe so long as I am good."
- "It is important to me to spend time in private thought and prayer."
- "I try hard to live all my life according to my religious beliefs."
- "Although I believe in my religion, many other things are more important in life."

Scoring: average items 1, 3, 4, 5 and 9; reverse items 3 and 9 ($6 - x$). At least three valid responses. High score: > 3.2; ties low.

Prompt and response options | Extrinsic items

Back to results | Survey questions

Extrinsic religiosity: four survey items

- "I go to church because it helps me to make friends."
- "What religion offers me most is comfort in times of trouble and sorrow."
- "I go to church mostly to spend time with my friends."
- "I go to church mainly because I enjoy seeing people I know there."

Scoring: average items 2, 6, 7 and 8, with at least three valid responses. High score: > 2.75; ties low. Includes social motives and comfort in difficult times.

Prompt and response options Intrinsic items

Back to results Survey questions

Leader evaluations: survey statements

- "[person] (participant [number]) is generally hardworking."
- "[person] (participant [number]) is trustworthy."
- "[person] (participant [number]) inspires and motivates me personally."
- "[person] (participant [number]) consistently sets a strong moral example for our community."

Response options: 1 = Strongly disagree; 2 = Somewhat disagree; 3 = Neither disagree nor agree; 4 = Somewhat agree; 5 = Strongly agree

[person] and [number] identify the group member being evaluated.

Back to results Survey questions

Sanctioning authority and trust: survey questions

Authority
"[person] (participant [number]) has the authority and ability to punish non-cooperators and reward cooperators."

Options: 1 = Strongly disagree; 2 = Somewhat disagree; 3 = Neither disagree nor agree; 4 = Somewhat agree; 5 = Strongly agree

Trust
"How much do you trust [person] (participant [number]) to fulfill their responsibilities during the next salongo activity?"

Options: 1 = Not at all; 2 = A little bit; 3 = Mostly; 4 = Completely

Back to results Survey questions

Belief in divine enforcement: survey questions

- "How often does God help people or reward them for good behavior?"
- "How often does God punish people for improper behavior?"

Response options: 1 = Never; 2 = Rarely; 3 = Sometimes; 4 = Often; 5 = Always

Back to results Survey questions

Leader contact and mobilization: survey questions

- "How many times did you talk to [person] (participant [number]) since the first meeting of the program happened?"
- "Did [person] ask you to participate/contribute in Salongo?"
- "Did [person] mention any consequences—positive or negative— for participating or not participating?"

Responses: a count of conversations for the first question; Yes/No for the other two. The latter questions refer to a reported conversation about Salongo.

Types of consequences

Back to results Survey questions

Reported consequences: question and options

"You said consequences were mentioned. Which of these came up? Select all that apply."

- Supernatural reward – e.g., blessings, protection, good fortune
- Supernatural punishment – e.g., misfortune, sickness, bad luck
- Material reward – e.g., money, goods, services given for participating
- Material punishment – e.g., loss of goods, fines, withheld help
- Social reward – e.g., praise, status, public recognition
- Social punishment – e.g., blame, shame, exclusion from group

Multiple responses allowed. Asked after consequences were reported.

Back to results Survey questions

Social networks: name-generator questions

"I am now going to describe several situations. For each situation, please think of your close friends who you would count on. For the following questions, please think of people who do not live in your own home."

Confidant network
"Who are the people with whom you share your personal projects and future plans?"

Material-support network
"Who are the people with whom you share money, food, or other things when they or you are in need."

Response format: list names; the enumerator records the count. Follow-ups count members sharing ethnicity, umbrella ethnicity, denomination, church, or extended family.

Network-composition questions

Back to results Survey questions

Network composition: follow-up questions

Verbatim templates; populated fields in brackets

- "Out of all the friends you have cited: [names listed] How many of them are from [ethnicity]?"
- "Out of all the friends you have cited: [names listed] How many of them are [umbrella ethnicity]?"
- "Out of all the friends you have cited: [names listed] How many of them are from your church?"

Response format: a count. The same questions are repeated for material-support networks. Populated values identify the named friends and the respondent's ethnicity or umbrella ethnicity.

Back to results Survey questions

Leadership roles: survey questions

"Do you take upon any traditional leadership role in this locality? If yes, which one?"

Options: avenue = Chef d'avenue; avenue_adj = Chef d'avenue adjoint; capita = Capita; costumary = Chef coutumier; notable = Notable; no_leader = No; Other (specify).

"Do you take any leadership role in your church? If yes, which one?"

Options: no_leader = No; pasteur = Pasteur; pretre = Pretre; preacher = Predicateur; leader_jeune = Leader Jeune; Other (specify).

Back to results Survey questions

Does ethnic identification proxy for lower religiosity?

1. **Little evidence of lower religiosity among stronger ethnic identifiers.**

	Homogeneous	Mixed
Intrinsic religiosity	$+0.069$ ($p = .150$)	$+0.011$ ($p = .785$)
Extrinsic religiosity	$+0.067$ ($p = .123$)	-0.023 ($p = .636$)
Christian identity, adjusted ranks	$+0.173$ ($p < .001$)	$+0.025$ ($p = .591$)

Correlations with own baseline ethnic identification; group-clustered p-values. Christian and ethnic ranks each exclude the other identity.

2. **Group context, rather than majority status alone.** Pastors can mobilize the chief's coethnics in mixed teams; their effect is small in homogeneous teams. The direct difference is imprecise.

Majority status: context versus individual

◂ Homogeneous results ◂ Mixed-group results Rank adjustment and group measure

Identity ranks excluding the compared identity

- **Group ethnic score:** remove Christian identity; score ethnicity (5 = most important), then average across ordinary members.
- **Own Christian score:** remove ethnic identity, then score Christianity among the remaining five options.
- Direct competition for a rank is removed. Shared remaining options can still generate dependence.

Group ethnic score versus own religious measures:

	Homogeneous	Mixed
Intrinsic religiosity	-0.015 ($p = .756$)	-0.036 ($p = .373$)
Extrinsic religiosity	$+0.029$ ($p = .521$)	-0.076 ($p = .123$)
Christian identity, adjusted ranks	$+0.115$ ($p = .011$)	$+0.024$ ($p = .584$)

Back to alternative explanation Original survey question

Recruited leaders and residents: backgrounds

	Traditional leaders	Religious leaders	Residents
Mean per neighborhood	2.14	1.68	101.14
Ethnicity (territory of origin)			
Dibaya	45.0	44.7	38.0
Demba	21.7	29.8	25.5
Kazumba	15.0	6.4	17.7
Other origins	18.3	19.1	18.8
Religious denomination			
Pentecostal	30.0	40.4	50.2
Catholic	18.3	2.1	11.7
Neo-Apostolic	16.7	17.0	10.6
Kimbanguist	6.7	21.3	6.4
Non-Christian	3.3	2.1	2.2
Other Christian	25.0	17.0	18.9

Shares (%). Full recruitment, 28 wave 1–7 neighborhoods.